

# Automated Micro-CT Image Segmentation and Void Analysis of HDPE-HGM Syntactic Foams Using Machine Learning

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# Abstract

Syntactic foams are a subclass of composite materials in which the matrix is embedded with hollow microspheres, offering a high strength-to-weight ratio suitable for the aerospace, marine, and automotive industries. The microspheres create porosity, visually indistinguishable from voids caused by manufacturing during imaging, both of which influence mechanical performance. Quantification and classification are, therefore, crucial for quality control and predictive modelling. This thesis proposes an integrated pipeline utilizing micro-CT imaging and unsupervised learning to segment and quantify voids, particles, and matrix in HDPE-HGM syntactic foams. A hybrid segmentation technique leveraging slice-wise thresholding and K-means clustering resulted in an F1 score of 0.92 compared to LabKit for voids. Voids were further classified into raster voids and HGM cavities based on shape descriptors, with subsequent spatial analysis performed using the R-index and clustering coefficients. Morphological features were correlated to Ultimate Tensile Strength and Young's Modulus. A custom GUI enables 3D visualization and segmented mesh export, facilitating the creation of digital twins for enhanced material analysis and process optimization.

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# List of Abbreviations

|      |                                    |
|------|------------------------------------|
| HDPE | High-Density Polyethylene          |
| HGM  | Hollow Glass Microspheres          |
| μCT  | Micro-Computed Tomography          |
| CT   | Computed Tomography                |
| FFF  | Fused Filament Fabrication         |
| FDM  | Fused Deposition Modeling          |
| AM   | Additive Manufacturing             |
| NDT  | Non-Destructive Testing            |
| DT   | Destructive Testing                |
| UT   | Ultrasonic Testing                 |
| IRT  | Infrared Thermography              |
| RT   | Radiographic Testing               |
| AE   | Acoustic Emission                  |
| ECT  | Eddy Current Testing               |
| SHM  | Structural Health Monitoring       |
| SHPB | Split Hopkinson Pressure Bar       |
| DMA  | Dynamic Mechanical Analysis        |
| SEM  | Scanning Electron Microscopy       |
| TEM  | Transmission Electron Microscopy   |
| CLSM | Confocal Laser Scanning Microscopy |
| AFM  | Atomic Force Microscopy            |
| ROI  | Region of Interest                 |
| FOV  | Field of View                      |
| CSV  | Comma-Separated Values             |
| GUI  | Graphical User Interface           |
| NND  | Nearest Neighbour Distance         |
| ML   | Machine Learning                   |
| CNN  | Convolutional Neural Network       |
| FEA  | Finite Element Analysis            |

# List of Variables

|                                        |                                                                                               |
|----------------------------------------|-----------------------------------------------------------------------------------------------|
| $\sigma$                               | Stress                                                                                        |
| $\varepsilon$                          | Strain                                                                                        |
| $V_{\text{Void}}$                      | Void Volume Fraction                                                                          |
| NND                                    | Nearest Neighbor Distance                                                                     |
| $F_1$                                  | F1 Score                                                                                      |
| IoU                                    | Intersection over Union                                                                       |
| AR                                     | Aspect Ratio                                                                                  |
| C                                      | Circularity                                                                                   |
| E                                      | Young's Modulus                                                                               |
| UTS                                    | Ultimate Tensile Strength                                                                     |
| d                                      | Diameter                                                                                      |
| A                                      | Area                                                                                          |
| P                                      | Perimeter                                                                                     |
| $T_1, T_2$                             | Greyscale Thresholds                                                                          |
| $N_{\text{void}}, N_{\text{particle}}$ | Number of voids and particles                                                                 |
| c                                      | Cohesion ( $0 \leq c \leq 1$ )                                                                |
| $x_i, x_j$                             | Two data points in the dataset                                                                |
| $\delta\epsilon_p(x_i)$                | Neighbourhood of point $x_i$ (set of points within $\epsilon$ distance under attribute set P) |
| $\delta\epsilon_p(x_j)$                | Neighbourhood of point $x_j$                                                                  |
| $\cap$                                 | Intersection of two sets                                                                      |
| $\cup$                                 | Union of two sets                                                                             |
| $\theta$                               | Orientation angle                                                                             |
| h                                      | Height of bounding box                                                                        |
| w                                      | Width of bounding box                                                                         |

# 1. Introduction

## 1.1 Motivation

Destructive testing, although long considered the gold standard for assessing composite materials, suffers from considerable limitations in terms of cost, efficiency, and sustainability. Furthermore, manufacturing and test workflows can yield as much as 30% material waste through specimen creation for single-use destructive assessment [1]. This constitutes not only an immediate financial expense but also an opportunity cost, especially in applications where low-volume, high-value components, such as those used in aerospace or subsea infrastructure, are difficult or impossible to replace. Additionally, in many situations, tested specimens cannot be replaced with untested counterparts, since even identical fabrication parameters do not ensure identical internal microstructures. The environmental costs are similarly alarming, since composite waste disposal is energy-intensive and rarely sustainable. These constraints highlight the critical demand for sophisticated non-destructive testing (NDT) and automated analysis pipelines that can identify internal microstructural information such as void morphology, particle distribution, and defect clustering, without compromising sample integrity. Such techniques not only conserve valuable material for actual service applications but also produce high-fidelity data that informs predictive frameworks like finite element analysis (FEA) or machine learning (ML) models. The current study addresses this need by establishing an automated, micro-CT-based strategy for three-dimensional characterization of syntactic foams, enabling robust quantification of defects to inform process improvements and enhance the fidelity of downstream computational model predictions.

## 1.2 Research Objectives

This thesis focuses on developing a framework for non-destructive analysis of High-Density Polyethylene (HDPE) and Hollow Glass Microspheres (HGM) syntactic foams, utilizing micro-computed tomography ( $\mu$ CT) as the imaging technique to characterize the microstructure. The goal is to identify the voids and HGM particles, which are referred to as HGM shells, simultaneously and segment the dataset to extract meaningful morphological and spatial descriptors from CT image data, differentiate between defect-induced and HGM porosity, referred to as HGM cavities or HGM voids, and organize the results in a structured and reproducible format. These objectives aim to improve efficiency, consistency, and scalability in characterizing complex microstructures, particularly for additively manufactured foams.

Objectives:

- (1) To develop an unsupervised image segmentation algorithm using K-means clustering for distinguishing voids, microspheres, and matrix regions in  $\mu$ CT images of syntactic foams.
- (2) To classify void types, extract and quantify morphological and spatial features such as area, circularity, aspect ratio, sphericity, and nearest neighbour distance in both 2D slices and 3D volumes.
- (3) To correlate the morphological features with tensile properties.

### 1.3 Thesis Structure and Scope

This thesis presents a systematic approach to understanding the microstructure and mechanical behaviour of additively manufactured syntactic foams, specifically HDPE-HGM composites in the present instance. The research combines experimental methods, advanced image analysis, and quantitative modelling to establish structure–property relations that are essential to the development of lightweight and high-performance materials.

The scope of this work includes 3D printing syntactic foam samples,  $\mu$ CT scanning for microstructural analysis, and establishing correlations between microstructural features and mechanical performance through tensile testing. It encompasses, additive manufacturing techniques, high-resolution micro-computed tomography ( $\mu$ CT) imaging, automatic image segmentation using clustering methods, morphological feature extraction, and the correlation of these features with tensile performance. The methodology developed is computationally efficient, unsupervised, and suitable for high-throughput quality control and design optimization. The work utilizes machine learning to identify threshold values for image segmentation and phase classification, after which voids and particles are quantified and their morphological features extracted using numerical image analysis methods.

Chapter 1 of the thesis provides an overview of the background on syntactic foams, existing testing techniques (both destructive and non-destructive), the rationale for utilizing  $\mu$ CT-based characterization, and the objectives and significance of the research. Chapter 2 then summarizes the previous research on syntactic foam fabrication, void analysis methods, and image segmentation methods. The advantages and limitations of various approaches are discussed to address the gap that this research aims to fill. Chapter 3 covers the HDPE-HGM material system, 3D printing via FFF, specimen preparation,  $\mu$ CT scanning protocol, and image processing pipeline. It also details the physical parameters and imaging settings essential for reproducibility. Chapter 4 is dedicated to the development of the segmentation algorithm, specifically the unsupervised K-means segmentation method with neighbourhood enhancement, and testing its performance against standard metrics, such as the F1 score and Intersection-over-Union (IoU). Following this, Chapter 5 presents the calculation of void and particle characteristics, such as circularity,

aspect ratio, nearest neighbour distance, orientation, and sphericity, in 2D and 3D using custom Python-based code with PyVista visualization, followed by correlation analyses that relate these morphological descriptors to the mechanical properties of the samples. Trends are identified for different infill densities and HGM compositions, offering insights into key design parameters. Chapter 6 summarizes the contributions and findings of this work, discusses limitations (such as resolution constraints and dataset size), and proposes potential extensions, including the integration of machine learning, real-time segmentation, and multi-scale modelling.

This thesis presents an end-to-end framework for microstructural characterization and mechanical analysis of syntactic foams, contributing to both a fundamental understanding and practical proficiency in material development and process optimization.

## 1.4 References

- [1] Tafesse, S., Girma, Y. E., & Dessalegn, E. (2022). Analysis of the socio-economic and environmental impacts of construction waste and management practices. *Heliyon*, 8(3), e09169. <https://doi.org/10.1016/j.heliyon.2022.e09169>

## 2. Literature Review

### 2.1 Introduction to Syntactic Foams

Syntactic foams are a family of composite materials characterized by their porous structure, created by embedding hollow microspheres, usually made of glass, ceramics, or polymers, into a continuous matrix. These foams offer low weight, high strength-to-weight ratios, and excellent thermal and acoustic insulation properties, making them particularly promising for various engineering applications, especially in the aerospace, marine, automotive, and defense industries. Unlike traditional foams, which achieve porosity through gas expansion during polymerization, syntactic foams develop their cellular structure through the intentional inclusion of pre-formed hollow particles. This controlled process results in a more uniform and predictable internal architecture. Syntactic foams differ from expanded foams because the enclosed voids are surrounded by solid particle walls rather than a continuous gas-filled network, thereby reducing moisture absorption and enhancing mechanical resistance to hydrostatic pressure [1]. Their properties can be engineered by a choice of matrix, microcell structure, and their concentration, as well as reinforcement, tougheners, etc. [2]. The innovation of syntactic foams was initially spurred by the demand for lightweight materials for deep-sea applications, where traditional buoyancy systems, such as those using gasoline, were constrained by their flammability and compressibility. Syntactic foams, which are inherently buoyant and compressively stable at high pressure, offered a reliable alternative to the submersible vehicle design [1]. The application of syntactic foams has grown steadily over the years, often extending far beyond the area of marine engineering. These are currently integral parts in thermal insulation systems for undersea pipelines, electronics packaging, aircraft structural parts, and lightweight cores for sandwich panels [3].

### 2.2 Manufacturing of Syntactic Foams

Syntactic foams are manufactured differently from regular foams. The primary source of the syntactic foams' porosity is prefabricated hollow microspheres. Coating microspheres is one method for combining a binder with microspheres. Syntactic foams are produced by compressing a polymeric or metallic matrix with hollow microspheres, and the processing techniques are customized based on the end-use application, matrix type, and microsphere diameter. Conventional processes include compression molding, injection molding, and extrusion, each with its own merits in controlling microstructure and mechanical properties.

One of the commonly applied processes for syntactic foams based on thermoplastics is compression molding. It allows for uniform distribution of microspheres and minimizes their damage during processing. Maharsia et al., document successful production of HDPE-glass microballoon syntactic foams by compression molding, noting that good control over temperature and pressure is essential to maintain structural integrity and avoid crushing of spheres [1]. Injection molding, which is suitable for production

on a large scale, has also been used for syntactic foams. The industrial processability of HDPE-based syntactic foams was demonstrated by Kurapati et al. They pointed out problems such as microsphere degradation due to high shear rates and cited process parameters to prevent this from happening [2]. Extrusion is optimally suited for continuous processing and filament production. Lawrence et al. used extrusion to produce polymeric syntactic foams with expandable microspheres, followed by stereological characterization to establish cell shape and distribution [3]. Shear flow in the process yielded aligned microballoons, which affected anisotropic mechanical properties. Additionally, pre-molding techniques such as dry blending and pre-impregnation have also been explored. Rawlings et al presented various pre-mold processing strategies that enhance microsphere dispersion, reduce agglomeration, and provide greater control over void content [4].

### 2.2.1 Additive Manufacturing

Additive Manufacturing (AM) has emerged as a transformative technique for fabricating complex structures with tailored properties, offering flexibility that traditional molding methods cannot match. Its integration into syntactic foam production has opened avenues for enhancing microstructural control and reducing material waste, which is particularly advantageous given the composite nature of syntactic foams. Early studies highlight the viability of AM in fabricating polymer-based syntactic foams with tailored architecture and mechanical performance.

Research has demonstrated that AM enables the precise placement of reinforcement particles, such as hollow glass microspheres or ceramic fillers, within a polymer matrix, which is crucial for customizing density and mechanical behaviour. For example, Singh et al. emphasized the potential of AM in developing multiscale, hierarchical microstructures in syntactic foams, which enables enhanced energy absorption and lightweight performance [5]. Furthermore, the feasibility of using various thermoplastics such as PLA, ABS, and HDPE in AM processes broadens the material selection for syntactic foam applications [6].

Fused Filament Fabrication (FFF), as illustrated in Figure 1.2.1, is a cost-effective and widely accessible AM method that utilizes thermoplastic filaments extruded through a heated nozzle. FFF offers control over infill density, layer height, raster angle, and print speed, which are parameters that directly affect porosity, interfacial bonding, and ultimately the mechanical properties of the printed syntactic foam [7]. The ability to tune these variables makes FFF especially suitable for research-oriented and small-scale production of syntactic foams, where flexibility in prototyping is key.

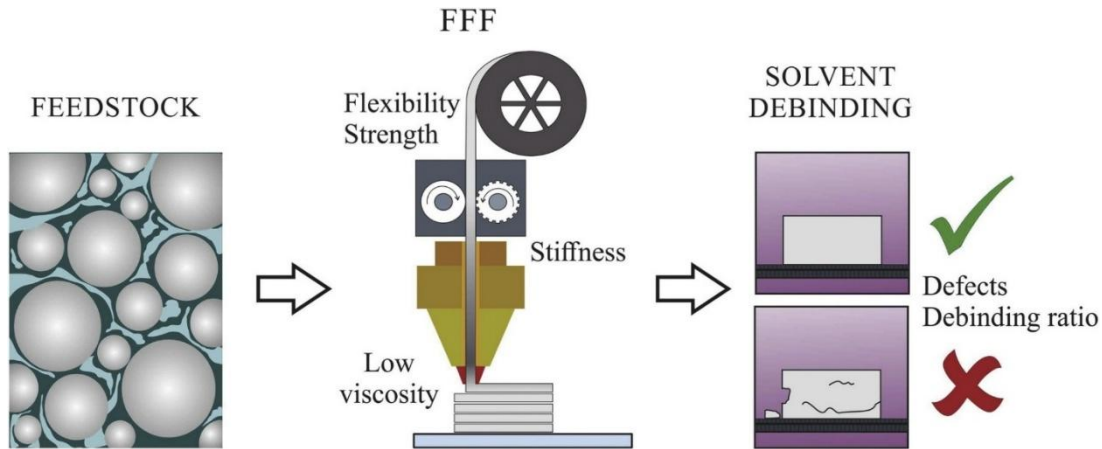


Figure 1.2.1 : Schematic representation of the FFF process, highlighting feedstock composition, extrusion parameters, and the influence of solvent debinding on defect formation [8].

The process involves melting a filament and extruding it through a nozzle, which then deposits the material onto a build platform following a predetermined path. This layer-by-layer method continues until the object is fully formed. In this thesis, FFF is chosen because it works well with thermoplastic matrices, such as HDPE, offers low initial costs, and is flexible enough to incorporate reinforcement fillers. Additionally, its ability to control each layer facilitates the investigation of relationships between microstructure and properties in syntactic foams, aligning with the main aims of this research.

## 2.2.2 Voids and Defects

Additive Manufacturing, specifically fused filament fabrication (FFF), has become popular for producing polymer parts due to its affordability and accessibility. The printed components, however, do possess internal and external defects that can reduce their strength [9].

One of the most common issues faced in FFF is voids that can appear between or within the printed layers. These voids are caused by uneven extrusion of the filament, incorrect print parameters such as insufficient temperature or excessive speed, or poor inter-layer adhesion during cooling [10]. In addition to voids, other typical flaws often present in prints include stringing, where fine strands of filament link printed features, and warping, where pieces bend away from the print bed, resulting in deformation and size discrepancies. One of the primary reasons for the formation of these flaws is the nature of layer-by-layer printing, which does not allow for seamless bonding throughout the structure. As each new layer is deposited, the quality of adhesion is vitally dependent on the mechanical and thermal interface conditions. If the final deposited layer is very cold or material flow is uneven, voids are trapped, and interlayer separation occurs [11]. Such defects not only compromise the structural integrity of the finished product but also impart random behaviour under load. Hence, choosing appropriate printing parameters is imperative.

Recent advancements in research have introduced methods to monitor and detect such flaws as the printing process progresses. For example, machine learning-based computer vision systems have been trained to detect patterns that indicate the early signs of developing flaws, enabling process correction or intervention at the right time [9]. These systems can detect surface issues, such as poor filament deposition or unexpected material buildup, which are often precursors to more significant internal malfunctions. Additionally, the consolidation of information from various sensors, including optical, acoustic, or current flow sensors, has enabled improved detection sensitivity and reliability for different printing processes [10]. Even with these monitoring methods being an improvement, many difficulties remain. Models trained to identify defects typically fail if the printing material, printer model, or part geometry is altered. In addition, there is no standard dataset or testing regime that is widely applied, which makes it very difficult to compare results or training systems that work in every environment [11]. Thus, as detection and quality control methods improve, their current limitations continue to necessitate the prevention of voids and defects largely depend on adequate process planning and frequent equipment calibration. This highlights the need to account for voids when evaluating the strength of printed samples, as voids have a significant influence on mechanical performance. Moreover, even when identical printing parameters are used, no two samples are truly the same, which reinforces the importance of non-destructive testing methods. It is essential to test the actual sample intended for use rather than relying on the assumption that it shares identical microstructural characteristics with a destructively tested counterpart.

## 2.3 Destructive and Non-Destructive Techniques

Syntactic foam testing, particularly for void detection and mechanical analysis, involves both destructive and non-destructive approaches, each applied to different objectives in quality control and research. The approaches provide information regarding microstructural integrity, defect morphology, and the influence of voids on mechanical performance. Destructive testing (DT) is widely used to measure the mechanical performance of syntactic foams directly under tensile, compressive, and flexural loading. Standard tests include quasi-static tension and compression testing, split Hopkinson pressure bar (SHPB) testing for high-strain-rate response, and dynamic mechanical analysis (DMA) for viscoelastic behaviour [4]. The techniques mentioned above are utilized as standards for verifying non-destructive test results. Kumar et al. conducted quasi-static tensile testing on syntactic foam samples having variable microsphere volume fractions and wall thicknesses. The results showed that failure strength significantly relies on wall thickness and spatial distribution of microspheres, and the clustered microspheres affect crack initiation at an early stage [5]. Furthermore, post-failure Scanning Electron Microscope (SEM) scanning analysis confirmed that fracture patterns typically trace interfacial debonding between the microspheres and the matrix, which cannot be easily identified with non-destructive methods. Further destructive testing used microscopy

combined with sectioning techniques to study failure surfaces and void dispersion. Optical microscopy and scanning electron microscopy (SEM) are standard techniques for identifying interfacial flaws, micro void coalescence, and shell break. However, they can cause sample damage and limit repeat testing [6].

Destructive test methods, while beneficial for specific tests, have several drawbacks that limit their use, especially in high-value or safety-critical applications. First, destructive testing permanently degrades the test article, making it unsuitable for expensive components, those in limited supply, or those intended for active operation, such as aerospace composites [7]. Secondly, these methods are typically applied to small sample sizes and do not permit an extensive examination of large or complex structures. Thus, they may occasionally fail to detect localized or hidden defects that could compromise the integrity of the entire structure [8]. Lastly, destructive testing is not compatible with in-service monitoring, which is crucial for ensuring the sustained safety and reliability of key infrastructure, such as aircraft, bridges, and wind turbines [6]. Instead, non-destructive test methods offer a strong alternative for ascertaining material integrity without destroying it. NDT methods such as ultrasonic testing (UT), infrared thermography (IRT), radiographic testing (RT), acoustic emission (AE), eddy current testing (ECT), and stereography are utilized to find both surface and subsurface flaws, delaminations, voids, and misalignments of fibers, without altering the structure or the material properties. These methods are widely used in aerospace, automotive, civil infrastructure, and energy industries, providing a low-cost, safe, and reliable inspection tool for quality control and structural health monitoring. NDT technology has enabled real-time inspection, damage detection at an early stage, and integration with intelligent monitoring systems, allowing them to play a fundamental part in engineering applications today [6][7][8]. Non-destructive testing provides a substitute for destructive techniques, offering internal structural evaluation without compromising sample integrity. Among the NDT techniques used, ultrasonic testing (UT), X-ray computed tomography (CT), and acoustic emission (AE) testing are most common in syntactic foam research.

Ultrasonic testing is especially valuable due to its high sensitivity to internal defects and its ability to estimate mechanical behaviour. Woldesenbet et al. used pulse-echo ultrasonic imaging to assess the dynamic Young's modulus and Poisson's ratio of syntactic foams with different microsphere volume fractions. These moduli were calculated using longitudinal and shear wave velocities measured with high-frequency transducers, following principles of physical acoustics. Attenuation behaviour was also examined, it increased with volume fraction up to 30%, but then decreased due to increased particle contacts, which enabled wave transmission over scattering. This approach proved effective in differentiating foam types and predicting performance trends without causing physical failure of specimens. Employing ASTM standards such as E664-93 and E494-95 to measure attenuation and velocity further supports reliability and repeatability in industrial settings [9]. In a separate study, Kalyan et al. demonstrated

the use of ultrasonic C-scan imaging to characterize clustering and void distribution within syntactic foams. Clustering and inhomogeneity in the packing of microspheres increased backscatter and reduced signal transmission, indicating areas of greater local porosity [10]. These results imply that ultrasonic techniques not only provide modulus estimates but also qualitative spatial maps of internal structure. X-ray computed tomography (CT) is a powerful non-destructive method which can give detailed three-dimensional reconstructions of foam microstructures at high resolution. CT is capable of directly measuring void size and shape, volume, and spatial location. CT imaging, while more costly and specialized, is commonly regarded as the ‘gold standard’ for internal defect analysis, primarily when used in concert with digital image processing for void segmentation [11]. Another NDT technique is acoustic emission monitoring, where the material is subjected to stress, and microcracking or debonding events are tracked in real-time using sensors. The technique helps predict initial damage build-up and failure thresholds [12]. Ultrasonic-based dynamic modulus measurements have the advantage of applying to field applications where components may not be subjected to thorough mechanical testing. The capability to estimate modulus rapidly from shear and longitudinal velocities without damaging a component is essential in structural health monitoring (SHM) systems [13]. Comparative studies have demonstrated the comparative advantages and disadvantages of destructive and non-destructive methods. Destructive testing provides ground-truth mechanical properties and absolute failure modes but is unsuitable for in-service material or high-cost prototypes. Non-destructive methods yield repeatable, spatially resolved, real-time data, but without direct relation to failure measures unless properly calibrated [4][9][11]. The newer developments address hybrid approaches, in which non-destructive testing methods, such as CT and ultrasonics, are used first, before destructive testing, to allow for direct correlation between the internal structure and mechanical performance. This multifaceted strategy enables the development of more accurate predictive models for syntactic foam behaviour underload.

## 2.4 Microstructural Imaging

Microstructural imaging has become a routine method for characterizing the internal structure of complex composite materials, including syntactic foams, which exhibit heterogeneous microstructures comprising embedded voids and reinforcement phases. Microstructural feature distribution, morphology, and evolution mapping are necessary to correlate them with mechanical performance and durability. A variety of imaging methods, including optical, scanning electron, and computed tomography, have been applied across various material systems to provide high-resolution images of internal geometries. NDT methods such as ultrasonic testing (UT), infrared thermography (IRT), radiographic testing (RT), acoustic emission (AE), eddy current testing (ECT), and stereography are utilized to find both surface and subsurface flaws, delaminations, voids, and misalignments of fibers, without altering the structure or the material properties. These methods are

widely used in aerospace, automotive, civil infrastructure, and energy industries, providing a low-cost, safe, and reliable inspection tool for quality control and structural health monitoring. NDT technology has enabled real-time inspection, damage detection at an early stage, and integration with intelligent monitoring systems, allowing them to play a fundamental part in engineering applications today [6][7][8]. Non-destructive testing provides a substitute for destructive techniques, offering internal structural evaluation without compromising sample integrity. Among the NDT techniques used, ultrasonic testing (UT), X-ray computed tomography (CT), and acoustic emission (AE) testing are most common in syntactic foam research.

Ultrasonic testing is especially valuable due to its high sensitivity to internal defects and its ability to estimate mechanical behaviour. Woldesenbet et al. used pulse-echo ultrasonic imaging to assess the dynamic Young's modulus and Poisson's ratio of syntactic foams with different microsphere volume fractions. These moduli were calculated using longitudinal and shear wave velocities measured with high-frequency transducers, following principles of physical acoustics. Attenuation behaviour was also examined, it increased with volume fraction up to 30%, but then decreased due to increased particle contacts, which enabled wave transmission over scattering. This approach proved effective in differentiating foam types and predicting performance trends without causing physical failure of specimens. Employing ASTM standards such as E664-93 and E494-95 to measure attenuation and velocity further supports reliability and repeatability in industrial settings [9]. In a separate study, Kalyan et al. demonstrated the use of ultrasonic C-scan imaging to characterize clustering and void distribution within syntactic foams. Clustering and inhomogeneity in the packing of microspheres increased backscatter and reduced signal transmission, indicating areas of greater local porosity [10]. These results imply that ultrasonic techniques not only provide modulus estimates but also qualitative spatial maps of internal structure. X-ray computed tomography (CT) is a powerful non-destructive method which can give detailed three-dimensional reconstructions of foam microstructures at high resolution. CT is capable of directly measuring void size and shape, volume, and spatial location. CT imaging, while more costly and specialized, is commonly regarded as the 'gold standard' for internal defect analysis, primarily when used in concert with digital image processing for void segmentation [11]. Another NDT technique is acoustic emission monitoring, where the material is subjected to stress, and microcracking or debonding events are tracked in real-time using sensors. The technique helps predict initial damage build-up and failure thresholds [12]. Ultrasonic-based dynamic modulus measurements have the advantage of applying to field applications where components may not be subjected to thorough mechanical testing. The capability to estimate modulus rapidly from shear and longitudinal velocities without damaging a component is essential in structural health monitoring (SHM) systems [13]. Comparative studies have demonstrated the comparative advantages and disadvantages of destructive and non-destructive methods. Destructive testing provides

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To investigate the microstructure of syntactic foams, several methods such as Optical Microscopy, Scanning Electron Microscopy (SEM), X-ray Computed Tomography (CT/ $\mu$ CT), Ultrasonic Imaging, Transmission Electron Microscopy (TEM), Confocal Laser Scanning Microscopy (CLSM), Atomic Force Microscopy (AFM), Stereo Microscopy, Image-based Stereology, Digital Image Processing & Segmentation. For instance, the application of X-ray computed tomography (CT) to syntactic foams enables non-destructive three-dimensional (3D) visualization of microsphere distribution, interfacial adhesion, and defect morphology within the matrix. The method provides volumetric data without damaging the sample and is highly efficient in evaluating internal damage mechanisms or void morphology for various loading conditions [14]. Elsewhere, Alawneh and Soliman highlighted microstructural imaging of asphalt concrete mixtures and the synergy of scanning electron microscopy (SEM), digital camera imaging, and CT scanning. Their study illustrated the effects of imaging resolutions and algorithms on the detection of pores, crack extension, and aggregate distribution. Although their focus was on asphalt systems, the imaging techniques described are broadly applicable to polymer-based syntactic foams in which local heterogeneity dictates mechanical response [15]. Davis et al further explored microstructural imaging specifically for syntactic foam systems. They emphasized the use of SEM to identify matrix-filler interactions, interfacial debonding, and fracture morphologies observed after mechanical testing. The study highlighted that surface high-resolution imaging, when combined with image analysis techniques, can yield quantitative descriptors such as void volume fraction, microsphere cracking, and matrix fragmentation, all of which are pertinent to micromechanical modelling [16]. Stereological methods, employed by Lawrence et al., utilize unbiased statistical sampling of optical cross-sections to obtain quantitative microstructural measurements. Ultrasonic testing provides an additional method for measuring mechanical properties in situ. Woldeesenbet et al. demonstrated the use of pulse–echo ultrasonic waves for determining dynamic elastic moduli (Young's modulus and Poisson's ratio) of syntactic and particulate foams [9].

Among all the available test methods, this study employed Micro Computed Tomography ( $\mu$ CT) to scan the samples because it can capture the entire 3-dimensional volume, a valuable characteristic for accurately determining the internal morphology of syntactic foams. Visualization and measurement of internal voids,

microsphere distribution, clustering behaviour, and interfacial debonding are enabled using this technique without physically sectioning the sample.  $\mu$ CT possesses greater depth penetration and spatial resolution than surface-based methods, making it especially appropriate for determining composite material structural integrity and heterogeneity. Its non-destructive nature preserves the specimens for additional mechanical testing or multiple imaging, allowing for the entire multimodal characterization workflow. The high-fidelity volumetric data obtained by  $\mu$ CT serve as the basis for accurate image segmentation, morphometric analysis, and microstructure-based modelling.

#### 2.4.1 $\mu$ CT Imaging

X-ray micro-computed tomography (micro-CT) has emerged as a powerful non-destructive tool for three-dimensional (3D) characterization of the microstructure of syntactic foams. Compared to conventional imaging techniques, micro-CT allows volumetric visualization and quantitative measurement of internal features, such as voids, microsphere distribution, and defects, without compromising the specimen integrity for further testing [12].

For syntactic foams, micro-CT enables the precise measurement of microsphere size distribution, packing structure, and porosity, which is crucial in predicting mechanical and thermal behaviour. For instance, Zhang et al. demonstrated its utility in describing fly ash/Al syntactic foams, imaging the heterogeneous microstructure, and facilitating image-based finite element models with a close representation of the actual geometry [12]. Likewise, in cementitious syntactic foams, in-situ micro-CT permitted the monitoring of microstructural development during compressive loading, with direct mapping between failure mechanisms and material architecture [13].

Apart from qualitative visualization, micro-CT also enables quantitative assessment of morphological parameters, such as pore volume fraction, ligament thickness, and connectivity. This quantitative capability, as illustrated in research on polymeric silicone foams, provides statistical insight into lot-to-lot variability, aging effects, and performance degradation, which is not obtainable by destructive sectioning techniques [14]. In metallic foams,  $\mu$ CT has also been utilized to quantify structural anisotropy and identify damage initiation sites, improving the mechanistic understanding of load-bearing capabilities [15]. Coupled with mechanical testing,  $\mu$ CT enables sophisticated investigations, such as real-time 3D deformation tracking and crack propagation, providing high-quality data for validating computational models [16]. Despite limitations such as the complexity of segmentation and the requirement for high-resolution scans to resolve fine features,  $\mu$ CT is unparalleled in its ability to provide accurate, reproducible, and comprehensive microstructural characterization of syntactic foams.

### 2.4.2 Segmentation Challenges in $\mu$ CT Data

Micro-CT provides valuable insight into the internal structure of syntactic foams, however, the processing of raw volumetric datasets into useful quantitative information is highly dependent on accurate segmentation. Segmentation involves classifying voxels into distinct phases, such as the matrix, microspheres, and voids. In practice, several factors make this process challenging for syntactic foams. A significant limitation is the partial volume effect, whereby the finite voxel resolution blurs material boundaries, resulting in mixed-intensity voxels that cannot be easily attributed to a single phase [15]. This is especially problematic for thin microsphere shells that are only a few voxels thick in the reconstructed volume. A further challenge regards the intensity overlap between different phases. In some instances, the attenuation difference between the polymer matrix and certain microsphere shell materials is minimal, making it hard to differentiate them based solely on greyscale values [13]. Such complexity can be compounded by beam hardening artifacts, which can corrupt attenuation measurements and reduce the accuracy of segmentation. Scanning artifacts, such as rings or streaks, and noise can introduce confusing patterns into the data that interfere with automatic thresholding [14]. Although noise-reducing filters can suppress such effects, over-smoothing can remove small yet significant features, like microcracks or tiny voids. The intricate geometry of syntactic foams introduces a further level of challenge. Fractured microspheres and irregularly shaped voids and pore networks that are interconnected commonly demand more advanced segmentation strategies than global thresholding. Accordingly, researchers often complement global and local thresholding with morphological operations and sometimes add supervised machine learning classifiers in pursuit of more robust segmentation [12]. While improving performance, these sophisticated approaches require meticulous validation against experimental ground truth for confident quantification.

## 2.5 Image Segmentation Techniques

The images obtained from  $\mu$ CT are to be segmented, i.e., the phases such as voids and particles must be identified and labelled before attempting to quantify them. Image segmentation is a crucial image processing technique that involves partitioning an image into meaningful regions to simplify its analysis and extract relevant structures. It plays a foundational role in numerous fields such as medical diagnostics, materials science, and remote sensing, where interpreting raw image data is insufficient for deriving structural or functional insights. In the context of microstructural imaging or biomedical analysis, segmentation enables the quantitative identification of features such as voids, tissues, or inclusions, allowing for precise measurement and classification [17]. The primary purpose of image segmentation is to facilitate higher-level analysis by grouping pixels into homogeneous regions based on characteristics such as intensity, texture, or shape [18]. This enables efficient feature extraction, object recognition, and further

quantitative evaluation in complex datasets. Several segmentation methods have been developed, each suited to different data types and objectives. Thresholding methods are among the simplest, relying on pixel intensity to delineate objects, and are particularly effective when there's high contrast between regions [18]. Edge-based segmentation methods detect boundaries by identifying discontinuities in image gradients and are widely used in applications where precise shape information is required [17]. Region-based techniques, on the other hand, group neighbouring pixels based on similarity measures such as intensity or texture, and help delineate larger homogeneous areas [19]. Clustering-based segmentation, including k-means and fuzzy c-means algorithms, classifies pixels in feature space and is particularly useful for images that lack clear edges or uniform regions. Model-based methods, such as Markov Random Fields, incorporate spatial relationships and probabilistic modelling to improve robustness in noisy environments [20]. In recent years, artificial intelligence and neural network-based approaches have gained prominence due to their ability to learn hierarchical representations and generalize across various image modalities.

## 2.6 Machine Learning

### 2.6.1 ML for Segmentation

Machine Learning for segmentation, especially the usage of Conventional Neural Networks (CNN), Artificial Neural Networks (ANN) or Deep Learning (DL) in general, has boomed over the last couple years, mainly due to their ability to perform better than traditional segmentation methods. Research has shown that these deep learning models reduce the over and under estimation of voids significantly, when applied for defect detection in  $\mu$ CT images of composite materials [21]. In recent research, the application of machine learning (ML) for image segmentation has been a breakthrough method in the field of non-destructive testing of composite materials. However, extracting useful information from  $\mu$ CT images heavily depends on efficient segmentation methods that can accurately differentiate between phases and damage areas in the material. Traditional manual or semi-automated methods are increasingly viewed as inefficient and prone to subjectivity, thus driving the transition towards data-dependent, ML-based segmentation techniques [1]. Early efforts in automated segmentation often relied on thresholding methods or clustering techniques such as statistical region merging. Although simple, these methods are susceptible to variations in image quality and intensity between different material phases. Their performance is drastically reduced when applied to complicated microstructures or composite systems with similar grayscale distributions between their constituent phases [22]. Consequently, efforts have shifted toward using supervised and unsupervised machine learning algorithms to automate the segmentation task and enhance accuracy. Supervised learning techniques, specifically CNNs, have been widely used for segmenting composite materials. CNNs directly learn discriminative features from annotated data and have demonstrated the capacity to address challenges such as noise, partial volume effects, and overlapping

textures. In a striking example, a deep CNN was trained to automatically segment CT scans of a silicon carbide (SiC) fibre-reinforced ceramic matrix composite [23]. These developments highlight the capability of deep learning models in simplifying segmentation tasks, although they are not exempt from limitations. Importantly, supervised models require large amounts of high-quality labelled data, which are typically not available in composite materials research.

To circumvent the dependence on labelled datasets, researchers have explored unsupervised machine learning approaches for segmentation. A recent study proposed an unsupervised method based on statistical distance measures such as Kullback-Leibler divergence, Helling distance, and Renyi divergence to determine optimal thresholds for segmenting grayscale micro-CT images of CFRP composites subjected to low-velocity impacts [24]. This intensity-based segmentation method utilizes the statistical distribution of pixel intensities to distinguish between damaged areas, eliminating the need for prior training. Among the considered distance measures, the Kullback-Leibler divergence was found to be the most effective in quantifying damage. This method proved its utility in cases where labelling data was inefficient, hence presenting a viable alternative for the automation of the segmentation process.

Despite these developments, many challenges remain. First, the lack of standardized datasets and benchmarking protocols impedes the reproducibility and comparability of machine learning-based segmentation methods across different research groups. Second, supervised models are constrained by the availability of labeled training data, which is both time-consuming and requires domain-specific knowledge. Third, while unsupervised approaches show much promise, they often face challenges in terms of generalization and robustness to varied imaging modalities and material systems. In addition, most current methods overwhelmingly focus on grayscale computed tomography (CT) images, therefore, there is an increasing need to incorporate multi-modal data (for instance, thermal and ultrasonic) within segmentation frameworks to enhance defect detectability and classification accuracy. In addition, there is a lack of research in explainability and uncertainty quantification in ML-based segmentation, which are imperative in high-stakes applications in aerospace, biomedical, and civil engineering fields. The interpretability of segmentation results, particularly in high-stakes settings, remains an ongoing issue. Therefore, future research needs to prioritize not just enhancing accuracy and efficiency but also generating confidence measures and physical justification of model predictions. To combat these gaps, recent attempts have investigated transfer learning and few-shot learning as promising remedies to reduce reliance on large training datasets. These methods take advantage of pre-trained models or meta-learning to transfer segmentation algorithms to new materials or imaging conditions using minimal labelled data. Their application in the context of composite material analysis is, however, in its nascent stage, pointing to rich opportunities for future investigation. In summary, machine learning, both supervised and unsupervised,

has made considerable progress in segmenting intricate microstructures of composite materials. Deep learning models such as CNNs achieve high accuracy and automation at the cost of large amounts of labelled data. Unsupervised approaches offer a data-efficient solution, but they must be carefully designed for reliability. Despite the encouraging advancements, the research is ongoing, with open issues regarding dataset availability, model generalizability, and interpretability. Overcoming these limitations will be key to seamlessly incorporating ML-based segmentation into the composite material analysis workflow.

## 2.7 Gaps in Current Research

Despite the significant progress made in characterizing composite materials using micro-computed tomography ( $\mu$ CT), some key research shortcomings remain in the characterization of syntactic foams, particularly for HDPE-HGM systems. Existing studies have largely focused on qualitative visual observations or manual analysis of voids and inclusions, with little attention given to the development of fully automated segmentation and quantitation routines. Manual and semi-automated approaches are prone to subjective interpretations, operator bias, and extensive time requirements, making them unsuitable for large datasets or high-throughput industrial environments. Furthermore, threshold selection for segmentation is often performed in an ad hoc manner without systematic optimization or verification, resulting in variable results between studies and limiting reproducibility.

In the case of particle and void characterization, prior work has tended to report only basic volumetric fractions, neglecting more advanced morphological descriptors such as sphericity, aspect ratio, orientation, nearest neighbor distance (NND), and clustering indices. These parameters have been shown to correlate strongly with mechanical performance but are rarely extracted in an automated and scalable manner. Moreover, while  $\mu$ CT has proven effective for capturing internal structures in three dimensions, its integration with computational modeling workflows such as finite element analysis (FEA) or machine learning (ML) remains underdeveloped. This gap hinders the translation of imaging data into predictive design tools for process optimization.

The literature also demonstrates a shortage of focused research tackling the specific defect mechanisms in additively manufactured syntactic foams. For instance, defects due to fused filament fabrication (FFF), such as interlayer voids, incomplete fusion, and irregular particle distribution, are rarely quantified in terms of directly correlating process parameters with defect creation. Furthermore, there is minimal standardization in data processing workflows, rendering cross-study comparisons challenging. Finally, there is a strong need for non-destructive, reproducible, and high-resolution characterization methods that not only preserve the structural integrity of the part but also provide actionable measurements of defects for quality assurance, process optimization, and performance prediction. The present study aims to address these gaps by

presenting an entirely automated  $\mu$ CT image segmentation and quantification workflow designed explicitly for HDPE-HGM syntactic foams, enabling the extraction of a comprehensive set of morphological descriptors and offering the potential for direct integration with downstream computational models.

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## 3. Materials and Methods

### 3.1 Material Composition (HDPE + HGM)

#### 3.1.1 Matrix Material: High-Density Polyethylene (HDPE)

High-Density Polyethylene (HDPE) is a semi-crystalline polymer and its mechanical properties, such as strength, stiffness, and toughness, are strongly influenced by its crystalline morphology. The mechanical behaviour of HDPE depends on the processing-induced morphology [1]. This polymer was selected as the matrix due to its widespread use and its products are preferable to metals, having properties such as lightweight, ease of manufacture, and being stronger and tougher [2]. HDPE, when embedded with HGMs, has been shown to improve tensile strength, stiffness, and energy absorption, while reducing density, making it suitable for advanced structural applications [3].

In the context of  $\mu$ CT, HDPE shows uniform X-ray attenuation and has good contrast with HGMs. This allows for clear differentiation between the features of interest.

#### 3.1.2 Filler Material: Hollow Glass Microspheres (HGM)

The matrix is embedded with Hollow Glass Microspheres (HGMs) specifically, sodium borosilicate hollow glass microspheres, which are low-density and have a wall thickness of approximately 0.5 to 2  $\mu\text{m}$ , ranging from 50 to 80  $\mu\text{m}$  in diameter, depending on the type of glass [4]. Figure 3. illustrates their actual appearance when embedded in HDPE matrix (Figure 3.a) and the idealized structure of HGMs in this work (Figure 3.b).

The hollow glass microspheres (HGMs) employed in this study have a wall thickness of approximately 0.37  $\mu\text{m}$ , as reported by the manufacturer. The actual diameters of the HGMs were determined directly from the calibrated micro-CT images using ImageJ, where line measurements across the widest cross-sections of individual microspheres yielded values close to 50  $\mu\text{m}$ . These measurements are consistent with the expected microsphere size range for HDPE-HGM composite filaments, thereby confirming the representative dimensions of the reinforcements analyzed in this work.

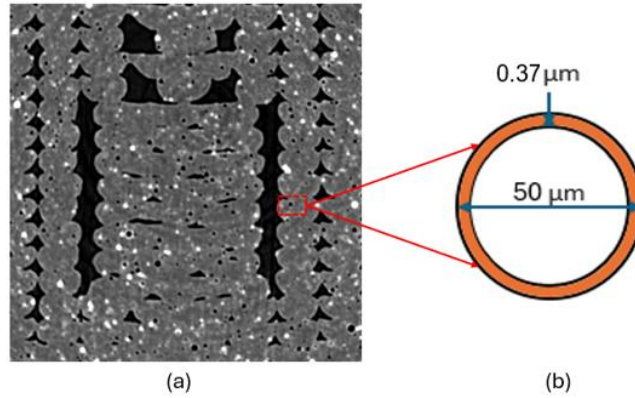


Figure 3.1 : Hollow glass microsphere (HGM) structure and distribution in HDPE matrix. (a) MicroCT image HGMs embedded within the polymer matrix. The white spherical features represent intact HGMs while darker regions indicate either hollow interiors of sectioned microspheres or processing-induced voids; (b) Schematic cross-section of a typical HGM showing the thin glass wall (orange,  $\sim 0.37 \mu\text{m}$  thickness) surrounding the hollow interior.

The HGMs act as rigid fillers, as their elastic modulus is higher than HDPE, hence increasing its stiffness and elastic modulus. The internal voids, thin glass wall structure created a foam structure that is lighter and has low density overall. When the interfacial bond is strong, the cracks cannot propagate well when load is applied, creating a strong material.

## 3.2 Additive Manufacturing

The samples were fabricated using Additive Manufacturing (AM). This section outlines the printing methodology and infill densities used. Fused Filament Fabrication (FFF) is a widely used AM technique wherein a thermoplastic filament is fed into the heated nozzle, where it melts, and is then extruded layer by layer onto a build platform. The material solidifies upon cooling [5].

### 3.2.1 3D Printing Process Description

All the samples were printed using a commercial Prusa i3 MKS FFF printer with a 4 mm diameter steel nozzle. A custom filament, composed of a blend of HDPE and HGMs, was fed into the printer. The material was tricky to work with and required an optimum bed temperature, good bed adhesion, and slow speeds. A layer height of 0.2mm was chosen, to ensure it is at least larger than twice the size of the largest feature (50 μm diameter of HGMs) The process parameters listed in

Table 3.1 were carefully selected to ensure high-quality printing of the samples.

Table 3.1 : Additive manufacturing parameters used during FFF 3D printing of HDPE-HGM syntactic foam samples.

| Parameter            | Value   |
|----------------------|---------|
| Bed Temperature      | 90°C    |
| Extruder Temperature | 230°C   |
| Printing Speed       | 10 mm/s |
| Filament Size        | 1.75 mm |
| Layer Height         | 0.2 mm  |

The raster-based deposition pattern inherent to FFF enabled the formation of aligned voids between adjacent paths, depending on the selected infill strategy. These voids served as a secondary porosity class, distinguishable from the spherical voids generated by HGMs.

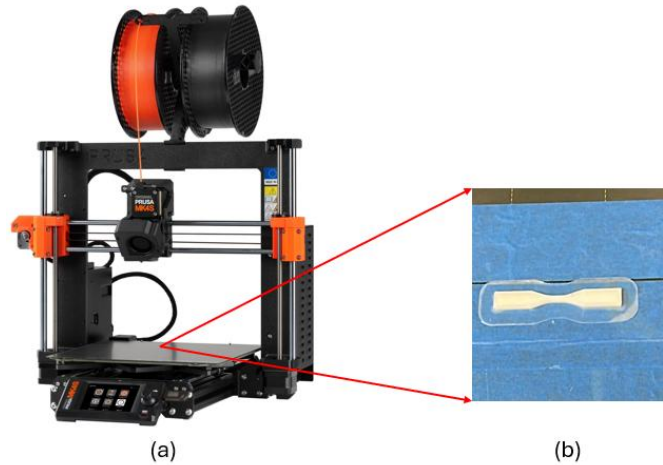
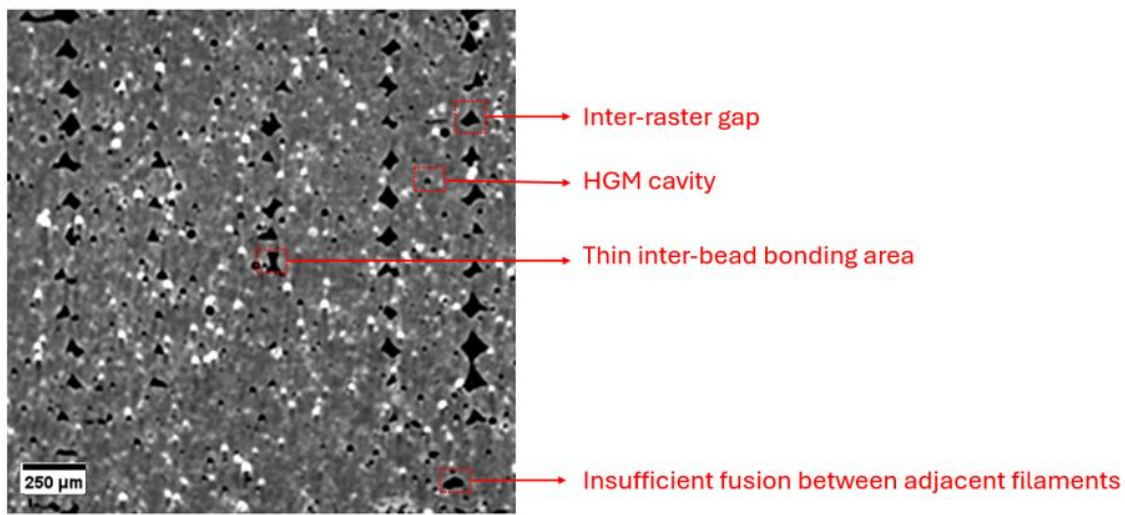
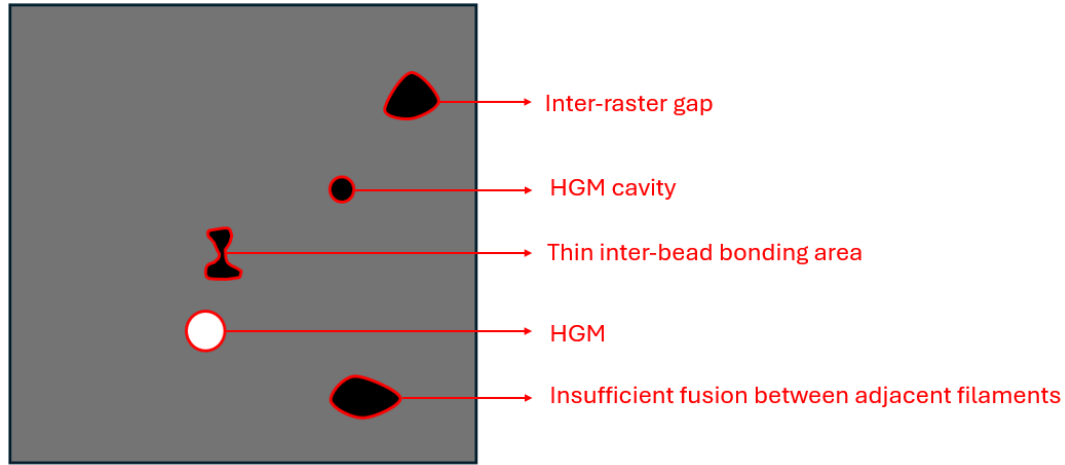


Figure 3.2: Additive manufacturing setup for HDPE-HGM syntactic foam fabrication. (a) Fused filament fabrication (FFF) 3D printer (Prusa i3 MK3) equipped with a hardened steel nozzle for processing glass-filled composite filament. The orange filament spool contains the custom HDPE-HGM blend used in this study. (b) Representative printed specimen showing the layered structure characteristic of FFF processing, with visible raster patterns that contribute to the complex void morphology analyzed in this work.

For this work, all such manufacture-induced voids, except the intended cavities of hollow glass microspheres, are grouped and referred to as raster voids for simplicity. Figure 3.3 : presents a schematic of the various void types in a 3D-printed HDPE-HGM syntactic foam to help illustrate these different void types. For syntactic foams, distinction of these raster voids from the intentional cavities of hollow glass microspheres is critical for proper microstructural characterization and performance analysis. This separation has a direct impact on mechanical properties, as raster voids are stress concentrators that undermine performance, whereas HGM cavities are intended features that aid in weight reduction. Raster voids have elongated or irregular morphology and follow raster deposition patterns, whereas HGM cavities are discrete circular features with high circularity, which allow shape-based differentiation.



(a)



(b)

Figure 3.3 : (a) Schematic of void types in a 3D-printed HDPE-HGM syntactic foam 100 % infill sample, illustrating raster voids from incomplete filament fusion and internal cavities of hollow glass microspheres (HGMs).; (b) Corresponding schematic representation of void morphologies observed in (a), highlighting the matrix (grey), voids (black), and intact HGM shells (white).

It is important to note that 3D printing parameters have an effect on the cooling rate, microsphere inclusion and mechanical properties, which is why, although according to Rule of Mixtures (ROM), the strength of the syntactic foam must be higher than that of its matrix counterpart, the results show otherwise. Printing speed, layer height and nozzle temperature affect void formation and the quality of inter layer bonding, which are important for structural integrity and performance, thereby affecting the microsphere inclusions. Lower printing speeds and smaller layer heights can improve inter-layer adhesion and result in better, more uniform distribution of HGMs, reducing porosity and enhancing mechanical properties. On the other hand, high printing speeds and larger layer heights can lead to weak adhesion and increased porosity, which negatively impacts composite strength, and larger filler diameters can cause further porosity issues.

### 3.2.2 Infill Strategies and Layer Deposition Parameters

Infill density, in other words, controlled porosity, significantly influences the microstructure of the syntactic foam. To examine this effect, specimens were printed with infill densities ranging from 60% to 100%, varying in increments of 10%. Furthermore, two compositions of HGMs were used, as shown in Table 3.2.

Lower infill densities promoted the formation of large raster-induced voids, while higher densities reduced these features and emphasized filler-induced porosity. Furthermore, a rectilinear infill pattern was used as it offers fast printing, good strength, especially along layer lines, no complex geometry waste and the ability

to easily tune and troubleshoot. This allowed the interpretation of void alignment and anisotropy in  $\mu$ CT images in further analysis.

Table 3.2 : Material Composition of Syntactic Foam Samples

| Composition | HGM Density (g/cc) | HGM Volume Fraction (%) |
|-------------|--------------------|-------------------------|
| A           | 0.14               | 55.8                    |
| B           | 0.32               | 37.6                    |

Composition A contains lower-density microspheres (0.14 g/cc) with thinner walls, its higher volume fraction (55.8%) leads to more uniform stress distribution and improved energy absorption, which ultimately results in better tensile strength compared to Component B. Component B, with denser microspheres (0.32 g/cc) and a lower volume fraction (37.6%), provides higher stiffness due to thicker walls but does not achieve the same strength retention as Component A.

In this work, different infill percentages were combined with both Composition A and B to analyse the interaction between process-induced porosity (raster voids from infill variation) and material-induced porosity (HGM cavities from different microsphere compositions). This design allowed the study to decouple how printing parameters and filler characteristics jointly influence microstructure and mechanical performance. Another alternative would be to print and analyze the influence of infill density and both compositions, by varying the composition and keeping the infill density the same. However, the chosen approach offered a broader mapping of combined effects within the time and resource constraints of this study, giving insights that a single-factor design may have missed.

### 3.2.3 Specimen Geometry and Printing Setup

All the test specimens were printed according to the ASTM D638 Type V standard for tensile testing of plastic materials. This geometry was selected for its compact size, which is compatible with both  $\mu$ CT imaging constraints and tensile testing. The dogbone-shaped profile, as shown in Figure 3., ensures stress concentration at the gauge section, promoting controlled fracture and accurate measurement of tensile properties. Figure 3. illustrates the dimensions of the printed specimen, with a total length of 63.5 mm and a narrow-gauge width of 3.18 mm. The thickness of 3.2 mm was consistent across all . Specimens were oriented horizontally on the build platform with the gauge length aligned along the raster path to maintain consistency in raster-induced void patterns. The build plate temperature and cooling fan speed were carefully controlled to avoid thermal warping, especially at the narrow-gauge section. Post-printing, specimens were inspected for geometric fidelity and dimensional accuracy before being selected for

mechanical testing or  $\mu$ CT scanning. The standardized geometry not only ensured compatibility with the tensile test setup but also provided a predictable internal void architecture for segmentation validation. The combination of ASTM-based geometry and additive manufacturing enabled reproducibility across batches and a reliable evaluation of morphology–mechanical property relationships.

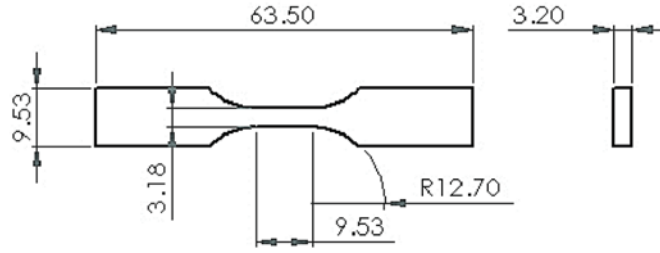


Figure 3.4 : Tensile test specimen size for type 5 ASTM D638

### 3.3 $\mu$ CT Imaging

Micro-computed tomography ( $\mu$ CT) was used in this study due to its ability to capture internal structure without requiring sample cutting in a 3D fashion [6]. The primary goal was to visualize and quantify internal voids, both those resulting from the printing process and those formed by embedded hollow glass microspheres. The  $\mu$ CT datasets provided the basis for 3D segmentation, feature extraction, and statistical modelling, making imaging quality and processing fidelity critical to downstream analysis.

#### 3.3.1 Imaging Equipment and Scanning Parameters

The scans were carried out on the Bruker Skyscan 1172, located at the MakerSpace in New York University. The scanner is equipped with an 11-megapixel X-ray camera and a tungsten target source, providing sub-micron resolution capability for close examination of polymer composites and particulate fillers.

Scanning was conducted using a source voltage of 71 kV and a current of 139  $\mu$ A, which were determined through pre-scan trials to provide optimal penetration and contrast for the HDPE matrix and embedded microspheres. A total of 3600 projections were acquired over a 360-degree rotation, with a rotation step of 0.1 degrees. To improve signal-to-noise ratio, 10-frame averaging was applied, resulting in an exposure time of 179 milliseconds per projection. The parameters are detailed in Table 3.3.3.

The source-to-object and source-to-detector distances were maintained at 58.3 mm and 214.9 mm, respectively, resulting in a magnification factor of approximately  $3.7\times$ . No metallic filter was used to retain the natural contrast between the low-density polymer matrix and the high-density glass shell of the

microspheres. The scans were all conducted in a temperature-controlled room to minimize thermal expansion impacts during data collection.

Table 3.3 : Micro-computed tomography scan settings for syntactic foam sample.

| Parameter                           | Value           |
|-------------------------------------|-----------------|
| Camera Type                         | SHT 11Mp camera |
| Camera Pixel Size ( $\mu\text{m}$ ) | 9.00            |
| Voltage (kV)                        | 71              |
| Current ( $\mu\text{A}$ )           | 139             |
| Image Pixel Size ( $\mu\text{m}$ )  | 4.8             |
| Image Width (pixels)                | 832             |
| Image Height (pixels)               | 740             |
| Exposure per Projection (ms)        | 179             |
| Rotation Step (degrees)             | 0.1             |
| Rotation Range (degrees)            | 360             |
| Frame Averaging                     | 10              |
| Filter Name                         | No filter       |
| Object-to-Source Distance (mm)      | 58.339          |
| Camera-to-Source Distance (mm)      | 214.889         |

| Parameter                       | Value          |
|---------------------------------|----------------|
| Field of View (FOV) Width (mm)  | 4.07           |
| Field of View (FOV) Height (mm) | 3.62           |
| Reconstruction Software         | NRecon 1.7.4.6 |
| Ring Artifact Reduction         | 7              |
| Misalignment Compensation       | 14.5           |
| Beam Hardening Correction (%)   | 50             |

### 3.3.2 Resolution, Field of View, and Voxel Size

The reconstructed voxel size was  $4.8\ \mu\text{m}$ , providing adequate resolution to resolve voids with diameters as small as  $10\ \mu\text{m}$ , thereby enabling accurate quantification of voids. This was done following the Nyquist-Shannon sampling theorem, which states that a signal can be perfectly reconstructed from its samples if the sampling rate is at least twice the highest frequency component of the signal [7]. A cross-sectional image of  $832 \times 740$  pixels was obtained for each projection, and the reconstructed volume covered a field of view of about  $4.07\ \text{mm} \times 3.62\ \text{mm}$ .

The chosen voxel size was a trade-off between acquiring fine morphological detail and retaining a reasonably sized dataset for image processing. This spatial resolution was critical for resolving boundaries of microspheres and small raster-induced gaps, which would be difficult to discern at lower resolutions. The isotropic voxel size enabled accurate 3D rendering and calculation of features, including surface area, volume, sphericity, and aspect ratio. The setup is as shown in Figure 3..

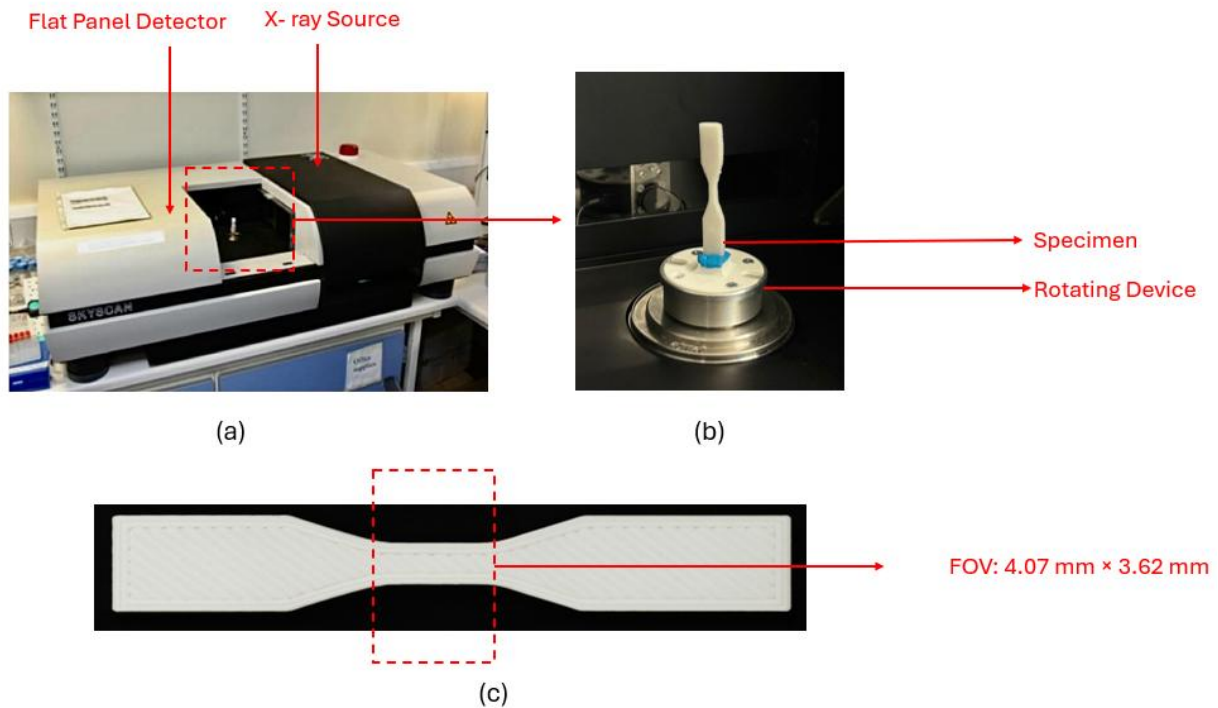


Figure 3.5 :  $\mu\text{CT}$  apparatus and imaging setup: (a) Bruker Skyscan 1172 system showing flat panel detector and X-ray source configuration, (b) specimen mounting on rotating stage with custom fixture, and (c) field of view positioning over the gauge section of the ASTM D638 Type V specimen; Field of View (FOV):  $4.07\ \text{mm} \times 3.62\ \text{mm}$

### 3.3.3 Image Reconstruction and Preprocessing Steps

After obtaining the raw projection data from the scans, reconstruction of these images was carried out using NRecon 1.7.4.6 software. The scans revealed ring artifacts and misalignment, which appear as rings, as well as beam hardening, where the boundary region of the scan images is whiter than the central region. These artifacts were corrected as shown in Table 4. Following this, the datasets were cropped to a standardized Region of Interest (ROI) to ensure uniformity across samples and to remove the background effectively. The reconstructed grayscale images were exported as 16-bit TIFF images. The  $\mu$ CT scans took about 50 hours, for all the samples and 2.5 hours for the reconstruction of the projection images obtained and image processing.

Table 4.1 : Reconstruction parameters used for the syntactic foam samples dataset.

| Parameter                          | Value        |
|------------------------------------|--------------|
| Image Pixel Size ( $\mu\text{m}$ ) | 4.89         |
| Filter                             | No filter    |
| Image Format                       | TIFF         |
| Depth (bits)                       | 16           |
| Smoothing                          | 5            |
| Smoothing Kernel                   | 2 (Gaussian) |
| Ring Artifact Correction           | 20           |
| Beam Hardening Correction          | 45           |
| Rotation (degrees)                 | 22.5         |

Gaussian smoothing was applied to reduce noise and artifacts, A smoothing strength of 5 was used, with a smoothing kernel of 2, defined by standard deviation,  $\sigma = 2$  voxels, which determines the spread of the Gaussian function. A small  $\sigma$  means only the immediate neighbors (close voxels) influence the smoothing strongly, while a large  $\sigma$  means even distant neighbors contribute, so the averaging (blurring) extends further. It also, like smoothing, allows the user to decide the effect of smoothing in the dataset. These parameters were selected to balance noise reduction with edge preservation, ensuring that void boundaries and microsphere shells remained distinct for accurate segmentation.

### 3.4 Tensile Testing

Tensile testing was conducted to evaluate the mechanical performance of the HDPE-HGM syntactic foam specimens and to establish relationships between void quantity and morphology, as well as macroscopic strength. The results of these tests provided quantitative benchmarks for validating image-based predictions and for understanding the influence of dual-scale porosity on structural performance.

#### 3.4.1 Sample Preparation and Dimensions

Tensile specimens were fabricated using fused filament fabrication by ASTM D638 Type V standards. This geometry was selected due to its compact size and suitability for small-scale 3D printing while maintaining consistent stress distribution within the gauge section. Each specimen featured an overall length of 63.5 mm, a narrow-gauge width of 3.18 mm, and a thickness of 3.2 mm, with a fillet radius of 12.70 mm connecting the grip and gauge regions. These dimensions were carefully chosen to ensure compatibility with the  $\mu$ CT scanner field of view and to promote uniform tensile loading during testing. Figure 3. (Section 3.2.3) provides the schematic of the specimen geometry.

After printing, the specimens were visually inspected for surface defects and dimensional deviations. Callipers were used to verify gauge width and thickness across multiple locations. Only specimens with consistent geometry without observable warping, stringing, or under-extrusion were retained for mechanical testing.

#### 3.4.2 Testing Procedure and Standards

Uniaxial tensile tests were conducted using the Instron 5980 Series universal testing machine. All specimens were checked for dimensions and visually inspected before testing to ensure consistency and spot any defects that could affect the mechanical behavior. A quasi-static testing method was utilized to capture the mechanical response without any rate-related effects. As HDPE-HGM syntactic foams are ductile in nature, the tests were automatically stopped at 150% strain, as the necessary strength parameters could be captured within this range. Five specimens were tested for each combination of composition and infill density, in a random order, to reduce systematic bias. Only specimens that showed gauge section failure with uniform deformation were included in the analysis.

For each composition and infill density, five specimens were printed, measured along their gauge section and tested. Their average of all their respective Ultimate Tensile Strengths and Young's moduli was chosen. This strategy adheres to standard testing procedures and yields robust statistical data. All load-displacement curves were computed using Instron's Bluehill Universal software and the UTS was extracted as the highest value of the engineering stress-strain response. A labeled image of the tensile testing setup is presented in Figure 3., showing the load cell, pneumatic grips, ASTM Type V specimen, and control interface.

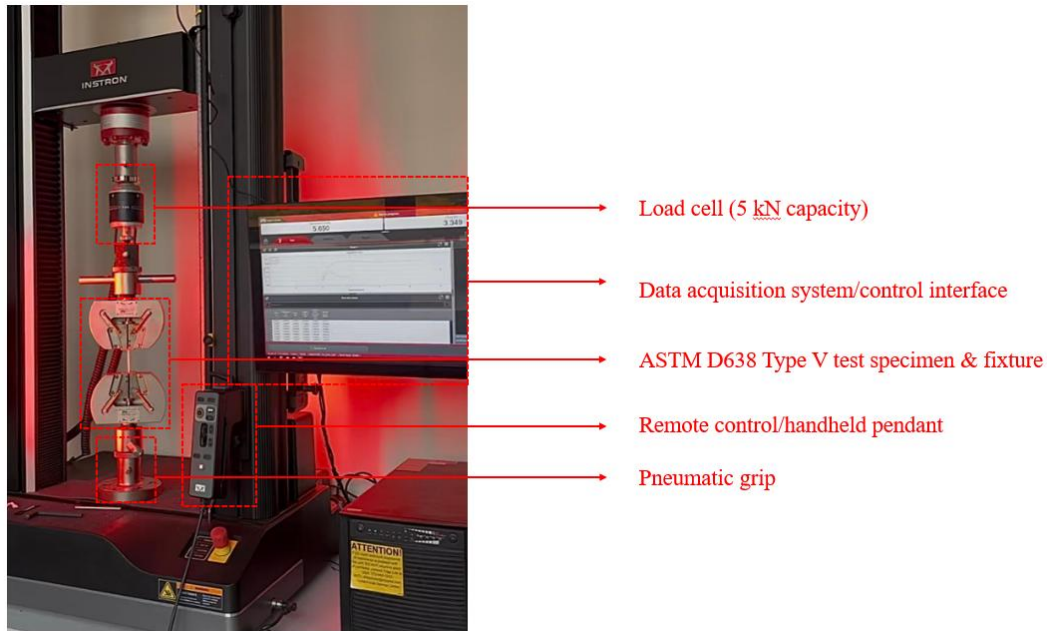


Figure 3.6 : Instron tensile testing system used in this study, equipped with a 5 kN load cell, pneumatic grips, ASTM D638 Type V test specimen, and real-time data acquisition interface.

### 3.4.3 Stress-Strain Data Acquisition and Analysis

Engineering stress strain curves were generated by normalizing the applied force by the initial cross-sectional area and dividing grip displacement by the original gauge length. From these curves, Ultimate Tensile Strength (UTS), defined as the maximum stress sustained before fracture and Young's Modulus, calculated from the slope of the initial linear portion of the curve were derived. A summary of tensile testing parameters, specimen geometry, and analysis methods is provided in Table .

Table 4.2 : Summary of tensile testing parameters, specimen geometry, and analysis methods.

| Parameter                | Description                                                                             |
|--------------------------|-----------------------------------------------------------------------------------------|
| Specimen Dimensions      | Length: 63.5 mm                                                                         |
|                          | Gauge Width: 3.18 mm                                                                    |
|                          | Thickness: 3.2 mm                                                                       |
| Testing Equipment        | Instron Universal Testing Machine with 5 kN load cell and pneumatic grips               |
| Displacement Rate        | 10 mm/min                                                                               |
| Replicates Per Setting   | 5 specimens per material and infill density                                             |
| Testing Conditions       | Room temperature (23°C), ambient lab conditions                                         |
| Recorded Data            | Real-time load and displacement until failure                                           |
| Derived Properties       | Ultimate Tensile Strength (UTS)                                                         |
|                          | Young's Modulus (slope of linear region)                                                |
| Data Analysis            | Engineering stress = Force / Cross-sectional Area                                       |
|                          | Strain = Grip Displacement / Gauge Length                                               |
| Integration with Imaging | Used to correlate $\mu$ CT-derived void content and morphology with tensile performance |

In summary, Chapter 3 details the selection of material, fabrication of the syntactic foam samples using additive manufacturing, followed by  $\mu$ CT imaging and tensile testing. Next, in Chapter 4, a fully automated image segmentation, classification and analysis pipeline is discussed, based on the images obtained through  $\mu$ CT scans, and is discussed in detail.

### 3.5 References

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## 4. Automatic Multi-Modal Segmentation

### 4.1 Introduction

Combining micro-computed tomography ( $\mu$ CT) with microstructure characterization and additive manufacturing (AM) has contributed significantly to optimizing the development of syntactic foams. Recently, research has utilized  $\mu$ CT imaging with computational image analysis techniques to effectively determine particle and void phases, thereby optimizing microstructure evaluation for performance analysis. Automated segmentation is crucial for quantifying microstructural features in materials science, particularly in composite materials like syntactic foams. This chapter presents a robust, unsupervised image segmentation framework designed to identify and differentiate voids, polymer matrix, and embedded hollow glass microspheres (HGM) within high-resolution micro-computed tomography ( $\mu$ CT) datasets of 3D-printed HDPE-HGM syntactic foams.

The goal of this chapter is to outline the computational workflow that enables the accurate and repeatable segmentation of multi-phase features in  $\mu$ CT images, thereby facilitating downstream morphological analysis and structure-property correlation. This includes the development of adaptive thresholds for phase boundary identification, refinement using local neighbourhood coherence, and shape-based post-processing for void classification. The segmentation pipeline serves as the analytical backbone of the study, linking

additive manufacturing parameters to the resulting microstructural configurations and mechanical performance.

#### 4.1.1 Greyscale Thresholding

Each pixel in an image has an intensity value ( $I$ ), with grayscale images ranging from 0 to 255, where 0 represents black and 255 represents white. Image segmentation, which involves dividing an image into distinct regions, is a crucial initial step in image analysis and interpretation. The aim is to separate the image into meaningful regions for a specific task [1]. Segmentation methods can be classified based on the features used and the type of technique applied. Features include pixel intensities, gradient magnitudes, or texture measures. Techniques for segmentation based on these features fall into one of three main categories region-based, edge-based, or classification [2].

A substantial amount of research has been conducted in this field, focusing on finding ways to automate this segmentation process. The need for this can be explained as follows. Figure 4.1 is a slice of the micro-CT scanned sample. It can be visualized that there are some voids ( $I$  close to 0), particles ( $I$  close to 1), and matrix ( $I$  in between 0 and 1) roughly. These features need to be categorized automatically.

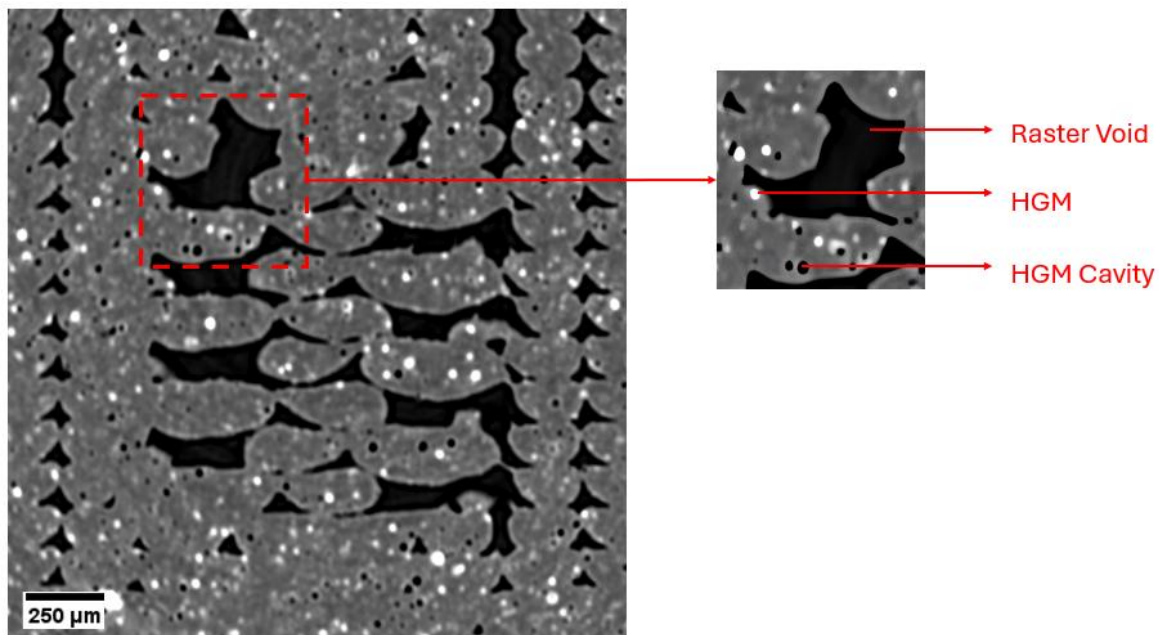


Figure 4.1 : Micro-CT image of HDPE – HGM Syntactic foam 100 % Infill Component A showing hollow glass microspheres (HGM) in matrix material, highlighting raster voids, HGM particles, and HGM cavities.

Thresholding techniques, especially for medical image segmentation, have evolved significantly over the past decades, particularly in the context of bone microstructure analysis using X-ray microtomography

( $\mu$ CT). It is relevant to investigate this field because the approach is relatively similar. These approaches can be broadly categorized into three main types: manual/visual, semiautomatic, and automatic techniques, each with distinct advantages and limitations.

#### 4.1.2 Conventional Methods

Manual thresholding techniques represent the earliest approaches to medical image segmentation, relying primarily on operator expertise and visual assessment. These methods assume that gray-level histograms of  $\mu$ CT images are bimodal, allowing for empirical threshold selection. Martin-Badosa et al. demonstrated this approach by setting the gray-level threshold to 100 for all slices of  $\mu$ CT data [3]. While these techniques offer the advantages of being fast and straightforward to implement, they present significant limitations in clinical practice. The primary concern is that fixing a single threshold value across all image slices can lead to substantial underestimation of the actual bone volume, particularly when dealing with uneven illumination, noise, and varying distributions of trabecular bone within individual slices [4]. Semiautomatic techniques emerged as a response to the limitations of manual approaches, incorporating more sophisticated algorithms while still requiring operator input for parameter optimization and tuning. Ding et al. developed an adaptive method requiring user selection of 50 different thresholds to segment  $\mu$ CT data, ultimately choosing the threshold that produces the least change in volume fraction [5]. Similarly, Antoniadis et al. implemented a probabilistic relaxation labelling technique that analyzes local features of trabecular bone images. However, its success depends heavily on the operator's selection of the tolerance limit parameter ( $\epsilon$ ) [6]. Helterbrand et al. proposed an alternative approach that analyzes the first derivative of the image histogram, selecting the threshold at the pixel intensity where the derivative returns to zero, with the operator determining the number of consecutive intensity points that satisfy this criterion [7]. Aslan et al. further advanced this category by developing a locally adaptive volume-growing technique that employs a voxel similarity criterion based on the difference between mean intensity values and a fraction of the standard deviation, utilizing a user-defined parameter  $\alpha$  to scale the standard deviation value [8]. In the case of syntactic foams, Awaja et al. used X-ray micro-CT to investigate thermal degradation in epoxy/glass microsphere syntactic foams up to 200°C, identifying glass microsphere fracture, void enlargement, and resin microcracks that increased sharply with temperature. However, their reliance on manual segmentation methods was time-consuming and potentially subjective, particularly for complex damage geometries at higher temperatures [9].

These thresholding techniques still share several limitations because they rely on user expertise. Moreover, a specific threshold value is fixed across all slices, which means that slight variations in the user-selected threshold can produce significantly different geometric models, leading to underestimation or overestimation of structures. The same is reported by Upadhyay et al. when they applied iterative greyscale

thresholding; two systematic errors were observed: over-segmentation, where resin-rich composite regions were misclassified as voids, and under-segmentation, where parts of large voids were missed because the grey level inside the void overlapped with fibre and matrix grey values [10]. Furthermore, simple global thresholding assumes uniform illumination and separable foreground and background intensities, which is not a practical assumption. Semi-automatic methods, such as interactive thresholding and adaptive thresholding or locally adaptive volume growing, attempt to overcome these issues but still depend on user-defined parameters. These approaches are more computationally intensive and require careful tuning of neighbourhood size, similarity criteria or tolerance limits.

Otsu's method, while effective for images with bimodal or multimodal histogram distributions, is less effective when applied to images with unimodal or near-unimodal distributions, particularly in defect detection applications where minor defects do not create distinct peaks in the histogram. This limitation results in incorrect threshold selection as the method attempts to split single peaks rather than identifying appropriate valleys or rims for defect isolation [11]. Due to these constraints, extensive research has focused on machine learning-driven methods, specifically supervised learning methods for microstructural analysis, which have shown promising applications. This is discussed in detail in section 4.1.3.

#### 4.1.3 Supervised Learning Methods

Supervised learning methods, such as decision trees, linear regression, naive Bayes, and logistic regression, utilize annotated training datasets to develop predictive models that can automatically classify or segment images based on learned feature relationships, rather than predefined threshold values [11]. Deep learning, a branch of machine learning, is a widely used framework that is a subclass of supervised learning methods. Using multiple layers of neural networks, these models directly learn feature representations from raw data such as images and texts without requiring manual feature engineering [12]. This approach is widely used in characterizing voids and defects.

Upadhyay et al. demonstrated that deep learning models achieved a precision and recall of 0.99 in void identification within filament-wound composites, outperforming grayscale thresholding by 15.4%. In contrast, traditional methods underestimated void morphology by up to 84% [10]. Pedro et al. demonstrated that deep learning, specifically a U-Net architecture, significantly surpasses traditional thresholding in segmenting voids and dry fibre regions in uncured carbon fibre/epoxy prepregs, particularly in low-porosity conditions where conventional methods often fail. However, their approach requires extensive manual annotation and lengthy training times on CPU, highlighting the computational cost and data preparation challenges associated with applying deep learning to micro-CT segmentation, especially for broader use in multi-phase or additively manufactured composites. Zhang et al. developed a supervised deep learning framework using a CBAM-enhanced U-Net for 2D microstructure classification and segmentation of

titanium alloys, achieving high accuracy (MIoU = 0.82) but requiring large, annotated datasets and retraining for new materials [13]. Lawrence et al. employed stereological sampling protocols to characterize hollow polymeric microballoons in polyethylene matrix syntactic foams, utilizing design-based methods that necessitated extensive manual intervention and expertise [14]. Similarly, Chen et al. quantified particle gradients in aluminum syntactic foams using quantitative image analysis on optical micrographs, estimating particle volume fraction and porosity distribution manually [15]. Although these studies have provided valuable insights into syntactic foam microstructures, they also highlight the limitations of manual analysis methods, such as long processing times, operator bias, and the difficulty in distinguishing between functional microsphere cavities and voids produced during manufacturing. Recent advances in machine learning for foam characterization have focused more on fibre-reinforced composites than syntactic systems, with limited use of automatic segmentation techniques tailored for the specific conditions of hollow microsphere-infused composites. The absence of automatic multi-phase segmentation methods designed explicitly for syntactic foams presents a significant gap in current characterization techniques, particularly in additive manufacturing, where rapid and reproducible analysis is crucial for quality control and process optimization. Moreover, a standard limitation across all these methods is that user expertise is essential, manual annotation is required, which is not only time-consuming and labour-intensive but also significantly affects the model's performance. Once a model is trained for a specific dataset, it must be retrained to work with a different set of images, indicating that these models are typically not versatile. Additionally, these approaches are computationally demanding. To address these issues, which constitute a notable gap in current literature, this work employs an unsupervised method. As shown in Table , existing segmentation approaches are either dependent on manual input, require large, annotated datasets, or perform poorly when grayscale contrast is low. These drawbacks limit their suitability for rapid, reproducible analysis of syntactic foams in additive manufacturing. To address these gaps, this work introduces an unsupervised K-means clustering method with neighbourhood refinement, providing a fully automated, training-free solution for accurate three-phase segmentation in  $\mu$ CT datasets.

Table 4.3 : Summary of existing  $\mu$ CT segmentation methods with advantages and disadvantages.

| Method type                         | Existing Studies                                                                    | Methodology                                                                                 | Advantages                                                | Disadvantages                                                                                                                                                                                         |
|-------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Manual thresholding                 | Martin-Badosa et al. [3]                                                            | User picks a fixed grayscale value for all slices                                           | Quick and easy to apply                                   | <ol style="list-style-type: none"> <li>1. Assumes clear grayscale separation</li> <li>2. Inaccurate when lighting or contrast changes</li> </ol>                                                      |
| Semi-automatic thresholding         | Ding et al. [5],<br>Antoniadis et al. [6], Helterbrand et al. [7], Aslan et al. [8] | User sets parameters (e.g., tolerance, neighbourhood size) and the computer adjusts locally | More accurate than manual and can handle local variations | <ol style="list-style-type: none"> <li>1. Reliance on user input</li> <li>2. Results depend on chosen parameters</li> <li>3. Slower processing</li> </ol>                                             |
| Automatic global thresholding       | Otsu's method [11]                                                                  | Computer analyzes the grayscale histogram to find optimal thresholds                        | No user input and reproducible for clear histograms       | <ol style="list-style-type: none"> <li>1. Fails with unimodal or low-contrast images</li> <li>2. Overlooks small defects</li> </ol>                                                                   |
| Manual analysis for syntactic foams | Awaja et al. [9],<br>Lawrence et al. [14], Chen et al. [15]                         | User manually traces voids/particles                                                        | Complete control and high confidence for small datasets   | <ol style="list-style-type: none"> <li>1. Time consuming</li> <li>2. Prone to human bias</li> <li>3. Unsuitable for large datasets</li> <li>4. Computationally heavy</li> </ol>                       |
| Supervised machine learning         | Upadhyay et al. [10], Pedro et al. [13], Zhang et al. [26]                          | Computer learns from labelled examples (e.g., U-Net)                                        | High accuracy and can detect complex features             | <ol style="list-style-type: none"> <li>1. Requires large, labelled datasets</li> <li>2. Must retrain for new materials</li> <li>3. Long training times</li> <li>4. High computational cost</li> </ol> |

#### 4.1.4 Unsupervised Learning Methods – K Means Clustering

Unsupervised machine learning methods identify latent patterns and hidden structures in high-dimensional data and can help simplify complex datasets. K-means clustering and hierarchical clustering are two classical unsupervised machine learning techniques that partition data into groups based on similarities in the data [16]. K-means clustering is an iterative algorithm that divides data points into distinct, non-overlapping clusters. The user defines the number of clusters ( $C_n$ ), and the process begins by randomly

selecting  $C_n$  points. For each data point  $p \in P$ , its squared Euclidean distance to each centroid is computed and assigned to the cluster with the nearest centroid. Based on the smallest distance, the points are assigned to their respective clusters.

Randomly selecting the points as centroids is known to be problematic because random seeds often produce different partitions, which affect the result; in our case, this can lead to over- or under-estimation of particles and voids. This can also make different results every time it is run. Furthermore, the algorithm can converge to a poor local optimum, resulting in low-quality results [17]. To solve this issue, a neighbourhood method was used, as discussed in section 4.1.5.

#### 4.1.5 Neighbourhood Method

Neighbourhood-based initialization methods use local density and proximity relationships to select more representative initial centroids. Cao et al. formalized Astrahan's density-based method within a neighbourhood-based rough-set model. They measured the cohesion of each data point's neighbourhood and the strength of its coupling with other neighbourhoods, then selected points with the highest cohesion and weakest coupling as the initial centroid [18].

This approach starts by normalizing the dataset and computing pairwise distances using the Euclidean method. Epsilon ( $\epsilon$ ) is set as the average distance between points. Following this, cohesion is calculated for each point, and the point with the highest cohesion is selected as the first cluster and so on. The clusters are iteratively chosen by identifying the next point with the highest cohesion, which measures how tightly packed or similar the data points are within the neighbourhood of a given point (Eq. 1). This is done only if the neighbourhood has low coupling, which is the IoU of two neighbourhood sets. (Eq. 2), with the selected clusters. The clusters are iteratively chosen by identifying the next point with the highest remaining cohesion, provided its neighbourhood has low coupling with already chosen centers. This is repeated until K centers are selected.

$$\text{Cohesion}(\delta_{\epsilon_p}(x_i)) = \frac{|\text{Lower Approximation of } \delta_{\epsilon_p}(x_i)|}{(|\text{Upper Approximation of } \delta_{\epsilon_p}(x_i)|)} \quad (1)$$

Where,

$\delta_{\epsilon_p}(x_i)$  is the neighbourhood of point  $x_i$  concerning attribute set  $P$  and threshold  $\epsilon$ . Lower Approximation is the number of points that belong to the neighbourhood. Upper Approximation is the number of points that possibly belong to the neighbourhood. The value of cohesion lies between 0 and 1.

$$\text{Coupling}(\delta\epsilon_p(x_i), \delta\epsilon_p(x_j)) = \frac{|\delta\epsilon_p(x_i) \cap \delta\epsilon_p(x_j)|}{|\delta\epsilon_p(x_i) \cup \delta\epsilon_p(x_j)|} \quad (2)$$

Where,

$x_i, x_j$ : Two data points in the dataset.

$\delta\epsilon_p(x_i)$ : The neighbourhood of point  $x_i$ , the set of points within distance  $\epsilon$  from  $x_i$  based on attribute set  $P$ .

$\delta\epsilon_p(x_j)$ : The neighbourhood of point  $x_j$ .

$\cap$ : Intersection, which is the set of points common to both neighbourhoods.

$\cup$ : Union, which is the set of all unique points in either neighbourhood.

$|\cdot|$ : Cardinality, which is the number of points in the set.

Taking inspiration from their work, the neighbourhood method is integrated into our k-means algorithm to cluster pixels based on their intensity into three clusters, thereby simultaneously quantifying voids, particles, and matrix. By directly addressing the automation, adaptability, multi-phase segmentation, and efficiency challenges summarized in

Table , the proposed approach provides a tailored solution for syntactic foam  $\mu$ CT analysis. The methodology for implementing this segmentation pipeline is detailed in Section 4.2.

Table 4.4 : Summary of critical research gaps in  $\mu$ CT segmentation methods for syntactic foams and the proposed approaches.

| Key Aspect             | Current state in the literature                                                                 | Gap identified                                                             | Proposed Approach                                                       |
|------------------------|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Automation             | Many existing methods (manual, semi-automatic) require user input for thresholds or parameters. | High user dependence slows analysis and introduces subjectivity            | Fully automated pipeline, no threshold tuning or annotations needed     |
| Adaptability           | Fixed thresholds or dataset-specific trained models.                                            | Poor transferability to different scan conditions, materials, or contrasts | Adaptive clustering with neighbourhood refinement works across datasets |
| Multi-phase capability | Most methods target 2-phase separation (matrix vs. void).                                       | Limited ability to segment voids, matrix, and microspheres simultaneously  | Three-phase segmentation tailored for syntactic foam $\mu$ CT datasets  |

| Key Aspect                     | Current state in the literature                                                     | Gap identified                                        | Proposed Approach                                                                                        |
|--------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Void type classification       | Rarely distinguish between functional microsphere cavities and manufacturing voids. | Inability to quantify functional vs. defect porosity  | Shape-based classification integrated into the pipeline                                                  |
| Computational cost             | Deep learning requires long training and GPU resources, manual tracing is slow.     | High cost and time for large dataset analysis         | CPU-friendly workflow suitable for batch processing                                                      |
| Application to syntactic foams | Most work on fibre composites, metals, or bone.                                     | Limited syntactic foam-specific segmentation research | Method optimized for AM syntactic foams, with validation on varied infill densities and HGM compositions |

## 4.2 Methodology

The methodology presented in this section outlines a comprehensive, automated image segmentation pipeline tailored for dual-phase void and particle identification in  $\mu$ CT images of syntactic foams. Building upon the neighbourhood-refined K-means clustering approach introduced in Section 4.1.5, this workflow integrates grayscale preprocessing, unsupervised classification, and local neighbourhood refinement to achieve precise, reproducible, and training-free segmentation of voids, matrix, and hollow microspheres. The pipeline is optimized for batch processing, ensuring consistent performance across varying scan conditions and providing a solid foundation for all subsequent morphological quantification and structure-property analyses.

### 4.2.1 Segmentation Workflow

The segmentation process developed in this study was designed to identify and quantify dual-scale voids and microspheres within  $\mu$ CT scans of HDPE-HGM syntactic foams. The workflow integrates grayscale image preprocessing, unsupervised clustering via K-means, and neighbourhood-based refinement, culminating in binary masks used for morphological quantification. Each step in this pipeline is outlined below and corresponds to the stages illustrated in the segmentation workflow figure (Figure 4.2).

OpenCV (cv2) was used to load grayscale images from the specified directory. The grayscale format was chosen to minimize computational complexity and emphasize pixel intensity variations, which are essential for distinguishing between different microstructural regions. The `cv2.imread()` function facilitated the efficient reading of images, while `cv2.IMREAD_GRAYSCALE` ensured that colour information, irrelevant

for this analysis, was excluded. Noise suppression was achieved using a Gaussian smoothing filter, which attenuated high-frequency scan artifacts such as ring effects and beam hardening. This filtering step enhanced the separability of features in the image histogram without blurring the critical boundaries between voids, matrix, and particles.

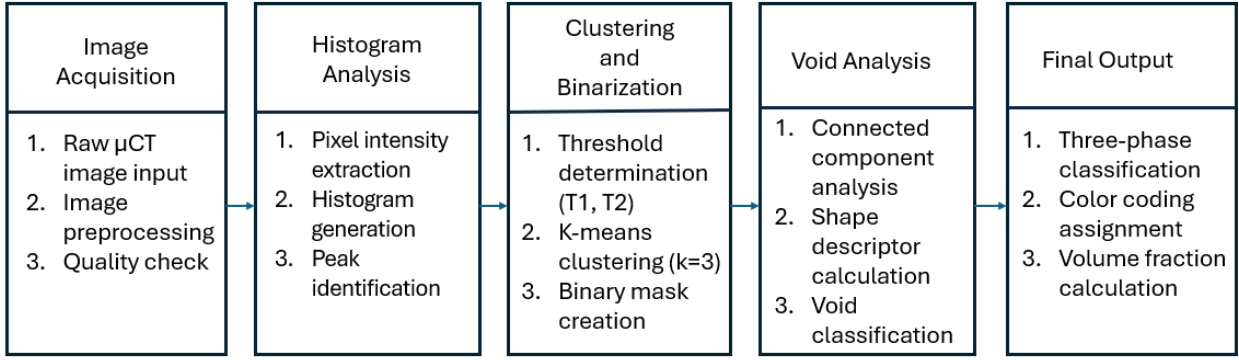
Following denoising, grayscale histograms were generated to analyze the distribution of pixel intensities across the image. These histograms typically displayed three distinguishable peaks, representing the void phase (low intensity), the HDPE matrix (mid-range), and the HGM particles (high intensity). The dynamic range of voxel intensities was normalized to 8-bit to ensure consistency across datasets. This normalization step was crucial for achieving reproducible clustering performance and accurate threshold estimation.

With intensity data prepared, unsupervised segmentation was performed using K-means clustering with three classes. This algorithm partitioned the image into three regions based on grayscale similarity, corresponding to voids, matrix, and microspheres. K-means offered adaptability to local variations in scan contrast and eliminated the need for user-defined threshold values, thereby increasing automation and reducing subjectivity in the segmentation process.

Initial K-means outputs often suffered from noise-driven pixel misclassification, especially in transition zones or regions with subtle intensity gradients. To address this, a local neighbourhood method was applied. Each pixel's label was reevaluated based on the intensity homogeneity of its surrounding neighbours within a fixed window.

The refined label map was converted into binary masks, isolating each class for independent morphological analysis. Connected component labelling was then applied to assign unique IDs to voids and particles, enabling measurement of volume fraction, shape descriptors, and spatial distribution.

The complete pipeline was implemented in Python using libraries including OpenCV for image processing, NumPy for numerical operations, and scikit-learn for clustering. The modular structure enabled the batch processing of multiple  $\mu$ CT stacks, providing consistent segmentation outputs across different material formulations and print parameters. Fig. 2 in the thesis visually summarizes each of these stages, from raw grayscale input through to the final labelled masks used for void and particle quantification. This workflow formed the foundation for all subsequent analyses, including void segregation, volume fraction computation, and mechanical correlation.



(a)

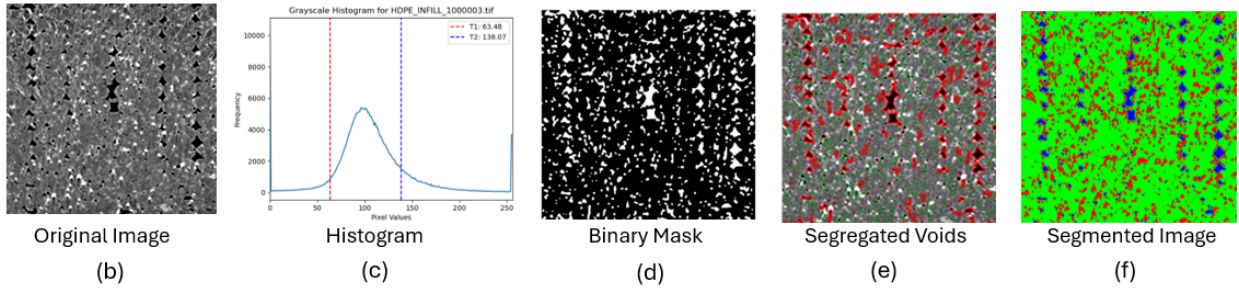


Figure 4.2 : Image segmentation methodology: (a) Complete processing workflow organized into five analysis stages, each corresponding to representative results shown in (b-f); (b) original  $\mu$ CT slice; (c) intensity histogram analysis; (d) binary mask generation; (e) void segregation; (f) three-phase segmented output.

First, image processing was performed in 2D rather than in 3D for both practical and methodological reasons. First, grayscale thresholds vary slice by slice due to local noise, artifacts, and intensity fluctuations, making adaptive 2D segmentation more accurate than applying a global 3D threshold. Second, pixel-level analysis in 2D is computationally less demanding and easier to validate than full voxel-based 3D segmentation, which requires significantly more memory and processing time.

For volumetric visualization and measurements requiring 3D continuity, marching cubes reconstruction was applied using the slice thickness specified by the user. Furthermore, certain analyses such as nearest neighbour distance (NND) were performed in 3D to preserve spatial relationships, while void shape categorization and morphological descriptors (e.g., area, aspect ratio, circularity) did not require full 3D treatment. This hybrid approach ensured accuracy, efficiency, and reproducibility, while still leveraging 3D methods where spatial fidelity was critical.

#### 4.2.2 Thresholds $T_1$ and $T_2$

Thresholds  $T_1$  and  $T_2$  are two critical grayscale intensity boundaries derived from the results of the K-means clustering algorithm. These thresholds demarcate the transitions between the three primary phases observed

in the  $\mu$ CT images of HDPE-HGM syntactic foams: air voids, HDPE matrix, and hollow glass microspheres (HGM). Specifically,  $T_1$  separates the void phase from the matrix, while  $T_2$  separates the matrix from the microsphere-filled regions. Accurate determination of  $T_1$  and  $T_2$  is crucial for ensuring precise label assignment and maintaining consistency in volume fraction and morphological quantification.

Once the K-means algorithm segments the grayscale image into three clusters based on voxel intensity, it assigns each cluster a centroid value that represents the average grayscale level of the pixels belonging to that cluster. These centroids are not static but are iteratively refined by the algorithm until convergence is achieved. For a given image, the three centroids ( $C_1$ ,  $C_2$ , and  $C_3$ ) are automatically ordered by intensity, as shown in the illustrative example in Figure 4.3. Based on histogram shape and prior domain knowledge, the centroids typically correspond to the following:

- a)  $C_1$ : Void (lowest intensity range)
- b)  $C_2$ : Matrix (intermediate range)
- c)  $C_3$ : Microspheres (brightest phase)

Thresholds  $T_1$  and  $T_2$  are computed as the midpoints between adjacent centroids. That is:

- a)  $T_1 = (C_1 + C_2) / 2$
- b)  $T_2 = (C_2 + C_3) / 2$

These thresholds are then applied to the label map to assign pixels to one of the three classes:

- a)  $\text{Intensity} \leq T_1 \rightarrow \text{Void}$
- b)  $T_1 < \text{Intensity} \leq T_2 \rightarrow \text{Matrix}$
- c)  $\text{Intensity} > T_2 \rightarrow \text{Particle (Microsphere)}$

This centroid-based thresholding allows the segmentation to adapt to grayscale variations across scans and maintains a data-driven separation of phases rather than relying on fixed, empirically set intensity values.

### 4.2.3 Quantification of Void and Particle Volume Fractions

While the initial thresholds  $T_1$  and  $T_2$  provide a global segmentation framework, certain regions of the  $\mu$ CT images exhibit subtle grayscale gradients or noise-induced inconsistencies that lead to misclassification near phase boundaries. To correct this, a neighbourhood refinement strategy is applied that incorporates a confidence margin parameter denoted as  $\epsilon$  (epsilon). This margin defines the allowable deviation from the threshold boundaries within which a voxel is considered uncertain or ambiguously classified. Figure 4.3 illustrates the simulated grayscale histogram used for initial threshold selection, showing the three cluster centroids (void, matrix, and particle)

For any voxel within the grayscale range  $[T_1 - \varepsilon, T_2 + \varepsilon]$  or  $[T_1 + \varepsilon, T_2 - \varepsilon]$ , the algorithm performs a neighbourhood check. A  $3 \times 3$  neighborhood window was selected because it is the smallest conventional mask used in spatial and morphological filtering, which balances local context and noise reduction [19]. This region around the voxel is evaluated to determine the dominant class in that region. If the voxel's class assignment disagrees with the majority of its neighbors, and if its intensity falls within the epsilon window, the voxel is relabeled to match the dominant class. This decision rule smooths class boundaries and corrects isolated misclassifications caused by fluctuations in intensity.

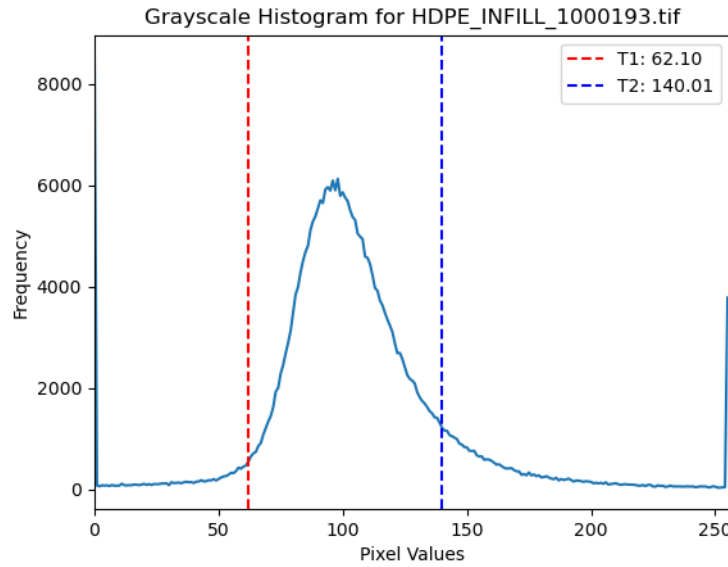


Figure 4.3 : Grayscale intensity histogram showing K-means thresholds  $T_1$  and  $T_2$ , calculated as midpoints between adjacent centroids, divides the intensity range into three regions corresponding to voids, matrix, and microspheres. Thresholds  $T_1$  and  $T_2$  are computed as the midpoints between adjacent cluster centroids and are used to segment the image into three material phases.

In this study,  $\varepsilon$  was empirically set to a value of 5 grayscale units after sensitivity analysis across several datasets. 5 grayscale units is a small one in the scale between 0 to 255, and this value has been implemented as a fixed parameter in the algorithm it need not be changed while using this approach for another material. This value provided a balance between correcting mislabeled edge voxels and preserving true microstructural features. A larger  $\varepsilon$  would risk over-smoothing and loss of detail, while a smaller value would reduce the effectiveness of the refinement. The combination of adaptive thresholds ( $T_1$ ,  $T_2$ ) derived from unsupervised learning and local  $\varepsilon$ -based reclassification significantly improves segmentation

accuracy. It ensures consistency across varying sample compositions and maintains fidelity to the actual void-matrix-particle interfaces observed in the  $\mu$ CT scans.

#### 4.2.4 Impact of Neighbourhood Refinement in K-Means Thresholding

The performance of K-means clustering in segmenting micro-CT images was found to depend highly on the initial placement of centroids. To improve the stability and reliability of segmentation across image slices with variable contrast, a neighbourhood-based refinement technique was integrated into the standard K-means algorithm. This approach applies an epsilon-based neighbourhood constraint during initialization, encouraging centroid selection that reflects local grayscale intensity distributions and reduces misclassification caused by random centroid seeding. To evaluate the effect of this refinement, two sets of threshold values were computed for each image slice. The first set includes  $T_1$  and  $T_2$ , which were obtained using conventional K-means clustering. The second set,  $T_{1K}$  and  $T_{2K}$ , reflects the corresponding thresholds generated using neighbourhood-refined clustering. Figure 4.4 illustrates the comparison between these values across more than 1,200 slices. In this study, each datapoint corresponds directly to a single  $\mu$ CT image slice. For every slice analyzed, two threshold values ( $T_1$  and  $T_2$ ) are computed based on the K-means cluster centroids. These thresholds are then used to classify all voxels in that slice into the three segmentation classes: void, matrix, and microsphere.

The right graph shows  $T_2$  and  $T_{2K}$ , which separate the matrix from the microsphere-filled domains.

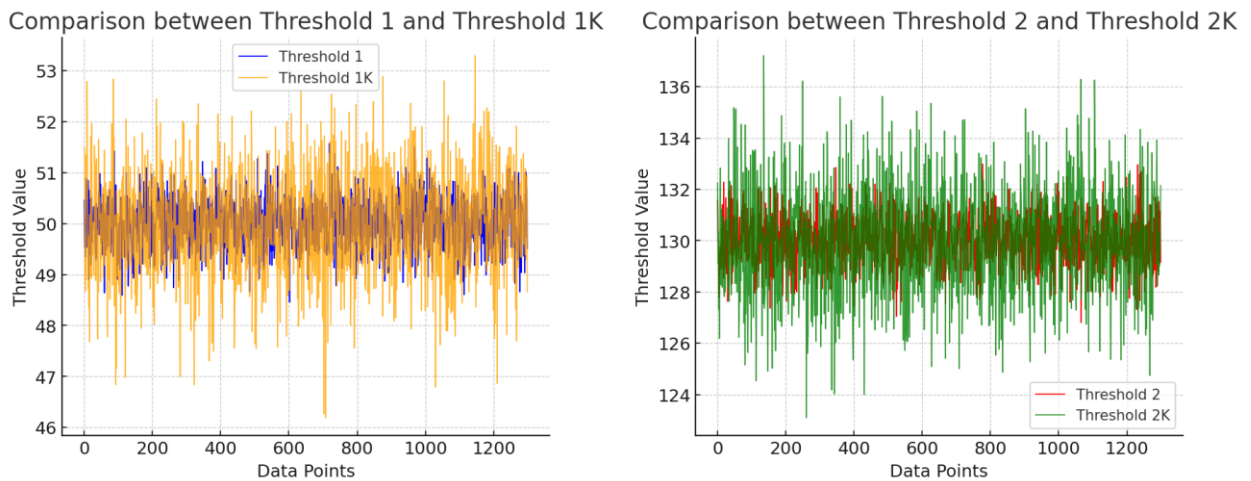


Figure 4.4 : Comparison of threshold values  $T_1$  and  $T_2$  with and without neighbourhood refinement for 100 % infill specimen labelled Composition A Sample 1. On average, the deviation between the standard and

refined thresholds was approximately 8 percent, which justifies the incorporation of neighborhood constraints to stabilize the segmentation process.

The convergence graph ([Figure 4.5](#)) represents the relationship between the average inertia of the K-means clustering algorithm and the number of iterations performed. Inertia, defined as the sum of squared distances between individual data points and their respective cluster centroids, serves as the cost function that the algorithm minimizes during the clustering process. At the beginning of the iterations, when the centroids are randomly initialized, the assignments are suboptimal and the inertia is comparatively high. During the first few iterations, there is a rapid decline in inertia as the centroids adjust toward the densest regions of the grayscale histogram. This sharp decrease reflects the substantial improvement in the compactness of clusters as the algorithm begins to stabilize. Following this initial phase, the curve reaches a plateau where further iterations result in negligible reductions in inertia. At this stage, the centroids fluctuate only slightly around stable positions, demonstrating that the clusters have effectively converged. The flat portion of the curve confirms that the maximum iteration limit set in the algorithm is more than sufficient, since the clustering stabilizes well before reaching that limit. Minor oscillations observed in the averaged curve arise from differences between individual  $\mu$ CT slices, each of which may reach convergence at slightly different iteration counts. These variations do not indicate instability but rather reflect natural differences in image data across the dataset. The convergence analysis therefore provides evidence that the K-means clustering procedure used for threshold determination is both stable and reproducible. It demonstrates that the segmentation routine consistently reaches a well-defined solution, validates the appropriateness of the iteration settings, and supports the robustness of the unsupervised segmentation method applied throughout this study.

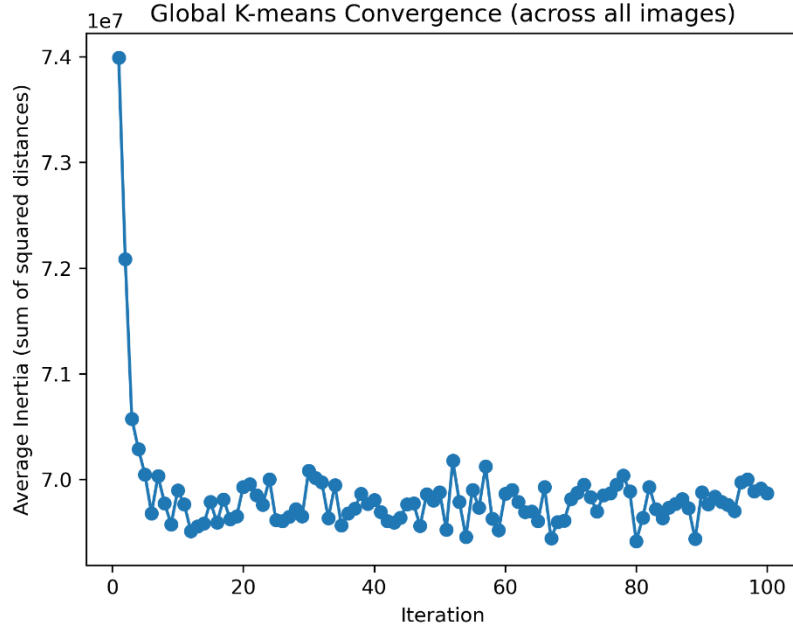


Figure 4.5: Convergence graph of the K-means clustering algorithm, for 100% infill Composition A, illustrating the rapid decrease in inertia during the first iterations and the subsequent plateau that confirms algorithmic stability and reliable segmentation outcomes.

#### 4.2.5 Quantification of Void and Particle Volume Fractions

Since each pixel corresponds to a physical size of 4.8  $\mu\text{m}$  and the slice thickness is also 4.8  $\mu\text{m}$ , the pixel area was calculated using Equation (1) and the voxel volume was obtained by multiplying the pixel area by the slice thickness, as shown in Equation (2). The total void volume was determined by multiplying the voxels identified as voids by the voxel volume Equation (3). In contrast, the particle volume was calculated similarly using Equation (4) for the particle voxel count. The total sample volume was obtained by multiplying the total number of segmented voxels by the voxel volume, as defined in Equation (5). Finally, the void and particle volume fractions were computed as percentages of the total sample volume using Equations (6) and (7), respectively. These calculations provided reliable quantitative measures of porosity and reinforcement distribution, enabling a direct link between processing parameters and microstructural outcomes.

$$A_{\text{pixel}} = R^2 \quad (1)$$

$$V_{\text{pixel}} = A_{\text{pixel}} \times T \quad (2)$$

$$V_{\text{void}} = N_{\text{void}} \times V_{\text{pixel}} \quad (3)$$

$$V_{\text{particle}} = N_{\text{particle}} \times V_{\text{pixel}} \quad (4)$$

$$V_{\text{sample}} = N_{\text{total}} \times V_{\text{pixel}} \quad (5)$$

$$VF_{\text{void}} = \frac{V_{\text{void}}}{V_{\text{sample}}} \times 100 \quad (6)$$

$$VF_{\text{particle}} = \frac{V_{\text{particle}}}{V_{\text{sample}}} \times 100 \quad (7)$$

Where,

$A_{\text{pixel}}$  is the area of a single pixel ( $\mu\text{m}^2$ ),

$R$  is the pixel resolution or the physical dimension of one side of a pixel ( $\mu\text{m}$ ), and thus  $(R)^2$

$V_{\text{pixel}}$  is the volume of each pixel computed by multiplying its area with the slice thickness  $T$  ( $\mu\text{m}$ ),

$V_{\text{void}}$  is the total volume occupied by void pixels ( $\mu\text{m}^3$ ),

$V_{\text{particle}}$  is the total volume occupied by particle pixels ( $\mu\text{m}^3$ ),

$V_{\text{sample}}$  is the total volume of the scanned region ( $\mu\text{m}^3$ ),

$N_{\text{void}}$ ,  $N_{\text{particle}}$  and  $N_{\text{total}}$  are the number of pixels classified as voids, particles, and total, respectively.

Finally,  $VF_{\text{void}}$  and  $VF_{\text{particle}}$  represent the void and particle volume fractions as a percentage of the total sample volume ( $\mu\text{m}^3$ )

The need for this correction is supported by the Nyquist-Shannon sampling theorem, which states that to accurately reconstruct a signal (or feature), the sampling frequency must be at least twice the highest frequency present [20].

#### 4.2.6 Void Segregation

Void segregation was implemented using a fully automated image analysis pipeline developed in Python, which incorporated grayscale conversion, global thresholding, and unsupervised clustering via the k-means algorithm. The input consisted of  $\mu\text{CT}$ -acquired 8-bit grayscale images, which were loaded using OpenCV's `cv2.imread()` function with the flag set to grayscale. This ensured that only pixel intensities were retained, minimizing memory usage while preserving intensity-based contrast essential for segmentation. Threshold ( $T_2$ ) was not included for further analysis, because, the focus is on differentiating types of voids and extracting their morphology, which corresponds to pixels lower than  $T_1$ .  $T_2$  however, was computed to

perform multimodal analysis and it defines the boundary between the polymer matrix and the microsphere phase.  $T_2$  can be used to isolate the reinforcements and analyze their morphology, for future studies.

Global thresholding was performed using pre-determined thresholds ( $T_1$ ) derived from the clustering of grayscale histograms, optionally refined using the neighbourhood method. The thresholded binary masks were generated using *cv2.threshold()*, resulting in black-and-white images where potential voids appeared as foreground pixels.

Subsequently, *cv2.connectedComponentsWithStats()* was applied to perform 2D connected component analysis, which labeled continuous foreground regions and extracted their size and location attributes. To distinguish between manufacturing-induced raster voids and inherent HGM cavities, morphological descriptors were computed for each connected region. Shape analysis involved calculating circularity and aspect ratio. Here, the area and perimeter were determined using *cv2.contourArea()* and *cv2.arcLength()*, respectively. This metric approaches 1.0 for perfect circles and decreases for irregular or elongated shapes. To supplement this, the aspect ratio was calculated as shown in Equation (9)

$$\text{Circularity} = \frac{(4 \times \pi \times A)}{P^2} \quad (8)$$

Where,

A is the area of the void ( $\mu\text{m}^2$ ),

P is the perimeter of the void ( $\mu\text{m}$ ),

$$\text{Aspect Ratio} = \frac{\max(\text{Major Axis}, \text{Minor Axis})}{\max(\text{Minor Axis}, 1)} \quad (9)$$

Where,

Major Axis is length of the longer axis of the fitted ellipse ( $\mu\text{m}$ ),

Minor Axis is the length of the shorter axis of the fitted ellipse ( $\mu\text{m}$ ),

Max() ensures the larger of the two values is used; the denominator uses a lower bound of 1 to prevent division by zero or instability.

Classification was then performed based on empirically validated thresholds. Figure 4.5 outlines the complete void classification and reconstruction workflow, from grayscale thresholding and morphological feature extraction to the classification of voids based on circularity and aspect ratio. Regions with circularity greater than 0.87 and aspect ratio close to 1 were classified as HGM cavities. These values are chosen based on the dataset and visual inspection. The area was calculated from pixel counts per slice. At the same time,

volumetric estimation was performed by summing the areas across slices and multiplying by the slice thickness, which was derived from the  $\mu$ CT voxel size. This data was then used to generate a 3D reconstruction of voids, differentiating spatially distributed raster voids from spherical microsphere cavities.

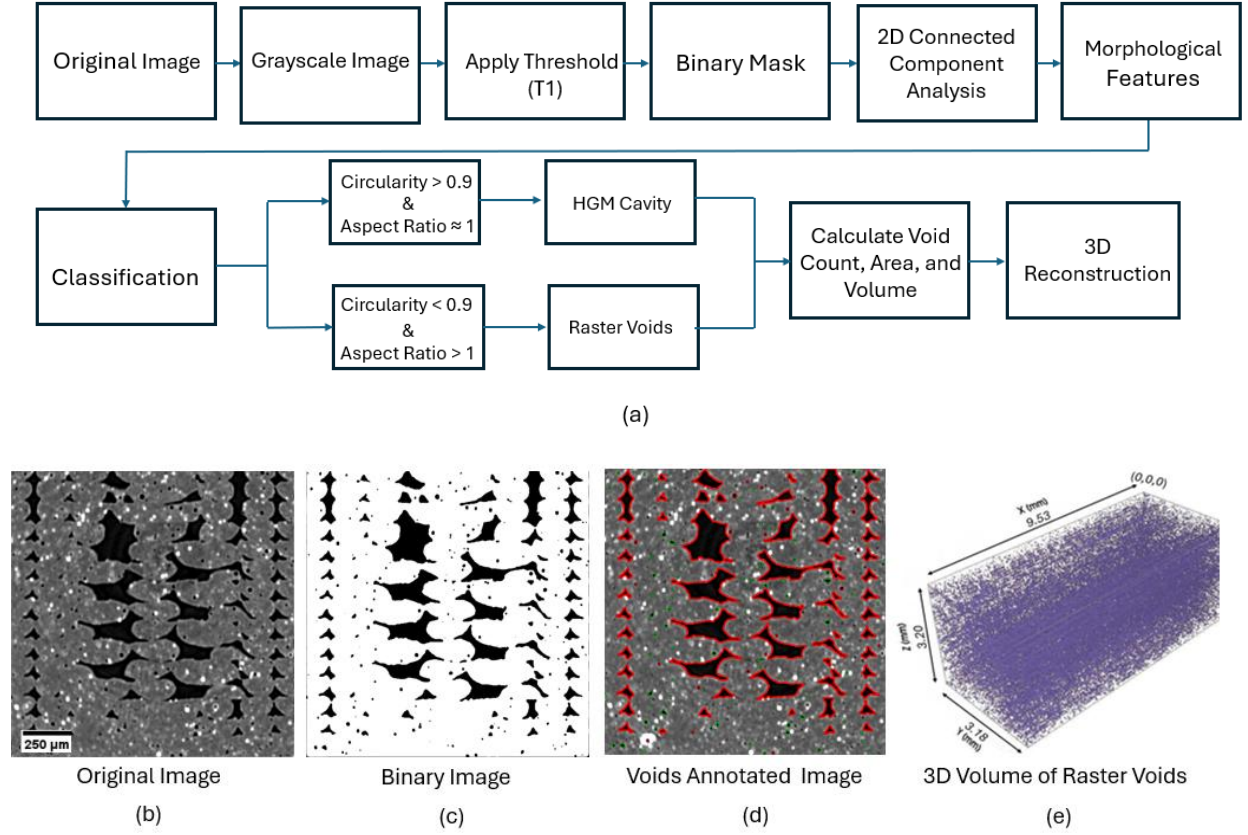


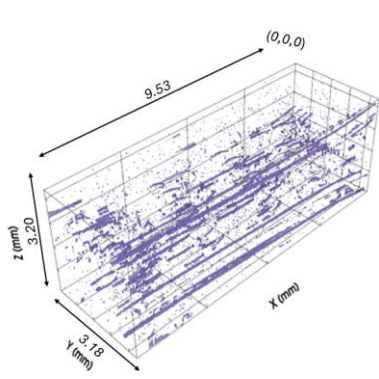
Figure 4.5 : Void classification and quantification workflow in 3D-printed syntactic foams: (a) Processing flowchart showing the sequential steps from original image through morphological analysis to final classification and 3D reconstruction; (b-e) Representative results demonstrating void identification: (b) original  $\mu$ CT slice; (c) binary mask generated using threshold T1; (d) annotated void classification showing raster voids in red and HGM cavities in green; (e) 3D volume reconstruction of classified void structures

## 4.3 Results and Discussion

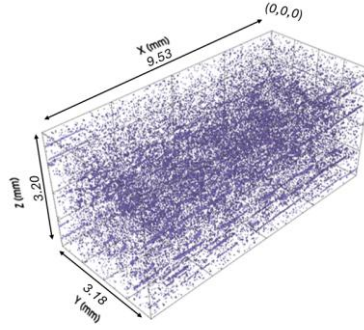
### 4.3.1 Output Visualizations

#### 4.3.1.1 Volumetric Representation of Voids

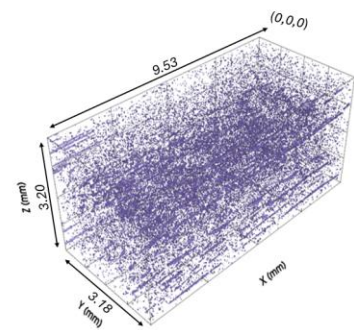
The 3D representation is shown in



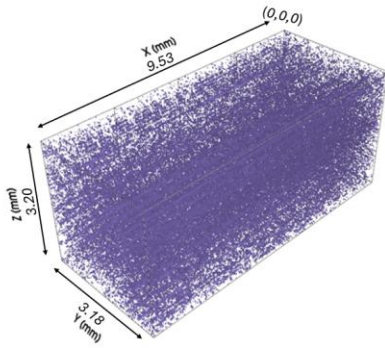
(a)



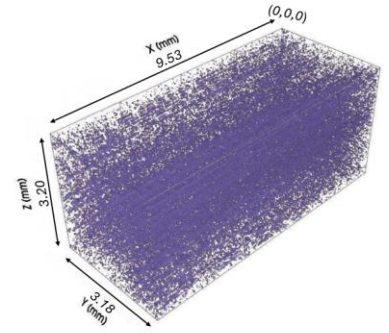
(b)



(c)

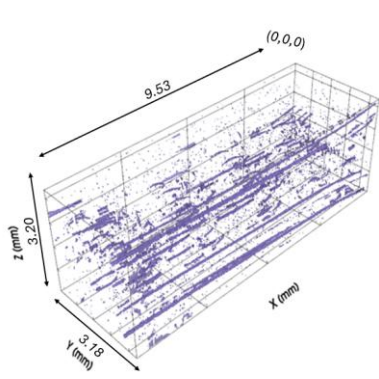


(d)

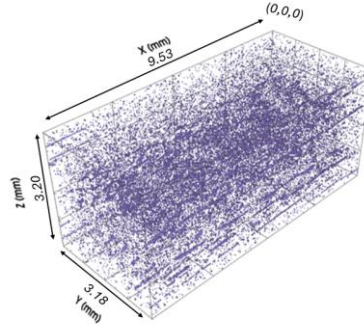


(e)

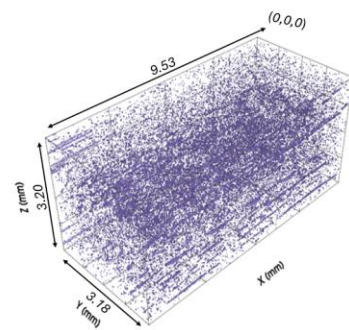
Figure 4.6, which visualizes voids segmented from  $\mu$ CT images as volumetric binary masks. Violet or blue regions correspond to void voxels (value = 1), and the enclosing box represents the full sample volume of interest (value = 0). This binary void mask, also referred to as an internal region of interest (iROI), was generated by stacking slice-wise void segmentations using the global or slice-wise T1 thresholds. Each connected component within the iROI represents a distinct void feature identified via morphological processing (connected components labelling and shape filtering).



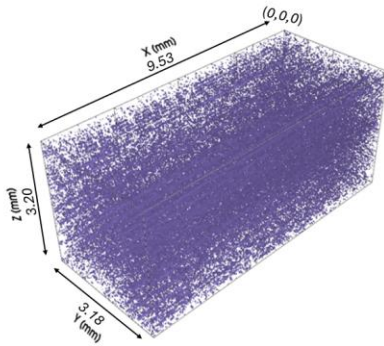
(a)



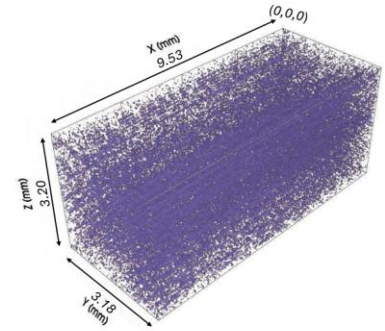
(b)



(c)



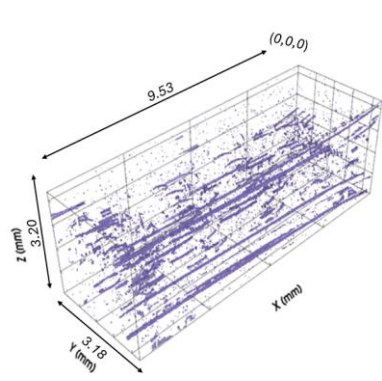
(d)



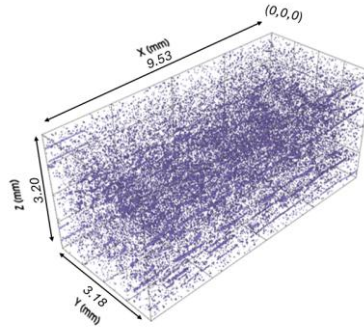
(e)

Figure 4.6 : Volumetric representation of raster voids in 3D-printed HDPE-HGM syntactic foams at varying infill densities and Compositions: (a) 100 % Comp. A, (b) 90 % Comp. B, (c) 80 % Comp. A, (d) 70 % Comp. B, and (e) 60 % Comp. A

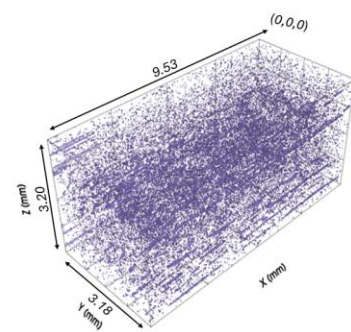
The 3D volumetric reconstructions using the Marching Cubes algorithm in



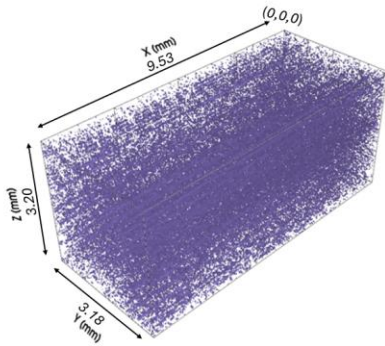
(a)



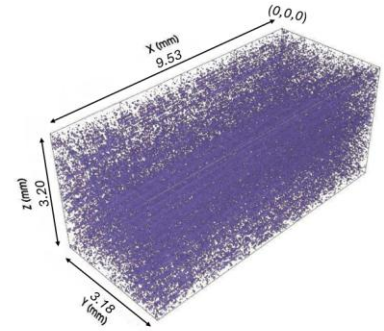
(b)



(c)



(d)



(e)

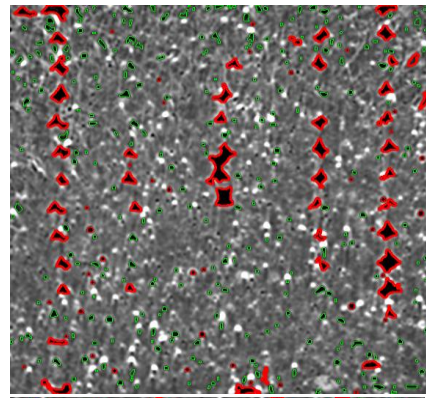
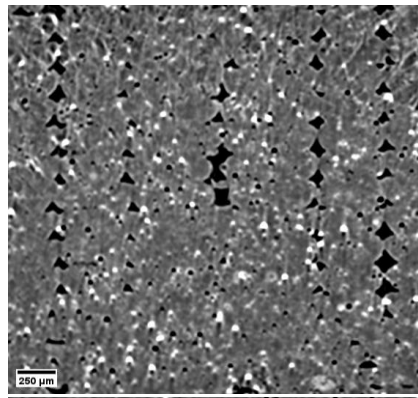
Figure 4.6 reveal a systematic evolution of void architecture with decreasing infill density and distinct composition-dependent behaviors across the HDPE-HGM syntactic foam samples. At high infill densities, samples (a) 100% Composition A and (b) 90% Composition B exhibit discrete, uniformly distributed voids that appear small and isolated, indicating optimal layer fusion with minimal inter-raster gaps and good structural integrity. As infill density decreases to medium levels in Composition A (c), a critical morphological transition becomes evident, with voids beginning to elongate and exhibit directional alignment corresponding to the raster printing direction, suggesting the onset of fusion-related porosity. This transition becomes more pronounced at lower infill densities, where (d) 70% Composition B and (e) 60% Composition A display dramatically increased void content with elongated, interconnected networks that form continuous porosity channels aligned with the printing direction. Notably, Composition A samples (a, c, e) consistently demonstrate more isolated void structures compared to Composition B samples (b, d), suggesting superior fusion characteristics or different thermal behavior that better contains porosity even at reduced infill densities. The progression from isolated to interconnected void networks indicates that infill

densities below 80% likely represent a critical threshold where mechanical integrity becomes compromised due to anisotropic void architecture, while the connected porosity channels at 70-60% infill densities suggest significantly altered permeability characteristics that could impact both structural performance and functional applications of these syntactic foams.

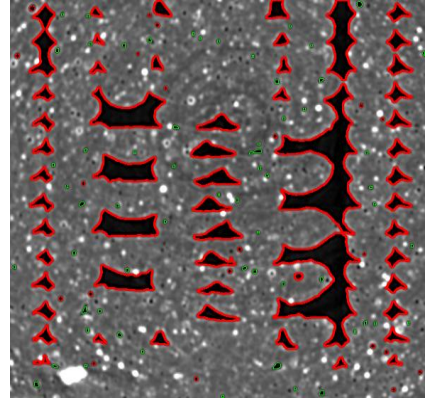
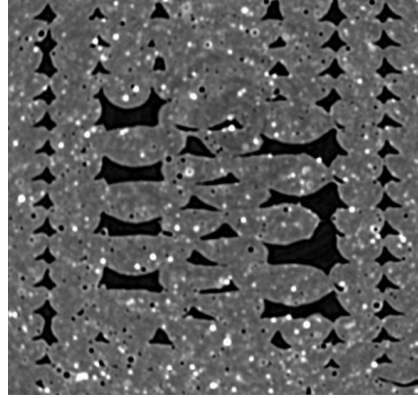
#### 4.3.1.2 Void Segregation and Morphological Classification Visualization

Shape descriptor-based analysis successfully differentiated circular HGM cavities from elongated manufacturing-induced raster voids. Figure 4.7 and Figure 4.8 illustrate the void segregation results, showing the evolution of void morphology with decreasing infill density. Raster-induced voids (highlighted in red) become increasingly prominent and interconnected at lower infill densities, while HGM cavities (marked in green) maintain their discrete, circular morphology. Due to their small size, green features may not be visible in low-resolution or printed formats. However, they become distinct when viewed digitally at higher magnification. The 3D connectivity analysis revealed that void structures transition from discrete, isolated features at higher infill densities ( $\geq 80\%$ ) to interconnected networks at lower infill densities ( $\leq 70\%$ ).

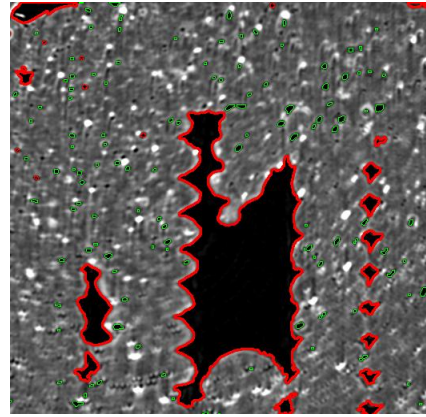
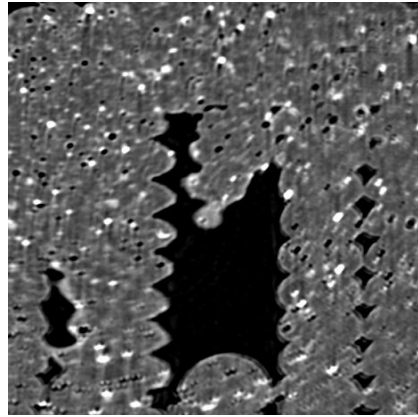
(a) 100 % Infill



(b) 80 % Infill



(c) 60 % Infill

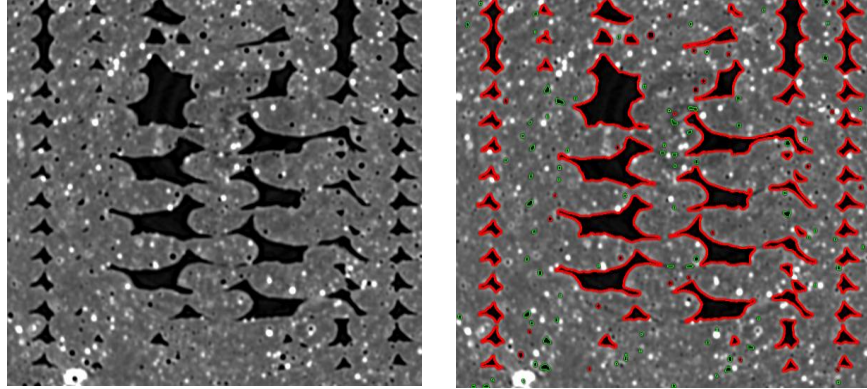


Original Image

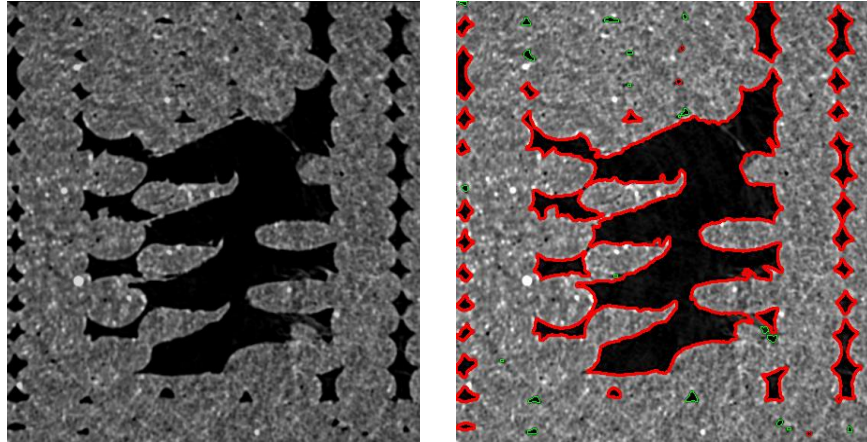
Voids Annotated Image

Figure 4.7 : Original  $\mu$ CT images and corresponding voids annotated images segmentation of 3D-printed HDPE-HGM syntactic foams at varying infill densities and compositions: (a) 100 % Comp. A, (b) 80 % Comp. A, (g) 60 % Comp. A. Raster-induced voids are highlighted in red, and hollow glass microsphere (HGM) cavities are marked in green, illustrating effective void segregation across different infill conditions.

(a) 90 % Infill



(b) 70 % Infill



Original Image

Voids Annotated Image

Figure 4.8 : Original  $\mu$ CT images and corresponding voids annotated images segmentation of 3D-printed HDPE-HGM syntactic foams at varying infill densities and compositions: (a) 90 % Comp. B, (b) 70 % Comp. B. Raster-induced voids are highlighted in red, and hollow glass microsphere (HGM) cavities are marked in green, illustrating effective void segregation across different infill conditions

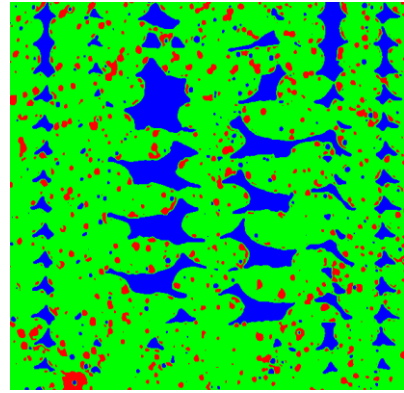
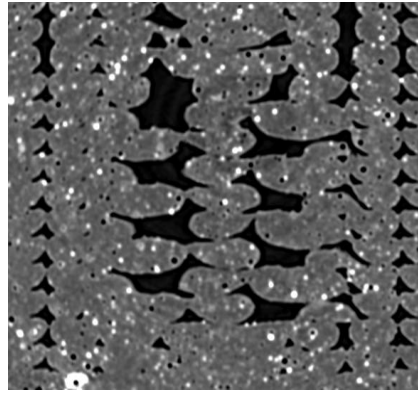
This finding has important implications for mechanical properties and permeability characteristics of the syntactic foams. The modular pipeline architecture supports broader applications across various porous material systems, with adaptable parameters for different composite types.

#### 4.3.1.1 Multimodal Segmentation Slice-wise Visualization

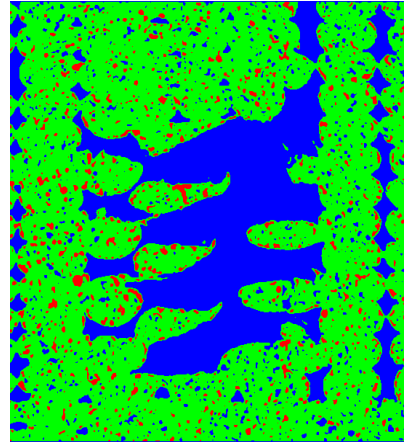
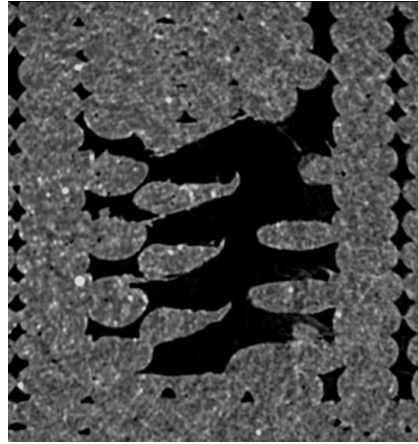
Representative 2D  $\mu$ CT slices were extracted from volumetric scans of syntactic foam specimens and processed using an optimized segmentation pipeline that incorporates K-means clustering with neighbourhood refinement for robust phase separation. Post-segmentation, morphological labels were assigned based on grayscale intensity thresholds and connectivity criteria, with colour mapping applied to visualize the HDPE matrix (green), HGM particles (red), and void domains (blue), as shown in Figure 4. and

Figure 4.. This enabled direct qualitative assessment of microstructural heterogeneity across varying infill densities and composite formulations.

(a) 90 % Infill



(b) 70 % Infill

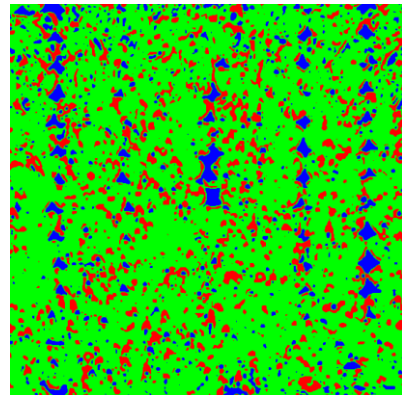
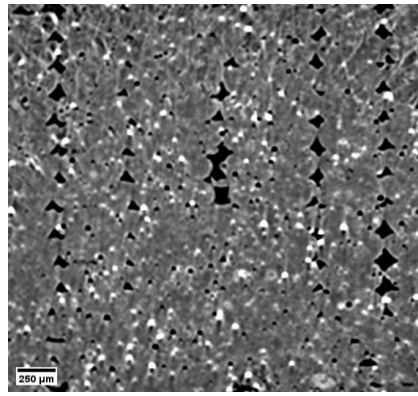


Original Image

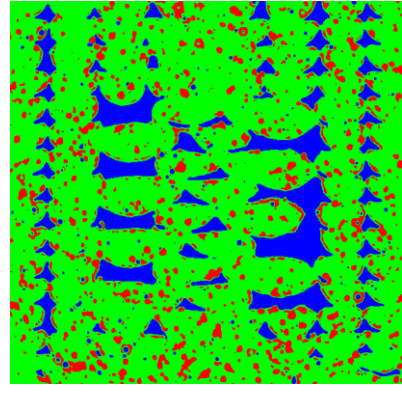
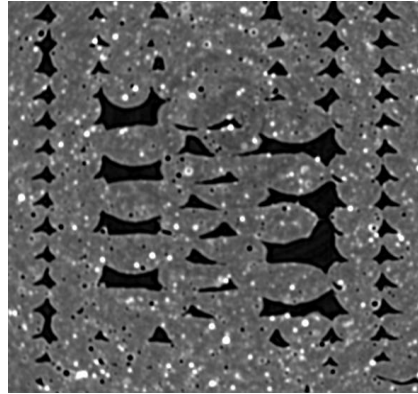
Segmented Image

Figure 4.10 : Original  $\mu$ CT images and corresponding segmented outputs of 3D-printed HDPE-HGM syntactic foams at varying infill densities and compositions: (a) 90% Comp. B, (b) 70% Comp. B. Segmentation highlights HGM particles in red, voids, including HGM cavities in blue, and the polymer matrix in green, demonstrating material distribution and void formation patterns across different infill conditions and compositions.

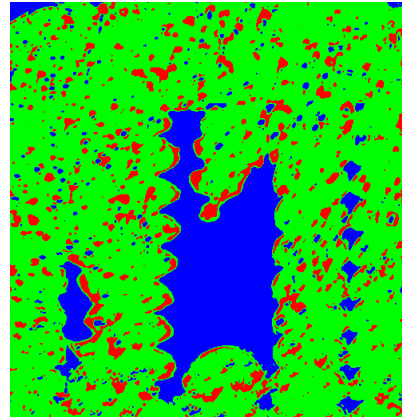
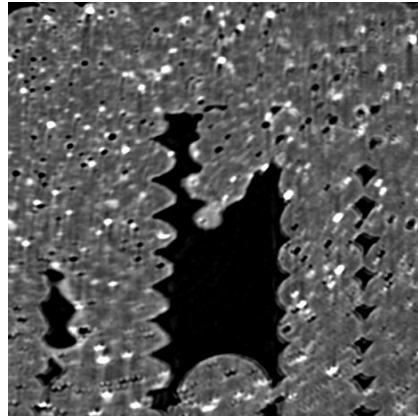
(a) 100 % Infill



(c) 80 % Infill



(e) 60 % Infill



Original Image

Segmented Image

Figure 4.11: Original  $\mu$ CT images and corresponding segmented outputs of 3D-printed HDPE-HGM syntactic foams at varying infill densities and compositions: (a) 100 % Comp. A, (b) 80% Comp. A, (c) 60% Comp. A. Segmentation highlights HGM particles in red, voids, including HGM cavities in blue, and the polymer matrix in green, demonstrating material distribution and void formation patterns across different infill conditions and compositions.

### 4.3.2 Void and Particle Volume Fractions by Infill Density

For Composition A, void volume fraction increased with decreasing infill density, from 3.28% at 100% infill to 20.98% at 60% infill (Figure 4.9). Even at 100% infill, inherent porosity was observed due to raster gaps characteristic of fused filament fabrication, consistent with findings by Gnanasekaran et al. [21]. The progression shows an accelerating void formation trend as infill density decreases, reflecting the combined effects of reduced material deposition and increased raster spacing.

Particle volume fractions for Composition A remained relatively stable (15.95% to 12.21%) across the infill range, though these values should be interpreted cautiously due to  $\mu$ CT resolution limitations discussed below. The matrix volume fraction decreased correspondingly from 80.77% to 66.81% as the void content increased. The sum of void, particle, and matrix volume fractions consistently totaled 100%, confirming complete pixel classification with no data loss. The computed void, particle and matrix volume fractions for each sample are summarized in Table , highlighting the inverse relationship between infill density and void content, as well as the relative stability of particle content across samples.

Table 4.5: Calculated particles, void, and matrix volume fractions for each sample.

| Infill Density (%) | Comp. | Void Volume Fraction (%) | Particle Volume Fraction (%) | Matrix Volume Fraction (%) |
|--------------------|-------|--------------------------|------------------------------|----------------------------|
| 100                | A     | 3.28                     | 15.95                        | 80.77                      |
| 80                 | A     | 15.85                    | 9.95                         | 74.68                      |
| 60                 | A     | 20.98                    | 12.21                        | 66.81                      |
| 90                 | B     | 15.37                    | 9.90                         | 74.25                      |
| 70                 | B     | 16.30                    | 24.67                        | 59.02                      |

Composition B samples exhibited void volume fractions of 15.37% at 90% infill and 16.30% at 70% infill, increasing as the infill density decreased. A visual representation of the void evolution with infill density is provided in Figure 4.9, which shows  $VF_{\text{void}}$  versus infill density. As shown, the  $VF_{\text{void}}$  decreases nonlinearly as infill density increases from 60% to 100%.

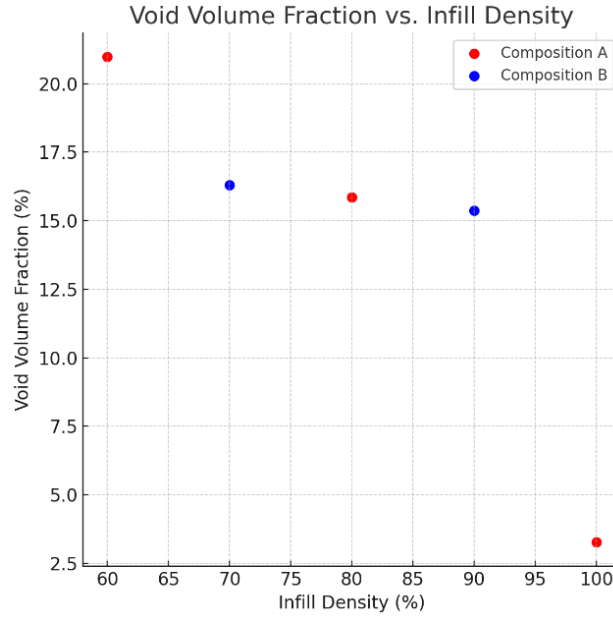


Figure 4.9 : Relationship between infill density and  $VF_{\text{void}}$  in HDPE-HGM syntactic foams, highlighting the differences between Comp. A (red) and Comp. B (blue). Lower infill densities correspond to higher void content

The experimental design enabled evaluation of composition performance across different void ranges, with one direct comparison possible at similar void levels: Composition A at 80% infill (15.85% voids) versus Composition B at 70% infill (16.30% voids). Notably, Composition A achieved slightly lower void content despite operating at higher infill density, indicating enhanced void suppression characteristics. This comparison, combined with the distinct void ranges explored for each composition, provides valuable insights into material performance across different porosity regimes.

The observed variations in  $VF_{\text{particle}}$  across different samples can be attributed primarily to the resolution limitations of the  $\mu\text{CT}$  scanner. The  $4.8\ \mu\text{m}$  voxel size is close to the average wall thickness ( $\sim 0.37\ \mu\text{m}$ ) of the hollow glass microspheres in Composition A, causing partial volume effects and underrepresentation of microsphere shells in the segmented data. This limitation, emphasized by Kaestner et al. [22], prevents reliable measurement of actual HGM content. Additionally, grayscale fluctuations due to beam hardening and intensity inhomogeneity can result in misclassification during segmentation [23]. From a manufacturing perspective, localized variation in particle distribution, possibly due to low flow pressures or manual feed inconsistencies, may contribute to inhomogeneities in Composition B. Similar challenges in achieving uniform particle embedding and interfacial bonding were observed by Obielodan et al. in ultrasonic consolidation processes involving dissimilar materials and embedded powders [24].

These findings confirm that discrepancies between nominal and calculated particle volume fractions represent broader challenges inherent to  $\mu$ CT-based microstructural analysis of syntactic foams. This shows that the accuracy of this model depends on the resolution of the scans, and it is acknowledged that voids smaller than 5  $\mu\text{m}$  are not accounted for, due to their small size.

### 4.3.3 Computed Global Thresholds

To ensure accurate segmentation of multi-phase  $\mu$ CT image stacks, slice-wise grayscale thresholds were first computed using unsupervised k-means clustering for each slice. These per-slice thresholds were used in all primary analyses and 3D reconstructions presented in this study, offering high local accuracy by adapting to contrast and intensity variations across slices. However, for reproducibility in external platforms such as Avizo or for streamlined batch processing, a set of representative global thresholds was also extracted for each dataset. Specifically, the maximum  $T_1$  (void threshold) and minimum  $T_2$  (particle threshold) from the slice-wise values were selected to ensure conservative yet consistent segmentation across the entire stack. While global thresholding introduces a slight reduction in segmentation precision compared to the adaptive slice-wise approach, it enables reproducible, scalable, and user-independent processing, which is particularly useful for third-party validation or cross-platform deployment. This dual strategy balances segmentation accuracy with practical reproducibility and is especially advantageous for large-scale additive manufacturing datasets that require consistent morphological and volumetric comparisons. The respective lower ( $T_1$ ) and upper ( $T_2$ ) global thresholds per dataset are listed in Table 6.

Table 4.6 : Recommended global grayscale thresholds  $T_1$  and  $T_2$  for void and particle segmentation, respectively, computed per sample.

| Infill Density (%) | Composition | $T_1$ | $T_2$  |
|--------------------|-------------|-------|--------|
| 100                | A           | 79.78 | 130.16 |
| 90                 | B           | 56.48 | 120.28 |
| 80                 | A           | 56.61 | 120.28 |
| 70                 | B           | 59.17 | 125.09 |
| 60                 | A           | 71.36 | 116.24 |

The thresholds show clear variation across infill densities and between Compositions A and B. As infill density increases (e.g., from 60% to 100%), the global  $T_1$  values fluctuate, reflecting differences in the image contrast of void regions. Composition A samples (100%, 80%, 60%) generally exhibit higher  $T_1$  values. Meanwhile, Global  $T_2$  increases consistently with infill, indicating that particle regions become brighter or more distinct, especially in the denser samples like 100% infill ( $T_2 = 130.16$ ).

The variation in  $T_2$  also suggests a composition-dependent brightness shift in the particle phase. This trend supports the hypothesis that both material composition and infill density influence  $\mu$ CT grayscale distributions and thus affect threshold selection.

## 4.4 Validation

Validation of the developed segmentation pipeline was conducted through two complementary approaches: visual agreement with manual segmentation and correlation with mechanical testing data. This dual validation strategy ensures both pixel-level accuracy in detecting microstructural features and macro-level consistency in predicting mechanical behavior. By comparing automated outputs with human-annotated references and evaluating trends against tensile properties, the robustness and reliability of the algorithm were assessed both quantitatively and qualitatively.

### 4.4.1 Validation against Manual Segmentation

To evaluate the performance of the proposed segmentation algorithm, manual ground truth masks were generated using Labkit, a user-friendly Fiji plugin for labelling and segmenting microscopy image data optimized to efficiently process very large (terabyte-range) images on consumer hardware. It couples manual drawing tools with automated pixel classification by random forest algorithms, enabling users to rapidly segment images by drawing sparse manual labels (scribbles) that train a classifier to compute dense segmentations automatically. The tool relies on the foundations of ImgLib2 and BigDataViewer, supports GPU acceleration, can be embedded in automated workflows through ImageJ macros, and is compatible with high-performance computing clusters for distributing the processing of extensive image data [24]. Segmentation accuracy was assessed using standard metrics such as precision, recall, intersection over union (IoU), and F1 score, defined respectively as the proportion of correctly predicted void pixels, the proportion of actual void pixels detected, the ratio of intersection to union of predicted and true void regions, and the harmonic mean of precision and recall. Across five representative samples, the algorithm achieved a mean precision of 0.945, recall of 0.908, IoU of 0.882, and F1 score of 0.926, demonstrating strong agreement with expert annotations (Table ), which apply exclusively to void detection and classification accuracy. These performance indicators demonstrate the method's effectiveness in accurately quantifying voids. The segmentation algorithm's strength lies in its ability to reliably identify and classify void structures, enabling a meaningful analysis of the effects of manufacturing parameters on porosity formation. However, these validation metrics do not extend to particle quantification accuracy, where the fundamental resolution constraints prevent reliable measurement of actual HGM content.

Performance variation in samples (d) and (e) was attributed to insufficient contrast, grayscale overlap, and the inherent subjectivity in manual boundary delineation. Additionally, beam hardening, intensity

inhomogeneity, and morphological filtering posed challenges for accurate segmentation of smaller voids. These findings are consistent with recent literature. The achieved F1 score of 0.926 compares favorably with recent machine learning approaches for composite characterization. Upadhyay et al. [10] demonstrated higher precision/recall (0.99) in void detection within filament-wound composites, but their deep learning approach required extensive training datasets and specialized computational resources. In contrast, our unsupervised K-means method achieves comparable performance (0.926 F1 score) without requiring annotated training data or computational overhead. While Zhang et al.'s supervised deep learning approach achieved a score of 0.82, it needed large, annotated datasets and retraining. The unsupervised K-means pipeline developed in this study achieved a higher mean IoU of 0.882 and an F1 score of 0.926 without any manual labelling, with minimal user involvement [25]. An F1 score above 0.9 is considered sufficient for image segmentation methods, with higher F1 scores showing better accuracy.

For syntactic foam-specific applications, this work represents a substantial improvement over existing approaches. Lawrence et al. [26], and Ferreira et al. [27] required significant manual intervention and expertise. The automated nature of our approach addresses these limitations while maintaining high accuracy.

Table 4.7 : Performance metrics of the automated void segmentation algorithm compared to manual ground truth annotations across five samples.

| Sample             | *Precision | *Recall | IoU*  | *F1 Score |
|--------------------|------------|---------|-------|-----------|
| 1                  | 0.975      | 0.925   | 0.904 | 0.950     |
| 2                  | 0.943      | 0.921   | 0.873 | 0.932     |
| 3                  | 0.971      | 0.905   | 0.881 | 0.937     |
| 4                  | 0.890      | 0.915   | 0.920 | 0.902     |
| 5                  | 0.946      | 0.874   | 0.833 | 0.909     |
| Mean               | 0.945      | 0.908   | 0.882 | 0.926     |
| Standard Deviation | 0.034      | 0.020   | 0.034 | 0.020     |

\*Intersection of Union – The ratio of the overlapping area between predicted and ground truth regions to the total area covered by both regions combined.

\*Precision – The fraction of predicted positive instances that are correct (true positives divided by all predicted positives).

\*Recall – The fraction of actual positive instances that were correctly identified (true positives divided by all actual positives).

\*F1 Score – The harmonic mean of precision and recall provides a balanced measure of model performance that accounts for false positives and false negatives.

The segmentation algorithm's strength lies in its ability to reliably identify and classify void structures, enabling a meaningful analysis of the effects of manufacturing parameters on porosity formation. However, these validation metrics apply exclusively to void detection and classification accuracy and do not extend to particle quantification accuracy, where the fundamental resolution constraints prevent reliable measurement of actual HGM content. This limitation is inherent to the  $\mu$ CT resolution (4.8  $\mu\text{m}$ ) relative to HGM wall thickness ( $\sim 0.37 \mu\text{m}$ ), as discussed in the methodology section.

#### 4.4.2 Comparison with Tensile Data

Tensile testing of the additively manufactured syntactic foam specimens was previously described in Section 3.5, where the experimental setup, specimen geometry, and testing parameters were detailed. In this section, the mechanical test results are compared with microstructural parameters extracted from the  $\mu$ CT analysis to investigate potential correlations between tensile performance and void or particle morphology. The ultimate tensile strength (UTS) exhibited a clear inverse relationship with void fraction ( $\text{VF}$ ), aligning with established composite theory. Composition A demonstrated strong tensile performance across its tested porosity range (Figure 4.10). At 3.28%  $\text{VF}_{\text{void}}$  (100% infill), the material reached a maximum UTS of 8.61 MPa. This trend is consistent with previous studies, such as the one by Nikishkov et al., which highlight the impact of even minimal void content on composite performance, both reporting significant reductions as void content increases [28]. The superior performance of Composition A is due to its higher microsphere volume fraction (55.8% vs. 37.6%) and lower-density HGMs (0.14  $\text{g/cm}^3$  vs. 0.32  $\text{g/cm}^3$ ). Lower-density HGMs generally have thinner walls and higher internal void ratios, which have been shown to improve matrix-filler interfacial bonding and reduce stress concentration effects compared to higher-density microspheres.

Composition B, evaluated at  $\text{VF}_{\text{void}}$  levels between 15.37-16.30%, exhibited UTS values ranging from 6.5-6.8 MPa. While this represents a narrow porosity range, the results provide critical insights when compared to Composition A at similar void levels. The 15-20% lower strength compared to Composition A at comparable porosity cannot be attributed to void content alone, indicating that microstructural architecture plays a dominant role.

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attributed to void content alone, indicating that microstructural architecture plays a dominant role. The inferior performance stems from Composition B's higher-density HGMs (0.32 g/cc vs 0.14 g/cc) and lower volume fraction (37.6% vs 55.8%).

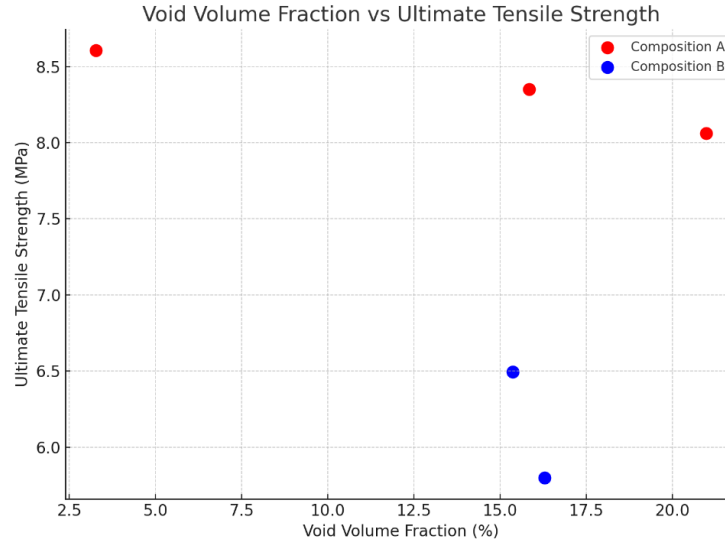


Figure 4.10 : The relationship between  $VF_{void}$  and UTS for all samples with varying infill densities and Comp. A and B. Red markers indicate samples with Comp. A, and blue markers represent Comp. B. A clear inverse trend is observed, with Comp. A maintaining higher tensile strength across a broader range of  $VF_{void}$ , highlighting the role of microstructural characteristics in governing mechanical performance.

The Young's modulus response of Composition A revealed more complex structure-property relationships than observed for UTS (Figure 4.10). Starting at 0.36 GPa at 3.28%  $VF_{void}$ , the modulus showed a non-linear degradation pattern: 22% reduction to 0.28 GPa at 15%  $VF_{void}$ , followed by a more pronounced 25% decrease to 0.21 GPa at 20.98%  $VF_{void}$ .

This accelerating degradation pattern suggests a transition in deformation mechanisms and provides evidence for a microstructural percolation threshold. Composition A's dense HGM network initially maintains effective load transfer despite increasing porosity. A study notes that syntactic foams made with low-density microspheres exhibited improved packing and reinforcement continuity, leading to enhanced specific tensile strength [29]. However, beyond 15%  $VF_{void}$ , the steeper decline indicates that void connectivity begins to compromise the structural network's ability to transmit loads efficiently.

Composition B exhibited significantly lower stiffness characteristics, with modulus values ranging from 0.18-0.24 GPa at  $VF_{void}$  levels of 15.37-16.30%. The substantial difference from Composition A at comparable void levels (0.28 GPa vs 0.18-0.24 GPa) demonstrates that matrix formulation and reinforcement efficiency are as critical as porosity control.

The lower modulus values reflect the inferior reinforcement architecture in Composition B. With fewer HGMs distributed less efficiently throughout the matrix, the composite shows a reduced ability to maintain structural continuity under load. The  $\mu$ CT-based segmentation approach, incorporating K-means clustering and void morphology classification, proved effective in isolating raster-induced porosity from microsphere-related cavities, a key advancement over global thresholding techniques. These results validate the utility of void volume as a predictive microstructural parameter and underscore the importance of microsphere dispersion and manufacturing precision in ensuring high tensile strength in AM syntactic foams.

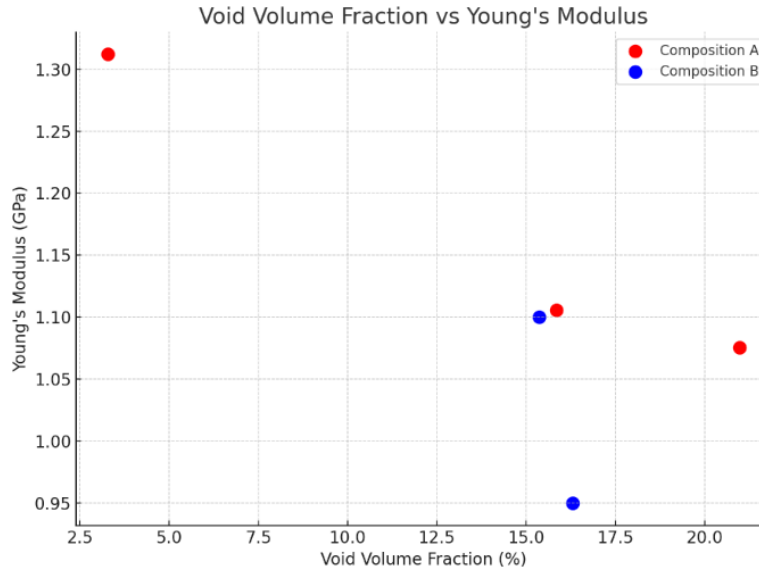


Figure 4.11 : Relationship between  $VF_{\text{void}}$  and Young's modulus for Comp. A and B. Red markers represent Comp. A and blue markers indicate Comp. B. A strong inverse relationship is evident, with Comp. A, maintaining higher stiffness over a broader  $VF_{\text{void}}$  range, indicating greater resistance to porosity-induced degradation.

The non-linear degradation pattern observed in Young's modulus (22% reduction from 0.36 to 0.28 GPa at 15% void fraction, followed by 25% decrease to 0.21 GPa at 20.98% void fraction) aligns well with established trends in syntactic foam literature. Colloca et al. demonstrated that elastic properties in hollow particle-filled composites follow power-law relationships with volume fraction and wall thickness, consistent with our observed non-linear response [30]. Their findings, that lower-density microspheres with higher volume fractions create more effective reinforcement networks, support our results showing the superior performance of Composition A (0.14 g/cc HGMs) compared to Composition B (0.32 g/cc HGMs). The critical threshold behavior around 15% void fraction where degradation mechanisms appear to transition is consistent with percolation theory applications in syntactic foams reported by previous researchers. Ozkutlu et al. observed similar stiffness reductions in PMMA syntactic foams with varying

HGM densities, reporting Young's modulus values that decreased non-linearly with increasing porosity [31]. Our results showing substantial stiffness differences between compositions at comparable void levels (0.28 GPa vs 0.18-0.24 GPa) validate their findings that HGM density and volume fraction are critical parameters governing mechanical performance. The quantitative relationships observed between void morphology and elastic response support the structure-property correlations established in the broader syntactic foam literature while providing enhanced precision through automated characterization.

## 4.5 Conclusion

This section of the study presents a robust, unsupervised K-means clustering-based image segmentation workflow designed for the microstructural evaluation of high-density polyethylene (HDPE) syntactic foams reinforced with hollow glass microspheres (HGMs). Leveraging micro-computed tomography ( $\mu$ CT), the methodology enables non-destructive, three-phase segmentation of matrix, microspheres, and voids, specifically assessing the effects of additive manufacturing (AM) parameters, particularly infill density and material composition, on phase distribution, void morphology, and microsphere dispersion. Threshold-based segmentation combined with connected component analysis facilitated quantitative void classification, while shape descriptor-based filtering enabled differentiation between microsphere cavities and raster-induced porosity. The framework demonstrated consistent performance across varying infill densities, effectively capturing microstructural heterogeneity despite  $\mu$ CT resolution constraints.

Critically, this work overcomes key limitations in existing methods. While conventional thresholding and supervised learning approaches rely on annotated datasets and struggle with phase similarity, this unsupervised pipeline achieves multi-phase segmentation without manual intervention. Shape-based classification resolves a significant challenge in distinguishing functional microspheres from defect-driven voids. The automated generation of 3D reconstructions, quantitative descriptors, and mesh-ready volumes directly support simulation workflows and bridges the gap between imaging and structural performance prediction. This makes the approach suitable for design optimization and high-throughput evaluation of AM composites.

The findings of this work are summarized as follows:

- (1) Raster-induced voids and microsphere cavities were successfully segmented using unsupervised clustering and shape filtering.
- (2) Composition A (higher HGM content) exhibited more uniform dispersion and fewer polymer-rich regions; Composition B showed increased microsphere clustering at lower infill.

- (3)  $\mu$ CT resolution limitations impacted particle volume fraction measurements, especially for thin-walled microspheres, requiring caution when interpreting PVF values.
- (4) Shape-based analysis revealed circular voids from intact microspheres and elongated raster voids due to incomplete fusion in low-density samples.
- (5) 3D visualization confirmed discrete voids at high infill (100–80 %) and interconnected networks at low infill (70–60 %), highlighting the influence of infill on permeability and strength.
- (6) The segmentation approach offers a reproducible, efficient framework for characterizing AM-fabricated syntactic foams and can be adapted to similar composites.

The segmentation approach offers a reproducible, efficient framework for characterizing AM-fabricated syntactic foams and can be adapted to similar composites. The current model performs well when matrix and particle phases have distinct grayscale intensities; however, future work could incorporate shape- or edge-based features, as seen in deep learning approaches, to extend applicability to composites with low contrast between phases. These results have significant implications for industries that demand lightweight, high-performance materials. In aerospace, the ability to produce syntactic foams with controlled porosity supports the fabrication of structural cores and impact-resistant enclosures with reliable mechanical behaviour. Marine applications benefit from the tunable void architecture, which enables buoyancy while minimizing water ingress. In the automotive sector, adjusting porosity through infill density and composition enables the creation of energy-absorbing structures suitable for crash protection. The segmentation pipeline developed in this study provides a practical and scalable tool for integrating microstructural evaluation into digital design and quality control workflows, aligning with the evolving needs of additive manufacturing in industrial environments.

The next chapter builds upon this segmentation framework by extracting detailed morphological features of the segmented voids, including their size, shape, spatial distribution, and orientation. This transition from segmentation to characterization enables a deeper understanding of the structure property relationships that govern mechanical performance in syntactic foams.

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## 5. Morphological Feature Extraction

### 5.1 Introduction

Building on the segmentation framework established in Chapter 4, this chapter focuses on the extraction and quantification of morphological features of the identified voids. Morphological features are quantitative descriptors derived from segmented  $\mu$ CT images that capture the geometry and spatial arrangement of voids and microspheres. While segmentation is essential for capturing material phases, a comprehensive understanding of the microstructure is required. Void shape, size, and distribution have a direct influence on mechanical performance. There is a need for a framework that can not only identify and separate voids and particles, but also provide insights into the shape, size, number, orientation, and distribution of these voids/defects.

This chapter addresses these critical limitations by presenting a comprehensive morphological feature extraction methodology developed for HDPE-HGM syntactic foams. The approach integrates three-dimensional image analysis, quantitative morphological measurements, spatial distribution analysis, and orientation characterization. A Graphical User Interface (GUI) was developed to ensure accessibility for researchers from diverse backgrounds, facilitating the broader adoption of advanced microstructural characterization techniques in composite materials research and industrial applications.

#### 5.1.1 Effect of Void Morphology on Strength of Syntactic Foams

Previous research has shown that voids have a significant impact on the mechanical strength of syntactic foams. Factors such as the size and orientation of voids relative to the loading direction, the shape of the voids (whether circular or elongated), and their spatial distribution, including clustering, influence the material's performance. A thorough understanding of these parameters is essential for effective quality control and for identifying potential failure mechanisms. Additionally, creating a digital twin of a sample is a crucial step in performing predictive simulations using finite element analysis.

A study by Hyde et al showed that larger voids cause more strength reduction than smaller ones, as they create higher stress concentration factors, and that void shapes affect the strength, with circular voids being the least detrimental [1]. Furthermore, Bandstra and Koss demonstrated that three-void clusters significantly accelerate both void growth and coalescence compared to isolated voids, with the most pronounced effects occurring under high stress triaxiality, low strain hardening, and small inter-void spacing (particularly when voids are spaced one diameter apart or less). The clusters cause strain to concentrate within the inter-void ligament, leading to a load-loss response that serves as the criterion for void coalescence onset [2]. A study by Gao et. al reveals that void orientation is critical for tensile strength

[3]. Voids positioned perpendicular to the loading direction grow significantly faster and lead to earlier failure. In contrast, voids aligned parallel to the loading direction tend to fracture differently through particle breaking rather than void expansion. The researchers found that void shape and orientation are more than just void volume, with elliptical voids showing larger effective surface areas that accelerate damage compared to round ones. The study directly links these microscopic void orientations to the macroscopic anisotropic strength properties engineers observe in real materials, providing a scientific explanation for why material strength varies with loading direction in processed metals. Moreover, the void aspect ratio has a significant impact on the development of stress triaxiality around voids [3].

### 5.1.2 Existing Methods on Voids Characterization

Void characterization techniques in syntactic foams and composites have advanced from basic volume-fraction measurements to sophisticated, multiscale, and multiphase modelling methods. Both computational and experimental approaches have been developed to understand the impact of voids, whether intentional or not, on mechanical behaviour. Some of the key methods highlighted here include representative volume element (RVE) modelling, microstructural simulation, finite element techniques, and progressive damage modelling. Hyde et al. developed a finite element model based on micromechanics to investigate the impact of different void shapes on unidirectional fibre-reinforced composites under transverse and shear loads [1]. They developed 3D RVEs and analyzed void orientation, fibre-to-void ratios, and void volume fraction. This approach provided a quantitative link between microstructural features and ply-level mechanical performance through the use of finite element analysis (FEA). Yu et al. used Random Sequential Adsorption (RSA) algorithms to generate three-dimensional microstructures of syntactic foams containing hollow ceramic microspheres [4]. Their models incorporated three phases: the matrix, microsphere shell, and internal voids, with the innovative inclusion of cohesive zone models (CZMs) at the microsphere–matrix interface to simulate interfacial debonding, microsphere fracture, and matrix cracking. They validated this microstructure-based model through tensile tests, which increased confidence in the FE predictions. Gao et al. examined ductile fracture in metal alloys by explicitly modelling spherical and ellipsoidal voids within small-scale yielding 3D unit cells [3]. Their FE approach accounted for evolving void spacing, shape, and orientation to assess fracture initiation toughness. They demonstrated that crack growth is influenced by interactions among multiple voids at high void volume fractions, revealing limitations of single-void models. Bandstra and Koss performed 3D FE simulations of three voids within a matrix under multiaxial stresses [2]. Their method employed a load–loss criterion to predict the onset of void coalescence, demonstrating that coalescence occurs more rapidly in matrices with low strain-hardening capacity, high stress triaxiality, and closely spaced voids. Unlike symmetric cell models, their approach captured the localized nature of coalescence due to clustering. Xiao et al. advanced void coalescence modelling by

incorporating Fleck–Willis strain gradient plasticity theory into unit-cell FE simulations. They modelled elliptical voids under intense shear loading to compare the effects of void size, shape, and orientation on failure at the micron scale. This method introduced a length scale into the plasticity formulation to account for strengthening caused by geometrically necessary dislocations (GNDs), which is essential for modelling failure in fine-grained or nanoscale features [5]. Suo et al. simulated particle-reinforced metal matrix composites using user-defined material subroutines (VUMAT) in ABAQUS. They developed multiparticle RVEs with varying aspect ratios and volume fractions, modelling the metal matrix as elasto-plastic and the brittle particles with Weibull statistics. Their approach provided a probabilistic prediction of particle cracking under strain, linking geometric parameters to the likelihood of crack initiation [6]. Zhu et al. combined experimental and analytical methods to evaluate the effects of voids in carbon/epoxy laminates. They employed ultrasonic attenuation, optical imaging, and mechanical testing on laminates containing different void contents. Their results indicated that triangular voids accelerate early crack initiation [7]. Although not computational, their research provided vital validation data for models. Gladysz et al. conducted experimental void characterisation of three-phase syntactic foams. They used helium pycnometry to measure volume fractions of the binder and microspheres and inferred the volume fraction of interstitial voids. Compression and flexural tests linked microstructural design with mechanical performance, demonstrating how microballoon packing density and organisation influence foam strength [8].

### 5.1.3 Research Gap and Problem Statement

Despite significant advances in void characterization methods for various composite materials, several critical gaps still exist in the field of syntactic foam analysis. While comprehensive methodologies are available for fibre-reinforced composites [1, 4, 6], syntactic foams lack a holistic characterization framework. Current approaches for syntactic foams are limited to either experimental methods [8] or simplified three-phase modelling [2,3], without capturing the complex interactions between intentional voids (hollow microspheres) and unintentional voids (manufacturing defects). Existing methodologies, particularly those using detailed microstructural simulation with cohesive zone models [2,3] and strain gradient plasticity [6], are computationally intensive and time-consuming. The Random Sequential Adsorption algorithms and multi-scale finite element approaches require considerable computational resources, restricting their practical use in design optimization and quality control scenarios. Current approaches mainly focus on single-scale analysis, either microscale void behaviour [4,5] or macroscale mechanical properties [8]. There is a notable absence of integrated multi-scale methods that can effectively link microstructural void features with overall performance predictions in syntactic foams. While some studies explore the effects of void shape and orientation in metallic matrices [4,6], the specific impact of

void orientation within the unique three-phase architecture of syntactic foams (matrix-microsphere-void) remains underexplored, especially concerning how manufacturing-induced void orientations influence mechanical anisotropy.

The lack of a comprehensive, computationally efficient method for void characterization in syntactic foams creates a significant obstacle to optimal material design and quality control. Current methods face three main issues: (1) fragmented approaches that examine individual aspects of void behaviour without considering the complete three-phase interaction unique to syntactic foams, (2) slow computation that makes detailed void analysis unsuitable for industrial use and design improvements, and (3) poor multi-scale integration that does not connect microstructural void features with overall mechanical performance in a unified way. This research aims to develop a unified, computationally efficient method that thoroughly characterizes void effects in syntactic foams across different scales. It will allow quick evaluation of how voids influence properties for better material design and manufacturing quality, and serve as a basis for FEA-based modelling by providing a framework that generates mesh files and other computational inputs for easy integration with existing simulation workflows.

## 5.2 Methodology and Feature Characterization

This study employed a structured image analysis pipeline in Python to segment, identify, and characterize voids and particles within HDPE-HGM syntactic foams using micro-computed tomography ( $\mu$ CT) data. The workflow integrates voxel-based clustering, 3D connected component analysis, and quantitative morphological measurements, implemented with Python libraries NumPy (v1.23.5), scikit-image (v0.19.3), and scikit-learn (v1.2.2).

### 5.2.1 Void Count & Area

The K-means clustering results were relabeled in ascending order of cluster centroids, ensuring that the lowest intensity values were consistently assigned to the void class. To isolate individual voids, a binary mask was created by selecting voxels labelled as voids, and 3D connected component labelling was applied using a 26-neighbour (connectivity = 3) scheme. This allowed discrete void structures to be uniquely identified across the entire volume. The total number of voids was determined by counting the number of connected components. For each void, cross-sectional areas were calculated on a slice-by-slice basis by projecting the 3D label volume into 2D slices and measuring the area of each region using the “*regionprops*” function, which computes geometric properties such as area and centroid for each labelled region. To filter out the measure for accuracy and subtract noise or false detections, a threshold region of at least 5 pixels was applied to eliminate voids with areas smaller than 5 pixels. The area computed for every valid region was then resized according to the known in-plane resolution ( $\mu\text{m}/\text{pixel}$ ) to give physical units ( $\mu\text{m}^2$ ). To

calculate the average void area, the mean of all valid cross-sectional areas across the entire volume was used. The frequency of voids per slice was also tracked to examine spatial clustering behaviour in the direction of the build (Z-axis). These area statistics were then employed to support volumetric estimates, as well as to contrast morphological features against mechanical performance findings.

### 5.2.2 Void Shape and Geometry

To determine the shape and form of the voids within the material, a series of measurements was taken for each void based on its appearance in two-dimensional image slices. One of the primary measurements was the orientation angle, which represents the angle each void makes with the Z-axis, corresponding to the loading direction that penetrates the depth of the sample. This was estimated by placing each void within a rectangle and measuring the angle of its longer side to help identify whether the voids tend to align in any direction. Next, the extent of elongation in each void was measured using its eccentricity, a value that indicates how circular or linear the void appears. Closer to zero signifies a more rounded shape, while closer to one indicates a more elongated or stretched form. Similarly, for aspect ratio, which is calculated by dividing the length of the longer side of a void by the length of its shorter side. This provides insight into how narrow or wide the void is. Finally, the solidity of each void was measured by dividing its actual area by the area of its boundary. A high solidity indicates that the void occupies a significant portion of its boundary and has a smooth, compact shape. In contrast, a lower solidity suggests a more irregular shape with indentations. Collectively, these measurements provide a clear picture of how the voids vary in orientation, size, and shape, which will influence the material's response under stress.

### 5.2.3 Orientation Analysis

It has been established that the orientation of voids relative to the loading axis influences the strength of the syntactic foam sample. Therefore, the objective was to assess the alignment and directional distribution of voids (Section 5.2.4). First, each void was identified using connected component analysis as described in Section 4.3.1. Following this, the orientation was calculated using geometric measurements extracted from its bounding box, as shown in Figure 5.1. To determine the alignment of each void, its orientation angle was computed based on the bounding box dimensions as demonstrated in Equation (10)

$$\theta = \tan^{-1} \frac{h}{w} \quad (10)$$

Where,

$\theta$  is the orientation angle in degrees ( $^{\circ}$ )

$h$  is the height of bounding box ( $\mu\text{m}$ )

$w$  is the width of bounding box ( $\mu\text{m}$ )

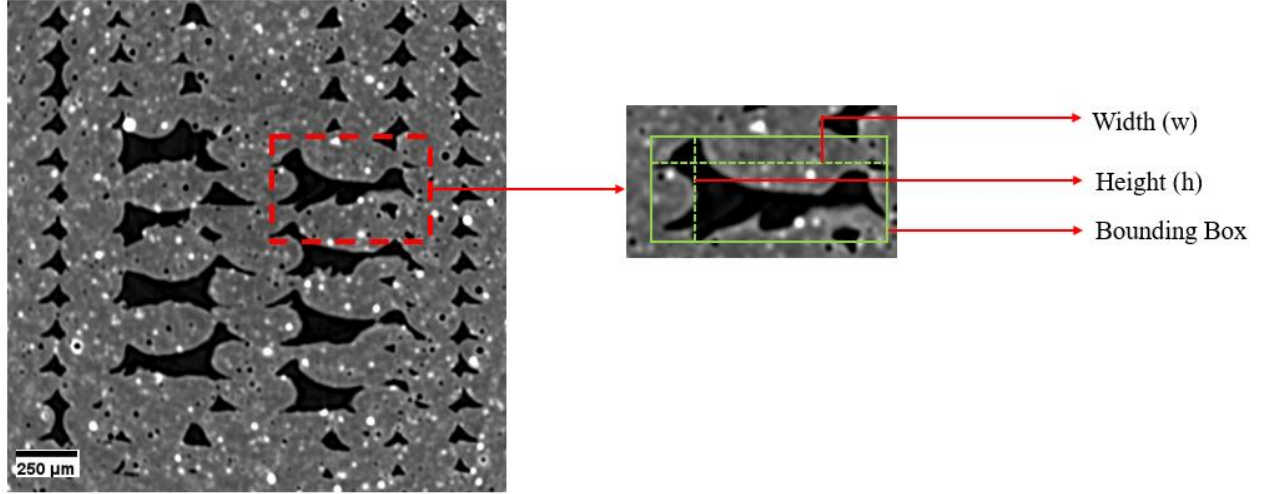


Figure 5.1 : Representative slice from 80% infill Component A showing elongated void morphology. The inset highlights a single void annotated with its bounding box. The bounding box width (w) and height (h), shown in micrometers ( $\mu\text{m}$ ), are used to compute the orientation angle and aspect ratio.

To demonstrate void directional bias, orientation values were sorted into three angular bins: 0–30°, 30–60°, and 60–90°. The proportion of voids in each bin was computed and presented for comparative assessment of directional orientation in the sample. This was also used to generate polar histograms (rose plots) that may be examined for orientation trends.

To quantitatively evaluate the alignment behaviour of voids across the entire image stack, an orientation tensor was constructed using the direction cosines of each void orientation according to Equations (11,12)

$$n_x = \cos \theta \quad (11)$$

$$n_y = \sin \theta \quad (12)$$

Where,

$\theta$  is the Orientation angle of a void's major axis in the image plane (radians)

$n_x$  is the direction cosine along the x-axis

$n_y$  is the direction cosine along the y-axis

The 2×2 orientation tensor T was then calculated as shown in Equation (13).

$$T = \frac{1}{N} \times (\sum_i = 1^n \times [ [n_{xi}^2, n_{xi}n_{yi}] \quad , [n_{xi}n_{yi}, n_{yi}^2] ]) \quad (13)$$

Where,

n is the total number of voids or features analyzed

T is the 2×2 orientation tensor capturing global alignment of void orientations

Eigenvalue decomposition of T yielded the principal directions of orientation. The alignment index A was defined as,

$$A = \frac{(\lambda_1 - \lambda_2)}{(\lambda_1 + \lambda_2)} \quad (14)$$

Where,

$\lambda_1$  and  $\lambda_2$  are the largest and smallest eigenvalues, respectively.

This scalar metric quantifies the degree of alignment. A closer 0 suggests a random orientation, while A closer to 1 indicates strong alignment in a preferred direction.

The alignment index A quantifies the degree of directional uniformity in void orientations, where values near 0 indicate random orientation and values near 1 reflect strong alignment. This metric is derived from the eigenvalues of the 2×2 orientation tensor, which aggregates the directional information of all voids. The larger, for example, eigenvalue  $\lambda_1$  represents the dominant alignment magnitude, and its associated eigenvector defines the preferred orientation direction. While  $\lambda_1 > \lambda_2$  confirms the presence of anisotropy, the actual preferred direction is not fixed to the X- or Y-axis but is determined by the direction of the eigenvector corresponding to  $\lambda_1$ .

This tensor orientation technique was selected for its ability to provide a firm, rotation-invariant, and quantitative measurement of void alignment throughout the sample. As opposed to relying on single orientation angles that are susceptible to noise or scatter, the tensor sums directional data from all voids throughout the dataset. This enables more statistically resistant characterization of anisotropy through eigenvalue decomposition. The derived alignment index A, from the eigenvalues of the tensor, is a scalar that differentiates randomly oriented voids from those with strong preferential alignment. Lastly, this method gives a comprehensive and interpretable account of microstructural organization, relating local geometric characteristics to bulk material behaviour.

#### 5.2.4 Nearest Neighbour Distance (NND)

The NND analysis method begins with the precise identification of individual voids, aided by image segmentation, and the computation of their centroid coordinates to construct a precise spatial mapping. The centroids form the foundation of constructing an efficient KD-tree data structure, which greatly enhances the nearest neighbour search algorithm by partitioning the spatial space. The KD-tree partitions the 3D space by recursively bisecting it along alternate dimensional axes, organizing void centroids into a balanced binary tree. For each identified void, the Euclidean distance to its closest neighbour is calculated, generating

a distribution of nearest neighbour distances across the entire sample. From this data, several critical spatial statistics are computed using Equation (14).

$$Expected\ NND = 0.554 \times \left(\frac{1}{\sqrt[3]{\lambda}}\right) \quad (14)$$

Where,

$\lambda$  is the void density (number of voids/ $\mu\text{m}^3$ )

In 3D, the mean NND observed from actual measurements and the R-index, which is the ratio of observed to expected NND. The R-index is an appealing measure for describing spatial patterns. Values near 1 indicate random (Poisson) distributions, values less than 1 suggest clustering tendencies where voids are more concentrated than by chance, and values greater than 1 signify regular or dispersed patterns where voids have more uniform spacing. This comprehensive spatial analysis provides crucial insights into the mechanisms of material formation, processing-structure relationships, and potential failure modes by revealing the underlying organizational principles that govern void distribution within the material's microstructure. To characterize the three-dimensional spatial distribution of voids in material microstructures, several defining measures were established. These are quantitative indices of void arrangement patterns that influence the behaviour and mechanical properties of materials. The mean NND was calculated by measuring the Euclidean distance from a void centroid to its nearest neighbour void centroid in 3D space and averaging these measurements. The interpretation of the coefficient of variation ( $C_V$ ) is especially valuable. The  $C_V$  is a standardized measure of dispersion of a probability distribution or frequency distribution, calculated using Equation (15).

$$C_V = \frac{\sigma}{\mu} \quad (15)$$

Where,

$\mu$  is the mean nearest neighbour distance across all voids ( $\mu\text{m}$ )

$\sigma$  is the standard deviation of those distances ( $\mu\text{m}$ )

Values close to 0 indicate very regular void spacing, and values greater than 1 indicate very extreme variation in spacing; mid-values suggest moderate variation in the spacing between voids. All the values of individual NNDs were recorded, which are used to calculate the NND standard deviation to estimate the spacing variability of voids. The range of void separation is determined by the minimum value of NND and the maximum value of NND. Additionally, the coefficient of variation provides a unitless estimate of dispersion that can be used to compare volumes with different void densities. These values indicate whether voids are grouped in specific locations, regularly spaced, or randomly distributed, which is critical data for

understanding material formation mechanisms and predicting mechanical behaviour under loading conditions.

### 5.2.5 Graphical User Interface

To visually reinforce the progressive disclosure model embedded in the interface, the GUI workflow is summarized in Figure 5.2, which outlines the sequential flow from threshold computation through image segmentation to result export. Each stage of the pipeline is clearly delineated, beginning with mode selection (K-means threshold generation or loading existing CSVs), followed by 2D segmentation and 3D analysis, and culminating in visualization and export. The diagram reflects the intuitive design of the interface, guiding users through each stage with minimal ambiguity and maximum transparency.

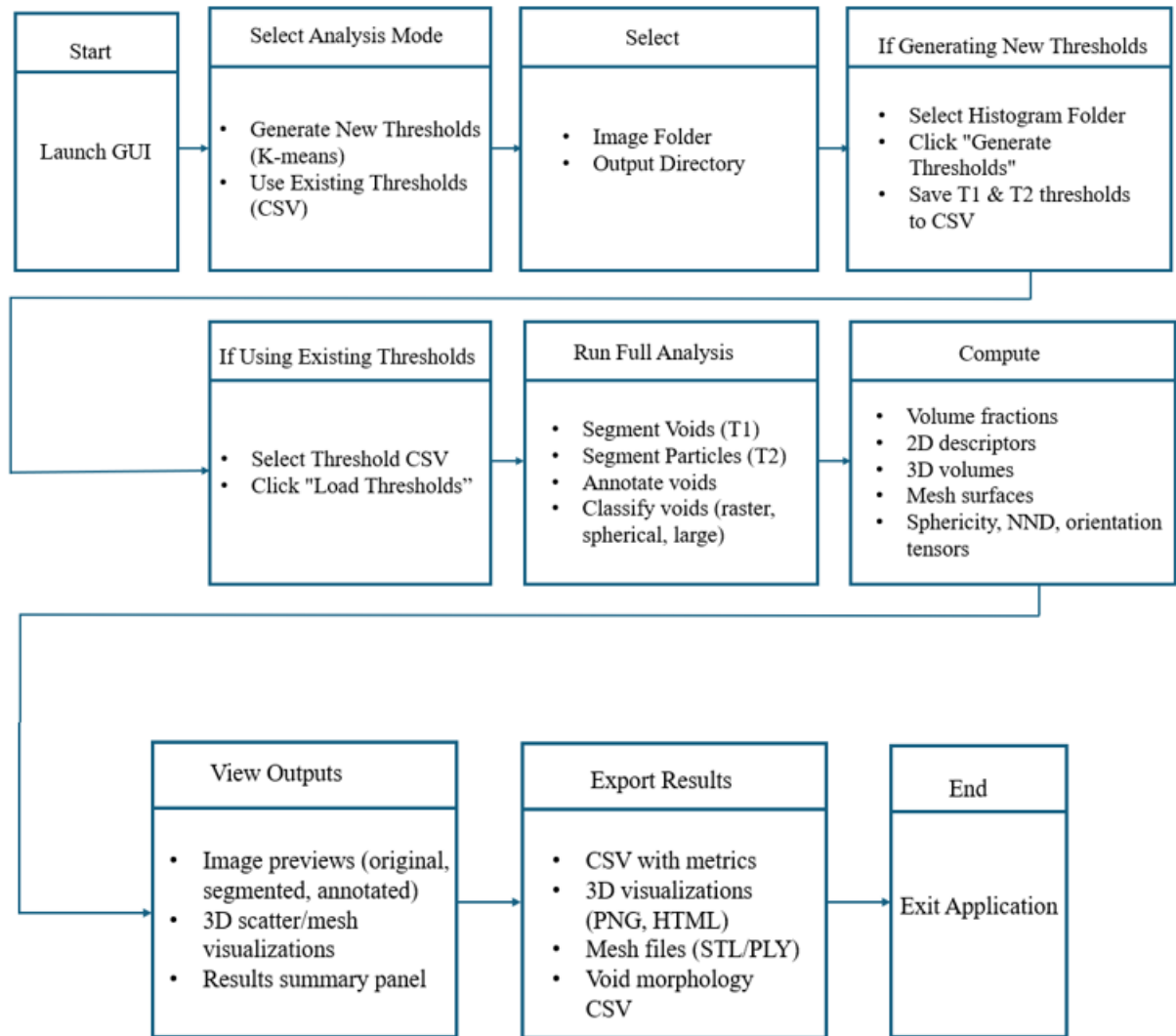
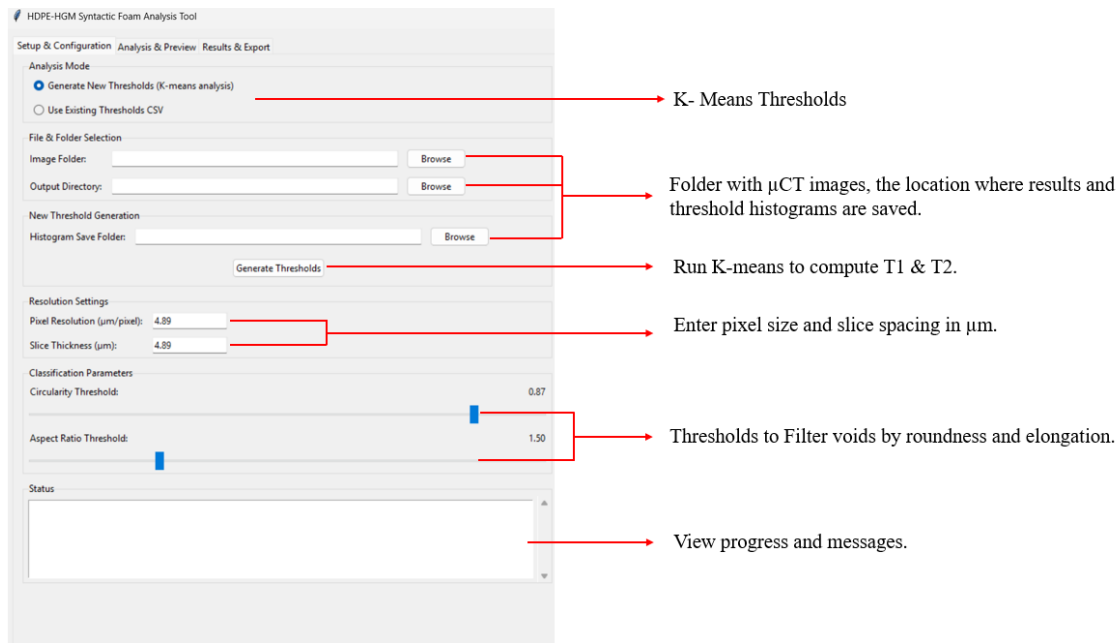


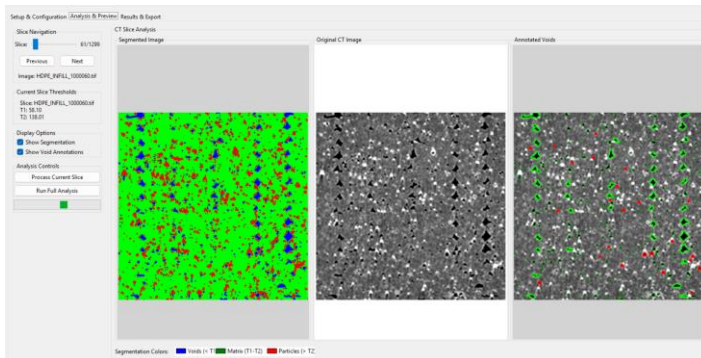
Figure 5.2 : This flowchart outlines the step-by-step process in the graphical user interface for analyzing HDPE-HGM syntactic foams. Users begin by selecting an analysis mode, followed by threshold generation

or loading. The tool then performs segmentation, void classification, and 2D/3D morphological analysis. Final outputs include visual previews, 3D visualizations, and exportable CSV and mesh files.

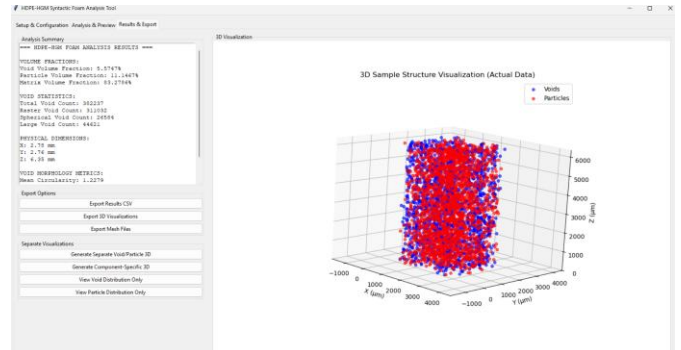
The analytical workflow is facilitated through a comprehensive graphical user interface (GUI) that streamlines the complex multi-modal segmentation and morphological analysis process for syntactic foam characterization. Figure 5.3 illustrates the three-stage workflow implemented in the HDPE-HGM Syntactic Foam Analysis Tool, demonstrating how the interface guides users through the complete analysis pipeline from initial parameter configuration to final 3D visualization and quantitative results. The GUI design follows a progressive disclosure approach, where each panel represents a distinct phase of the analysis: setup and configuration (Figure 5.3a), real-time slice analysis and segmentation (Figure 5.3b), and comprehensive results presentation with 3D visualization (Figure 5.3c). This integrated workflow enables researchers to conduct sophisticated void and particle characterization without requiring extensive programming expertise, while maintaining complete transparency in the analytical process through visual feedback mechanisms and comprehensive result documentation.



(a)



(b)



(c)

Figure 5.3 : HDPE-HGM syntactic foam analysis workflow: (a) analysis tool setup and configuration interface showing threshold generation parameters, pixel resolution settings, and classification parameters for void and particle identification; (b) CT slice analysis displaying segmented image with color-coded void classification (blue: voids <T1, green: matrix T1-T2, red: particles >T2), original CT image, and annotated voids overlay; (c) analysis results summary showing volume fractions, void statistics, physical dimensions, and 3D sample structure visualization with spatial distribution of voids (blue) and particles (red).

Real-time feedback mechanisms are integrated throughout the entire interface. Progress bars, status updates, and result displays at intermediate points prevent the application from appearing unresponsive during lengthy processing sequences. The interface design also considers that 3D reconstruction and

morphological analysis can take several minutes, especially with large image stacks. The three-panel layout is achieved by strategically utilizing Tkinter's hierarchical frame management system. The control panel at the top employs an ordered grid geometry manager that organizes input controls logically. The pack geometry manager with fill and expand properties is used to make effective use of the main container frame, providing a responsive layout across various window sizes and screen resolutions. To support user adoption and efficient use of the analytical pipeline, extensive documentation, including detailed algorithm specifications and user instructions provided. This documentation accompanies the software distribution and functions as both a technical reference for understanding the underlying methodologies and a practical guide for conducting foam characterization analysis. The instruction file contains step-by-step workflows, parameter descriptions, output interpretation guidelines, and troubleshooting procedures, ensuring accessibility for researchers from diverse institutional backgrounds. The “*setup\_ui*” function acts as the structural coordinator, initializing the main window properties such as title, dimensions, and visual style via “*ttk.Style*”. This approach automatically generates nested frame layouts from the top control frame, which contains options for image folder selection, threshold CSV output settings, and controls to initiate processing. The grid geometry manager positions these elements with specific row and column coordinates, ensuring equal spacing and alignment through the use of pad x and pad y arguments.

The use of the double-panel lower section involves the strategic application of side-based packing with the Pack Geometry Manager. The left frame panel employs *side='left'* specification with *fill='both'* and “*expand=True*” arguments to supply an adaptive image display area that adapts to window size changes. The right panel employs the complementary *side='right'* packaging to establish the 3D visualization and results pane. Inside the left panel, several sub-frames organize threshold display labels, an image display area, and navigation slider widgets. The “*threshold\_frame*” contains horizontally stacked labels with the computed T1 and T2 values having *side='left'* alignment. The “*image\_frame*” employs the *fill='both'* and “*expand=True*” arguments to utilize the entire available space to display the processed image, and the “*slider\_frame*” provides navigation features with a horizontal direction and full-width expansion.

The proper panel layout also mirrors this hierarchical organization, with the inclusion of “*button\_frame*” for controls in 3D visualization, a main *static\_3d\_label* for displaying 3D visualization, and “*yflabel*” to show quantitative results. These are each arranged using pack geometry management with appropriate fill and padding arguments to provide visual harmony and functional accessibility. The style configuration management provides visual harmony between all interface components through the “*ttk.Style*” system. Font parameters, background colours, and component-specific style parameters are used throughout to create a professional appearance that enhances usability and reduces visual fatigue during prolonged analysis sessions. The directory selection feature for the image folder is utilized using the “*browse\_image*”

function, which combines the file dialogue. The “*askdirectory*” system provides native operating system directory browsing capabilities. This feature includes error handling and validation logic to prevent users from selecting unavailable directories, providing real-time feedback via the “*img\_entry*” widget to represent the chosen path. The related algorithm documentation features explicit specifications for image preparation, including recommended file naming schemes, resolution guidelines, and format optimization processes. This documentation ensures users can properly prepare data for experimentation and understand the technical capabilities and constraints of the processing pipeline. The instruction file provides specific information on imaging parameters, file structure strategies, and quality control procedures that maximize analytical accuracy and reliability. File format checks occur within the “*load\_images\_from\_folder*” function, which recursively probes directory contents using “*os.listdir*” functionality in combination with filename extension filtering. The operation supports the complete list of extensions, including .tif, .tiff, .png, .jpg, and .jpeg formats, ensuring compatibility with various imaging equipment and data acquisition systems. Alphabetical ordering of filenames ensures that files are processed by the computer system in the same order, regardless of the file system organization. The CSV output specification is made possible through the use of the “*browse\_csv*” function, which utilizes the file dialog. “*asksaveasfilename*” functionality to enable users to specify output locations for calculated threshold information. This approach ensures user control over the data structure, with automatic default file extension handling and format validation. The function updates the CSV entry widget to display the selected path, providing immediate visual feedback on the output location.

Error handling throughout the file management system employs extensive exception handling and the presentation of polite error messages via the “MessageBox” system. Documentation of algorithms supports these errors by presenting preventative information and troubleshooting steps, enabling users to repair common problems independently.

The “*process\_images*” function controls the entire file processing chain, from input validation to confirming image directory and CSV output directory parameters. The “*process\_images*” function performs progressive error checking, verifying the presence of images before initializing computationally intensive processing operations. The “*load\_images\_from\_folder*” function provides a list of filename-image data tuples, enabling subsequent processing functions to maintain filename associations throughout the analysis pipeline. Memory management problems are addressed throughout the file handling operations using OpenCV's “*imread*” operations, which are balanced by appropriate memory cleanup operations. The software processes images individually during processing, unlike having entire datasets in memory simultaneously, which enables the processing of enormous stacks of images without bogging down system resources. Platform independence and path validation are obtained through the consistent use of

“*os.path.Join*” interface functionality for filename and directory manipulation. This method prevents platform-specific path separator issues and is compatible with the Windows, macOS, and Linux operating systems, which are extensively used in research settings.

### 5.3 Correlation

The previous chapters have enabled the separation of voids and particles, including a clear distinction of HGM cavities. The morphology of process-induced voids was also computed. Following this, it becomes essential to understand how these features affect the strength of the syntactic foam sample, particularly its tensile strength. That work is achieved in this chapter by performing a correlation analysis with the tensile data we obtained (Chapter 3). Void characteristics such as shape, size, orientation, and spatial distribution play a crucial role in defining the mechanical performance of composite materials, particularly syntactic foams. Numerous studies have investigated the impact of these microstructural features on tensile strength and stiffness, utilizing experimental data, micromechanical modelling, and finite element simulations. Hyde et al. emphasized that void volume fraction and void shape significantly impact the mechanical performance of unidirectional fibre-reinforced composites. Their study employed a micromechanics-based finite element framework to quantify the influence of different void configurations on strength under various loading conditions. Notably, pentagonal interfiber voids showed the most pronounced weakening effect, followed by square and triangular voids, with circular matrix voids being the least detrimental. They also noted that while void orientation has a limited impact, void size and fibre-to-void ratio were critical to load-bearing behaviour [9]. Specific to syntactic foams, Lawrence et al. conducted stereological analyses on polymer-based syntactic foams and reported that the aspect ratio and alignment of hollow microspheres (often used as void analogs in syntactic foams) directly correlated with the material's mechanical response. Their findings support the importance of orientation alignment and elongation characteristics when assessing foam strength. The study showed that elongated voids, especially those aligned with the loading axis, contributed to anisotropic weakening [10].

Furthermore, a study by Costa et al. highlighted the nucleation and propagation of cracks from pre-existing voids, demonstrating that voids could act as stress concentrators and trigger early failure in carbon/epoxy systems. They observed that elongated voids or those oriented perpendicular to the load were more detrimental than circular voids aligned with the load [11]. In syntactic foams with mixed void populations, studies have found that a higher proportion of circular voids leads to better tensile strength and stiffness. In contrast, irregular or elongated voids introduce local stress concentrations, reducing the effective modulus [12]. Additionally, the mean nearest neighbour distance (NND) of neighbouring voids has been associated with strength degradation; closer voids tend to interact under stress, leading to coalescence and premature failure [13].

The orientation of voids also influences mechanical behaviour. Voids aligned with the loading direction generally have a lesser effect on tensile strength than those randomly or perpendicularly oriented. This trend is supported by both experimental evidence and 3D computational models [14]. Lastly, particle-related parameters such as particle volume fraction and R-index, which quantifies the regularity or uniformity of particle distribution, have emerged as essential descriptors. A higher particle volume fraction often contributes positively to mechanical strength, provided that good particle dispersion and interfacial bonding are achieved. However, irregular distribution or poor alignment can negate this effect [15]. While many studies have examined the impact of individual features on the mechanical properties of syntactic foams, it remains unclear which features have a more substantial influence than others, and to what extent these differences are significant. There is also a need to examine how different features interact with each other, not just how they relate to the final properties. This chapter focuses on building a comprehensive picture by exploring these relationships in detail and clearly illustrating how the internal structure of the material relates to its performance.

### 5.3.1 Structure Property Correlation Matrix

A structure–property correlation matrix provides a practical and visual means to understand how various microstructural features influence the mechanical behaviour of composite materials. This approach enables the observation of both direct and indirect dependencies, highlighting which features have the most significant impact on material performance.

In a study on aluminum matrix composites, researchers investigated the relationships between particle size ratio, aspect ratio, and volume fraction and their impact on tensile strength, hardness, and stiffness through a comprehensive correlation matrix. It emphasized the value of using such correlation techniques to not only improve understanding of complex materials but also guide optimization of their design and processing conditions [16]. This kind of holistic mapping between structure and performance serves as an essential framework for advancing the development of syntactic foams and similar multiphase materials.

### 5.3.2 Pearson's Coefficient

In the study of syntactic foams and fibre-reinforced composites, researchers have consistently used correlation analysis to understand how structural features affect mechanical behaviour. The most commonly used correlation metrics include Pearson's correlation coefficient, Spearman's rank correlation, and Kendall's tau coefficient. Pearson's correlation coefficient is widely adopted for its ability to measure linear relationships between two continuous variables. Amin et al. employed Pearson's coefficient to assess the connection between fibre content and mechanical strength in kenaf fibre composites, revealing direct associations between density, fibre volume, and tensile properties [17]. Similarly, Syakir et al. analyzed the

thermal and mechanical performance of polypropylene composites reinforced with wood flour using Pearson's correlation to examine the relationship between filler dispersion and strength of the sample [18]. Other studies, such as those by Zhang et al. [19], focused on syntactic foams and utilized Pearson's correlation to analyze how microsphere characteristics, including shape and distribution, influenced stiffness and compressibility. Statistical analysis and correlation calculations were performed using Python libraries, including Pandas, NumPy, and SciPy. A custom GUI tool was developed to streamline the correlation analysis workflow and generate prediction model setups, though this auxiliary tool is not part of the core research contribution. For this study, where the features extracted from image analysis and mechanical testing are continuous and normally distributed, Pearson's correlation is the most appropriate choice. It enables a direct and interpretable quantification of the strength of association between each structural parameter and the resulting mechanical performance, making it both a conventional and practical choice in the research of composite materials.

Pearson's coefficient is used to measure the linear relationship between 2 variables. For the sample set used in this study, it is assumed that the variables have a linear relationship, i.e., as the void volume fraction increases, the strength of the sample decreases. This correlation coefficient indicates the strength of the correlation and its direction, i.e., whether the two variables are directly or inversely proportional. Pearson's coefficient is calculated using Equation (16)

$$r = \frac{\sum[(x_i - \bar{x})(y_i - \bar{y})]}{\sqrt{[\sum(x_i - \bar{x})^2 * \sum(y_i - \bar{y})^2]}} \quad (16)$$

Where,

$x_i$  and  $y_i$  are the individual values of variables  $x$  and  $y$ , respectively.

$\bar{x}$  and  $\bar{y}$  are the means of variables  $x$  and  $y$ , respectively.

$\Sigma$  is the summation symbol.

The Pearson correlation coefficients were calculated between all the features obtained in Chapter 5 and the experimentally measured Ultimate Tensile Strength and Young's Modulus presented in Chapter 3.

## 5.4 Results and Discussion

The analysis of void properties reveals that not all voids have an equal impact on the deterioration of mechanical properties in additive manufacturing. Classical techniques that rely on the absolute volume fraction of voids are unable to capture the complex relationship between void morphology and structural integrity. In this work, a general classification framework is presented that distinguishes voids based on both size and morphological demands, illuminating how different types of voids impact mechanical behaviour. The sample ID, corresponding infill density, and compositions are provided in Table for better understanding.

Table 5.1 : Sample identification, infill density, and component composition for the 3D printed specimens.

| Sample ID | Infill Density (%) | Component |
|-----------|--------------------|-----------|
| 1         | 100                | A         |
| 2         | 90                 | B         |
| 3         | 80                 | A         |
| 4         | 70                 | B         |
| 5         | 60                 | A         |

#### 5.4.1 Total Void Count and Geometric Summary

The total void count analysis revealed remarkable variation across samples, ranging from 71,951 voids (Sample 4) to 381,958 voids (Sample 1), representing more than a five-fold difference in void density. This substantial variation reflects the sensitivity of void formation to processing parameters and material composition. The observed patterns, as shown in Table 5, demonstrate that optimal void management does not necessarily correlate with maximum infill density, suggesting that intermediate infill levels may compromise microstructural integrity through competing processing mechanisms. The geometric summary indicates that void populations exhibit distinct size distributions. Sample 4, despite having the lowest total void count, exhibited the largest mean void area ( $511.18 \mu\text{m}^2$ ), while Sample 1 showed the smallest mean void area ( $51.56 \mu\text{m}^2$ ). This inverse relationship between void count and individual void size suggests competing formation mechanisms during processing, where conditions favoring numerous small voids suppress the growth of large voids, and vice versa. Moreover, it is found here too that different composition materials and their properties will behave differently with varying infill densities, supporting previous claims [20]. Void skewness here provides insight into how balanced or uneven the distribution of void sizes is within each sample. When the skewness is low, as observed in the 100% and 70% infill density samples, the sizes of the voids are relatively consistent throughout the material, with most voids clustering around the average size and very few being extremely large or small. This suggests that the manufacturing process at these infill levels was able to maintain a uniform flow and consolidation, leading to a more predictable and evenly distributed internal structure. On the other hand, the samples with infill densities of 60%, 80%, and 90% exhibit much higher skewness values, indicating that while many voids remain small, there are also a few significantly larger voids present. This uneven distribution could be a result of inconsistent material deposition or insufficient consolidation pressure during fabrication, potentially leading to areas of weakness that may affect the mechanical performance of the printed part. A higher skewness value, therefore, reflects a wider variation in void sizes and suggests less control over the internal microstructure.

of the material. The raster void count analysis revealed that manufacturing-induced voids represent the dominant void population across all samples, with counts ranging from 27,433 to 310,860. The large void count category showed more moderate variation (34,866 to 71,878), indicating that while small voids are highly process-dependent, large void formation follows more consistent patterns.

Table 5 : Summary of Void Characteristics Across Infill Densities Including Total Count, Mean Area, and Skewness.

| Sample ID | Void Count Raster | Void Count Large | Void Count Total | Void Area Mean | Void Skewness | Total Void Count | Void Area Mean ( $\mu\text{m}^2$ ) |
|-----------|-------------------|------------------|------------------|----------------|---------------|------------------|------------------------------------|
| 1         | 310860            | 44551            | 381958           | 51.55          | 6.34          | 381,958          | 51.56                              |
| 2         | 95232             | 71504            | 199721           | 314.1          | 18.06         | 199,721          | 314.03                             |
| 3         | 100968            | 71878            | 206984           | 300.71         | 18.51         | 206,984          | 300.72                             |
| 4         | 27433             | 44115            | 71951            | 511.17         | 9.42          | 71,951           | 511.18                             |
| 5         | 227576            | 34866            | 278237           | 330.51         | 17.74         | 278,237          | 330.51                             |

While void formation mechanisms in polymer processing are well-established in the literature [21][22][23], the specific phenomenon observed at maximum infill density in additive manufacturing represents a novel application of these principles. Although Classical Nucleation Theory (CNT) suggests that rapid consolidation under high pressure can promote the formation of numerous small void nuclei rather than fewer large voids [24], the voids observed in this study are created by mechanisms in 3D printing which are different from CNT. The voids here arise due to incomplete fusion, entrapped air and raster path discontinuities. Similarly, Roychowdhury et al. demonstrated that rapid heating rates hinder adequate volatile desorption, thereby facilitating void nucleation [25]. The mechanism we observe, where 100% infill promotes numerous small voids through trapped air and volatile compounds during rapid consolidation, extends these established polymer processing principles to the unique conditions of layer-by-layer additive manufacturing.

The 70% infill sample exhibits the lowest void count (71,951) but the largest mean void area ( $511.18 \mu\text{m}^2$ ), a tenfold increase compared to the 100% infill. This pattern likely results from void coalescence, a phenomenon in which inadequate material overlap leads to the merging of smaller pores into larger, more irregular voids. The result aligns with findings by Chen et al., who documented the tendency of subcritical voids to grow and coalesce when the matrix volume is insufficient to maintain consistent bonding between

raster paths [26]. These large voids are typically more detrimental to mechanical performance, especially in load-bearing directions, due to their stress amplification effects.

The observed skewness trends, where intermediate infill densities (60%, 80%, and 90%) exhibit high asymmetry in void size distributions, align with broader findings in the literature that emphasize the heterogeneous nature of void morphology in thermoplastic composites. For instance, in a micro-CT study of AS4/APC2 carbon-fibre reinforced prepregs, Bensadoun et al. reported that void populations were predominantly rod-like, with aspect ratios ranging from 2 to 150, and that only 6.5% of the voids were close to spherical. Their analysis demonstrated that such non-uniform geometries resulted in significant distribution tails, reinforcing the importance of characterizing not just average void size but also distribution skewness [27].

#### 5.4.2 Void Shapes and Geometry

The detailed quantitative analysis of void shape characteristics, such as circularity, aspect ratio, and the percentage of circular versus elongated voids, is summarized in Table 6. This table presents the mean values and standard deviations for both circularity and aspect ratio across all five samples, along with the proportion of circular and elongated voids identified in each case.

Table 6 : Void shape descriptors showing mean circularity, aspect ratio, and percentages of circular and elongated voids for each sample.

| Sample ID | Mean Circularity | Mean Aspect Ratio | Circularity * Std. | Aspect Ratio Std. | Circular Void Percentage | Elongated Void Percentage |
|-----------|------------------|-------------------|--------------------|-------------------|--------------------------|---------------------------|
| 1         | 1.22             | 1.28              | 0.35               | 0.36              | 86.35                    | 3.21                      |
| 2         | 1.1              | 1.28              | 0.46               | 0.49              | 67.6                     | 5.69                      |
| 3         | 1.11             | 1.29              | 0.46               | 0.5               | 68.88                    | 5.70                      |
| 4         | 0.94             | 1.35              | 0.48               | 0.71              | 44.06                    | 8.74                      |
| 5         | 1.2              | 1.31              | 0.4                | 0.56              | 82.26                    | 4.94                      |

\*Std. - Standard Deviation

The shape of the voids found across different samples exhibits a noticeable pattern that links the infill density and material composition to the regularity or irregularity of the voids. Sample 1, which was printed with 100% infill, had the most regular shapes, as indicated by its high average circularity value of 1.23 and a low average aspect ratio of 1.28. The minor standard deviations for both circularity (0.35) and aspect ratio (0.37) indicate that most of the voids in this sample were consistently shaped and not very different from one another. This sample also had the highest percentage of circular-shaped voids, at 86.36 percent, and the lowest number of elongated voids, which was just 3.21 percent. On the other hand, Sample 4, printed with 70 percent infill, showed the least regular shapes. It had the lowest average circularity of 0.95, the highest aspect ratio of 1.35, and the most extensive spread in shape values, meaning that its voids varied more in form. The percentage of circular voids in this sample dropped to 44.06 percent, while elongated ones increased to 8.74 percent.

Samples 1, 3, and 5 tend to have more circular voids than the others because they contain a higher amount of hollow glass microspheres (HGM). To verify that the code accurately detected circular shapes, the cavities created by these spheres were included in the analysis of shapes. These cavities are naturally spherical, and their presence helps confirm that the segmentation process correctly identifies round voids. However, for other measurements, such as total void volume, these HGM cavities were excluded to prevent them from being mistaken for raster voids. The fact that these HGM-rich samples still exhibited more circular voids, even when compared under the same conditions, supports the idea that the algorithm was functioning properly and illustrates how the material choice can influence the shape and consistency of voids formed during the printing process. Table 6 shows that samples with higher infill densities, particularly Sample 1 (100%) and Sample 5 (60%), have higher mean circularity and a greater percentage of circular voids, while lower infill densities, such as Sample 4 (70%), display more elongated voids and lower circularity. This suggests that extreme infill conditions produce more uniformly shaped voids, possibly due to improved material distribution and melt flow. Similar trends have been reported in previous studies, where circular voids were associated with stable processing conditions and uniform microstructure. In contrast, irregular voids were often linked to inconsistent deposition or thermal gradients.

The trends in void morphology observed in this study are consistent with earlier findings in the literature. Higher circular void percentages, such as those observed in Sample 1 (86.36%) and Sample 5 (82.27%), were associated with better microstructural uniformity and mechanical performance. This observation aligns with the work of Mehdikhani et al., who emphasized that voids with higher circularity often result from more stable processing conditions and lead to improved stress distribution in fiber-reinforced composites [28]. Conversely, Sample 4 exhibited the lowest circular void percentage (44.06%) and the highest percentage of elongated voids (8.74%), which correlated with poorer mechanical behaviour. Similar

findings were reported by Hyde et al., who showed that elongated voids increase local stress concentrations and compromise the structural integrity of unidirectional fiber composites [1]. Moreover, the relationship between void shape and failure mechanisms was further confirmed by Gao et al., who demonstrated that irregular or high-aspect-ratio voids act as stress risers, accelerating damage progression in composite materials [3]. These consistent observations across multiple studies reinforce the importance of promoting circular void formation through process optimization to enhance the performance of syntactic foams.

### 5.4.3 Orientation Tensor Analysis

The void orientation in every specimen was examined to determine how infill density affects their directional behaviour. In every specimen, voids were largely oriented at 30°–60°, reflecting a dominant prevailing orientation that might be related to the raster pattern of additive manufacturing. For instance, Sample 1 exposed over 91% of voids in this category, followed by Sample 2 and Sample 3, which had approximately 91% and 90% voids, respectively. The percentages indicate a high directional bias, indicative of high raster regularity and directional material deposition. Few voids in these samples were found below 30° or above 60°, adding credence to the argument that void orientation was not random.

The alignment index, a measure of directional bias, further supports these results. Sample 1 displayed the highest alignment index of 0.9624, indicating a strongly ordered void orientation, which is typically characteristic of uniform layer deposition and constant melt flow. Samples 2 and 3, at 0.859 and 0.858, respectively, displayed lower but still significant alignment, indicating some discrepancy in deposition. Significantly, Sample 4, which contained fewer total voids, had a high alignment index of 0.9466 and demonstrated that directional deposition can induce extreme void orientation even in samples with low void counts. These orientation patterns and alignment indices across all samples are presented in Table 7, highlighting the dominant void orientations and the degree of anisotropy introduced during the printing process

Table 7 : Orientation distribution and alignment index of voids across samples.

| Sample ID | Orientation 0-30<br>(%) | Orientation 30-60<br>(%) | Orientation 60-90<br>(%) | Orientation Alignment<br>Index |
|-----------|-------------------------|--------------------------|--------------------------|--------------------------------|
| 1         | 5.5034                  | 91.5775                  | 2.9192                   | 0.9624                         |
| 2         | 7.3391                  | 90.954                   | 1.7069                   | 0.8592                         |
| 3         | 7.2671                  | 90.6567                  | 2.0762                   | 0.8579                         |
| 4         | 9.0247                  | 86.4404                  | 4.535                    | 0.9466                         |

| Sample ID | Orientation 0-30<br>(%) | Orientation 30-60<br>(%) | Orientation 60-90<br>(%) | Orientation Alignment<br>Index |
|-----------|-------------------------|--------------------------|--------------------------|--------------------------------|
| 5         | 7.6391                  | 89.3781                  | 2.9827                   | 0.8603                         |

The overall trend is that increasing infill densities, such as Sample 1 (100%), exhibit a higher void orientation, that is, the voids are more consistently oriented in a specific direction rather than being randomly scattered. This most likely occurs due to the more uniform material distribution and regular toolpath during printing, which imposes a prevailing directional raster pattern. On the other hand, intermediate infill densities could introduce inconsistencies that reduce directional control, which is evident in the slightly reduced alignment indices of Samples 2 and 3. While void alignment is not directly related to mechanical properties with this dataset, the findings suggest that alignment can be a qualitative indicator of print consistency and raster fidelity in additively manufactured syntactic foams.

In contrast to earlier works that relied primarily on idealized simulation or finite element analysis to examine the effect of void orientation on material performance, the current work introduces a more straightforward image-based method grounded in experimental data. Past approaches, such as those of Hyde et al., Yu et al., Gao et al., and Bandstra and Koss, involved assuming ideal geometries or stress conditions and were computationally intensive, with extensive model calibration. In contrast, orientation information in this research is directly derived from 3D  $\mu$ CT scans by calculating bounding box angles and computing an orientation tensor, followed by eigenvalue decomposition to determine the measure of alignment as the alignment index. The above technique preserves the actual anisotropy of the fabricated parts without making any assumptions and accurately reflects the actual spatial distribution of voids that occur during fabrication. By binning orientation angles into given bins and correlating them with an alignment index, the method provides a robust, scalable, and statistically grounded alternative to bridge the gap between predicting material performance and morphological analysis.

The orientation analysis revealed significant directional preferences in void alignment across all samples, indicating that processing conditions induce anisotropic void formation patterns. The directional distribution analysis demonstrated that the majority of voids align within the 30-60° orientation range, with percentages ranging from 86.44% (Sample 4) to 91.58% (Sample 1). This pronounced orientation bias suggests that layer-by-layer additive manufacturing processes create preferential void alignment directions that persist throughout the material structure. The 0-30° orientation category showed relatively consistent but lower proportions across samples, ranging from 5.50% to 9.02%. Sample 4 exhibited the highest proportion in this category (9.02%), while Sample 1 showed the lowest (5.50%). This pattern suggests that

while most samples exhibit similar distributions of low-angle oriented voids, processing variations can affect the extent of near-horizontal void alignment. The 60-90° orientation range demonstrated the most substantial variation among samples, with percentages ranging from 1.71% (Sample 2) to 4.54% (Sample 4). Sample 4 consistently showed the highest proportions in both extreme orientation categories (0-30° and 60-90°), suggesting less controlled directional void formation compared to other samples. This broader orientation distribution may reflect processing instabilities or non-uniform consolidation conditions.

The orientation alignment index quantified the degree of directional preference across samples, with values ranging from 0.860 (Sample 5) to 0.962 (Sample 1). These consistently high alignment indices (all approaching 1.0) confirm strong preferential orientation in all samples, indicating that additive manufacturing processes inherently create anisotropic void structures. Sample 1 achieved the highest alignment index, suggesting superior processing control that promotes more consistent directional void formation. The orientation tensor analysis revealed that void alignment patterns are not random but follow systematic directional preferences established during layer deposition. The predominant alignment within the 30-60° range suggests that voids form preferentially at angles that reflect the interaction between layer boundaries, material flow directions, and consolidation stresses during processing.

These orientation characteristics demonstrate that void formation in HDPE-HGM syntactic foams is fundamentally anisotropic, with processing-induced directional preferences that significantly influence the three-dimensional void architecture. The systematic variation in orientation distributions across samples indicates that processing parameters can be manipulated to control not only the quantity and morphology of voids but also their directional alignment within the material structure.

The consistent observation of strong orientation alignment across all samples confirms that microstructural anisotropy is an inherent characteristic of additively manufactured syntactic foams, requiring consideration in both material characterization and performance prediction models.

#### 5.4.4 Nearest Neighbour Distance (NND) Analysis

The spatial distribution of voids within each sample was further analyzed using nearest neighbour distance (NND) metrics, which help evaluate how closely voids are spaced relative to each other. As illustrated in Table 8, Sample 1 exhibited the lowest mean NND of 17.61  $\mu\text{m}$ , suggesting a relatively uniform and tightly packed void distribution. In contrast, Sample 4 had the highest mean NND at 32.30  $\mu\text{m}$ , indicating that voids were generally farther apart in this sample. The standard deviation of the NND values followed a similar trend, with Sample 1 showing the lowest variation at 7.30  $\mu\text{m}$  and Sample 4 the highest at 13.77  $\mu\text{m}$ .

Table 8 : Spatial distribution analysis of voids showing nearest neighbor distance (NND) statistics, regularity index (R Index), and coefficient of variation ( $C_V$ ) for each sample.

| Sample ID | Mean NND | Std NND | R Index  | $C_V$ |
|-----------|----------|---------|----------|-------|
| 1         | 17.6062  | 7.3009  | 1.1515   | 0.415 |
| 2         | 25.4202  | 10.7893 | 16.12213 | 0.425 |
| 3         | 25.3353  | 10.7446 | 1.1706   | 0.424 |
| 4         | 32.3046  | 13.7694 | 88.1204  | 0.426 |
| 5         | 19.4537  | 12.346  | 63.34038 | 0.635 |

The  $C_V$  is a dimensionless ratio of the standard deviation to the mean, providing a measure of how dispersed or variable the void spacing is relative to the average distance. Generally,  $C_V$  values near zero suggest minimal variation and a highly uniform structure, while values above 0.5 indicate increasing irregularity and heterogeneity. For Sample 1, the  $C_V$  was 0.415, implying consistent void spacing throughout the material. Samples 2, 3, and 4 all showed  $C_V$  values close to 0.425, indicating moderate variation in void spacing. However, Sample 5 stood out with a much higher  $C_V$  of 0.635, indicating less uniformity and more chaotic spatial distribution, where tightly clustered voids coexist with larger void-free regions-

#### 5.4.5 Correlation Analysis

Table 9 : Comprehensive structure-property correlation matrix

| Morphological Feature    | UTS Correlation | E Correlation |
|--------------------------|-----------------|---------------|
| Void Volume Fraction     | -0.354          | -0.815        |
| Circular Void %          | +0.857          | +0.840        |
| Elongated Void %         | -0.840          | -0.762        |
| Mean NND                 | -0.835          | -0.673        |
| Void Area Mean           | -0.774          | -0.998        |
| Particle Volume Fraction | +0.968          | +0.586        |
| R-index                  | -0.781          | -0.468        |
| Orientation Alignment    | +0.652          | +0.584        |

The correlation results in Table 9 reveal essential relationships between the microstructural features and mechanical properties. The particle volume fraction used here to calculate the correlations is the one

received by the manufacturer of the filaments, to ensure no reliance on the micro-CT data, as it has already been discussed that it is not accurate.

Strong positive correlations can be observed for  $V_{\text{Particle}}$  and UTS (+0.968), indicating that higher particle content dramatically improves tensile strength. The same observation is made in Figure 4.10, Section 4.2.2, where samples 1, 3, and 5, which comprise higher  $V_{\text{Particle}}$ , exhibit higher UTS. This finding also aligns with established research on syntactic foam composites, where Yousaf et al. demonstrated that increasing the microsphere volume fraction from 2% to 40% in polyurethane-based syntactic foams consistently enhanced the tensile strength properties. This effect was attributed to the microspheres' ability to transfer load and minimize crack propagation through the matrix [12].  $V_{\text{Particle}}$  also showed a positive relationship with Young's Modulus (E), aligning with Huang et. al's work on epoxy-based syntactic foams. Our correlation analysis revealed a strong positive relationship ( $r = 0.586$ ) between particle volume fraction and Young's modulus, indicating that a higher particle content significantly enhances the material's elastic stiffness. This finding aligns with established research on syntactic foam composites, where Huang et al., proved that E consistently increases with glass microballoon volume fractions ranging from 10% to 40% in epoxy-based syntactic foams. The authors attributed this strengthening effect to microballoons acting as a hardening phase due to their considerably higher modulus (60 GPa) compared to the epoxy matrix (2.2 GPa). Finite element analysis showed that microballoons bear the majority of the applied loads, thereby confirming the rule of mixtures observed in our data [29].

Circular voids also showed positive correlations with both strength properties (+0.857 UTS, +0.840 E), while elongated voids demonstrated negative correlations (-0.840 UTS, -0.762 E). Similar results were reported by Yang et al., who discovered that stress concentration factors around elongated voids are significantly higher than those of their circular counterparts [30]. Other positive correlations include the orientation alignment index with E (+0.584) and UTS (+0.652). The Orientation Alignment Index showed strong positive correlations with E ( $r = +0.895$ ) and moderate correlations with UTS ( $r = +0.652$ ), indicating that better microstructural alignment significantly improves stiffness more than strength. This pattern suggests that elastic behaviour is highly sensitive to alignment, while failure mechanisms are controlled by multiple factors, including defects and interfacial bonding [31]. Moreover, this behaviour extends to void shape evolution, wherein spherical voids transform into ellipsoids with growth rates and eccentricity that depend on both the stress state and microstructural orientation [32].

Strong negative correlations were observed in the void area mean with both strength properties, as expected (-0.998 E, -0.774 UTS) [33]. The negative correlations of Mean NND with mechanical properties (-0.835 UTS, -0.673 E) define an unexpected yet mechanistically meaningful relationship: larger spacing between voids is correlated with weaker material properties. This is because sparsely spaced voids typically are a

sign of larger individual sizes of voids or more scattered defect regions that are not capable of effective load redistribution with adjacent material. When voids are small and closely spaced, the distribution of the stresses between the areas of defects can be made more effectively by the matrix. The extreme stress concentrations depend on the square root of the void size when individual large voids occur. These heterogeneous microstructures, characterized by large nearest neighbour distances, create localized regions that carry disproportionate loads, ultimately leading to premature failure. From a processing perspective, high NND values are likely to be indicative of improper control over the processing conditions, such as inadequate mixing, temperature gradients, or low consolidation, making the parameter both a quality measure and a microstructural descriptor of the processing. The differential mechanical property versus void volume fraction behaviour provides information regarding specific failure mechanisms.  $E$  is very strongly correlated with void content (-0.815), as is predicted by classical composite theory, where the elastic modulus acts by the rule of mixtures. Voids add zero stiffness, so any void content linearly decreases the overall modulus due to homogenization effects. Conversely, UTS is weakly related (-0.354) to total void volume so that tensile failure is governed by critical flaw size and stress concentration rather than bulk void content. This means that several large voids with equal total volume are much more detrimental than numerous small voids, as the material may redistribute stress around isolated minor flaws by bypassing loads. Practical significance is that higher void volume is acceptable if void size and distribution are under reasonable control, with emphasis placed on void quality rather than quantity in applications where strength is critical.

The R-index parameter is more closely related to UTS (-0.781) than to  $E$  (-0.468), indicating that this microstructural parameter assesses the geometric complexity or uniformity of particle distribution. It influences failure mechanisms rather than elastic response. This difference in sensitivity suggests that crack initiation depends more on local microstructural inhomogeneities than on overall elastic behaviour, with stress concentrations at particle/matrix interfaces and crack paths being affected by geometric configurations, as indicated by the R-index. The stronger link with strength properties implies that optimizing the R-index may be more critical in applications where tensile failure is a key concern. Meanwhile, the elastic modulus is less affected by such geometric changes.

The correlation patterns show that  $E$  and UTS respond to different microstructural features through separate physical mechanisms. Stiffness primarily depends on the volume fraction and aligns with traditional composite theory, being sensitive to overall void content and geometric shape factors through well-known homogenization methods. Tensile strength is influenced by the distribution of flaws and defects, highlighting the importance of defect placement over the total number of defects. The stronger correlation between UTS and particle content and void distribution indicates that these factors influence crack initiation

and growth under tensile stress. Meanwhile, E is more consistently affected by the overall composition. This relationship is further visualized in Figure 5.4, which presents a correlation heatmap highlighting the varying degrees of association between key microstructural metrics and mechanical properties such as ultimate tensile strength (UTS) and Young's modulus (E). Therefore, successful microstructural design involves balancing void quality over quantity, especially in strength-critical applications, and recognizing that different properties respond differently to microstructural changes.

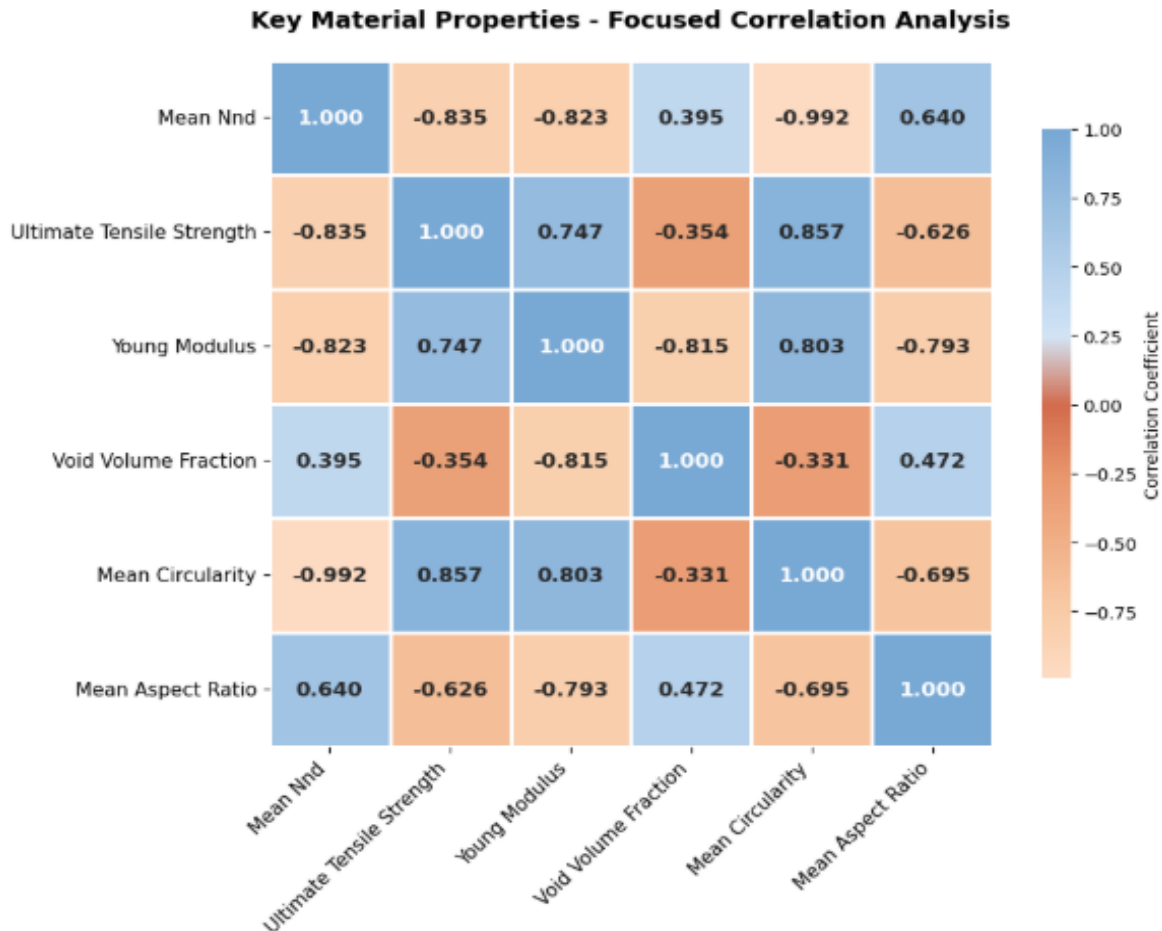


Figure 5.4 : Pearson correlation matrix illustrating the correlations between microstructural parameters (Mean NND, Void Volume Fraction, Mean Circularity, Mean Aspect Ratio) and mechanical characteristics (Ultimate Tensile Strength, Young's Modulus). The high negative and positive correlations are indicated and demonstrate the significant structure–property trends in HDPE-HGM syntactic foams. Colour saturation indicates the strength of the correlation between the two variables.

The microstructural feature versus mechanical property correlation plot, represented by the heatmap, reveals a variety of interesting correlations that enhance understanding of how void features influence material performance. The most notable finding is the extremely high negative correlation between mean

circularity and mean nearest neighbour distance (Mean NND),  $r = -0.992$ , which indicates that as voids become more dispersed within the matrix, their shapes become progressively less circular. This distribution also affects mechanical properties, as indicated by the strong negative correlations between Mean NND and ultimate tensile strength (UTS,  $r = -0.835$ ) and Young's modulus (E,  $r = -0.823$ ) [34]. This means that when voids are more spaced out, the material tends to be less stiff and strong, likely due to a more disordered stress transfer path through the matrix. The shape of voids is also critical for performance. Mean circularity is highly correlated with higher values of UTS ( $r = +0.857$ ) and E ( $r = +0.803$ ), indicating that the mechanical strength and stiffness are greater when voids are more circular. This is likely because circular voids produce more uniform stress fields, reducing stress concentrations, unlike elongated or irregular voids. Conversely, the mean aspect ratio, which measures void elongation, is inversely correlated with UTS ( $r = -0.626$ ) and E ( $r = -0.793$ ), further supporting the idea that elongated voids concentrate stress and weaken the mechanical integrity of the foam. Although void volume fraction, often used as a standard measure of porosity, has a less significant correlation with UTS ( $r = +0.354$ ), it shows a strong correlation with E ( $r = -0.815$ ). This suggests that while volume fraction does influence mechanical properties, its effect is less significant than that of spatial and shape-based void features. Additionally, analysis indicates that certain morphological features are correlated with one another. There is a moderate correlation between Mean NND and aspect ratio ( $r = +0.640$ ), and a negative correlation between circularity and aspect ratio ( $r = -0.695$ ), implying inherent structural interactions that can further inform material design. Overall, the work emphasizes that not all voids impact material properties equally, and that identifying the precise shape and distribution of voids, rather than just their quantity, is vital for optimizing the mechanical properties of syntactic foams and similar materials. This understanding forms the basis for developing a more targeted microstructural engineering approach, where characteristics such as circularity, spacing, and aspect ratio can be adjusted to tailor performance.

While the void morphology (orientation, aspect ratio, circularity, and NND) showed correlations with tensile properties, it is important to note that these trends cannot be attributed solely to morphology. Infill percentage, 3D printing parameters (bed temperature, raster orientation, cooling rate), interfacial bonding, crystallinity of HDPE, stress transfer mechanisms, among others, affect the strength of the material. It should also be noted that microsphere composition influences the cooling rate during printing, which in turn affects the crystallinity of the HDPE matrix, while infill percentage not only changes void morphology but also directly impacts interlayer bonding and overall mechanical performance. These factors interact with void structure, meaning that the observed correlations reflect a combined influence of morphology, cooling effects, and process-induced changes in the polymer matrix. For the scope of this study, morphology is analysed as it is one of the key contributors to the strength and stiffness of syntactic foams.

## 5.5 Limitations

One of the primary limitations of the current correlation analysis lies in the dataset size, which consists of only five distinct samples. While this is sufficient for delivering a proof-of-concept for the segmentation section of the work, it restricts the statistical power and the ability to generalize the findings. A larger dataset would enable more secure conclusions, minimize the potential for overfitting to unique trends, and provide statistical power to ascertain whether observed relationships exist through more general material variations.

## 5.6 Conclusion

This work outlines a comprehensive methodology for characterizing the morphology and spatial distribution of voids in HDPE–HGM syntactic foams, with the additional aim of developing initial structure–property correlations through statistical analysis. The combined methodology integrates 3D  $\mu$ CT imaging, automated segmentation, quantitative morphological analyses, and correlation analysis within a specially designed graphical user interface. The methodology allows simultaneous examination of void morphology, spatial distribution, and orientation characteristics, aspects that are rarely integrated within existing studies and makes advanced image analysis accessible to researchers without a deep programming background, thus enabling broader adoption of quantitative microstructural characterization.

The application of the framework revealed strong and sometimes counterintuitive correlations between microstructural features and mechanical properties. Elastic modulus (E) was strongly negatively correlated with void volume fraction ( $-0.815$ ), but ultimate tensile strength (UTS) was less strongly correlated to total void content ( $-0.354$ ). Void shape had a strong influence: circular voids were positively correlated with both UTS ( $+0.857$ ) and E ( $+0.840$ ), but elongated voids were negatively correlated ( $-0.840$  UTS,  $-0.762$  E). Higher infill densities yielded more regular void shapes than intermediate densities. Orientation alignment was high for all samples (indices  $0.860$ – $0.962$ ), with the fully dense samples having the most marked alignment, an aspect associated with enhanced load transfer behaviour but also indicating processing-induced anisotropy that may affect directional performance. The combination of spatial, morphological, and orientation metrics has not just corroborated previous computational predictions but also yielded experimental observations of void behaviour in intricate microstructures. These results highlight the relevance of infill density as an adjustable fabrication parameter for the optimisation of syntactic foam properties while also illustrating the generalisability of the framework developed to composite systems where defect morphology and distribution dictate mechanical response. Together, the calculated metrics present an integrated picture of the internal microstructure, illustrating how fabrication parameters control the number, size, shape, and spatial distribution of voids, and ultimately dictate the mechanical integrity of the final part.

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## 6. Conclusion and Future Work

### 6.1 Summary of Research Contributions

This thesis presents a comprehensive framework for the automatic characterization of void morphology in additively manufactured HDPE-HGM syntactic foams, achieved through the combination of advanced image segmentation methods, morphological analysis, and structure-property correlations. The research addressed outstanding shortcomings in methodologies for syntactic foam characterization by formulating an unsupervised, computationally lightweight pipeline that eliminates the need for manual annotation while achieving a high degree of segmentation precision.

The key contributions of this study can be summarized in the following aspects:

- (1) First, the development of a robust, unsupervised K-means clustering algorithm, refined by neighbourhood improvement, successfully addressed the problem of multi-modal segmentation in  $\mu$ CT images. The algorithm achieved an average F1 score of 0.926 and an IoU of 0.882 for void detection, demonstrating performance comparable to supervised deep learning approaches, without the need for annotated training data, high computational power and user intervention. This achievement represents a significant improvement over conventional thresholding techniques.
- (2) Second, shape-based void classification was able to distinguish between raster voids caused by manufacturing and intentional HGM cavities through morphological descriptors. The aspect ratio and circularity threshold enabled the effective segregation of these disparate void populations, providing key insights into the dual-scale porosity that characterizes AM syntactic foams. This is an important functionality for process optimization and quality control since raster voids and HGM cavities have different contributions to mechanical performance.
- (3) Third, the comprehensive morphological characterization framework quantified not only void volume fractions but also spatial distribution patterns, orientation alignment, and shape characteristics. The orientation tensor analysis revealed strong directional preferences (alignment indices ranging from 0.86 to 0.96) in void formation, confirming the inherently anisotropic nature of FFF-produced structures. The nearest neighbour distance analysis further identified clustering behaviours that vary significantly with processing parameters.
- (4) Fourth, quantitative structure-property relationships showed that the volume fraction of particles was most highly correlated with ultimate tensile strength ( $r = +0.968$ ). At the same time, the mean void area was most strongly related to Young's modulus ( $r = -0.998$ ). The findings indicate that mechanical properties are influenced by various microstructural features, with strength being more sensitive to particle concentration and void distribution, while stiffness is mainly dependent on the overall void content.

- (5) Fifth, a user-friendly GUI was developed using Python, which encompasses all the steps above and serves as a connection between image and property analysis, allowing the user to view voids and particles in static and dynamic image forms, with exportable files that can serve as a starting point to FEA and creating digital twins of syntactic foam samples.

## 6.2 Main Findings and Their Implications

### 6.2.1 Processing-Structure Relationships

The study uncovered multifaceted interactions between infill density and void formation mechanisms. Against traditional expectations, maximum infill density (100%) yielded the most significant void number (381,958 voids) but with the lowest mean void area ( $51.56 \mu\text{m}^2$ ). Composition-dependent behaviour between Component A (0.14 g/cc HGMs, 55.8% volume fraction) and Component B (0.32 g/cc HGMs, 37.6% volume fraction) revealed that microsphere properties play an essential role in void suppression ability. Component A demonstrated improved void management and mechanical response, with a 15-20% increase in strength at equivalent void levels, confirming the validity of HGM selection for syntactic foam design.

### 6.2.2 Morphology-Performance Correlations

Correlation analysis showed that void morphological parameters often have stronger correlations with mechanical properties than traditional void volume fraction measurements. The proportion of circular voids was positively correlated with ultimate tensile strength (UTS) ( $r = +0.857$ ) and Young's modulus ( $r = +0.840$ ), while elongated voids showed negative correlations ( $r = -0.840$  for UTS,  $r = -0.762$  for E). These finding highlights that void quality, not quantity, dictates mechanical performance in syntactic foams.

The differential sensitivity of UTS and Young's modulus to microstructural characteristics has important implications for material design. Whereas Young's modulus adheres to classical composite theory with high sensitivity to void volume fraction ( $r = -0.815$ ), tensile strength is more weakly correlated ( $r = -0.354$ ) and is instead controlled by critical flaw size and stress concentration effects. This allows focused optimization strategies based on whether stiffness or strength is the main design requirement.

### 6.2.3 Practical Applications

The framework developed has direct applications in various industries. For aerospace applications, quantification and control of dual-scale porosity allow the design of structural cores with predictable mechanical properties and reduced weight. The segmentation pipeline that has been automated allows fast quality control in production, which is critical to ensure consistency in safety-critical parts.

In marine applications, the description of the void connectivity patterns, and more specifically the discrete to interconnected network transition below 70% infill density, is vital in predicting water ingress and long-

term durability. In the automotive industry, the capability to engineer energy absorption behavior through controlled void morphology is exploited, where the GUI-based analysis environment facilitates short prototyping and optimization cycles.

### 6.3 Limitations and Challenges

Despite the significant advancements outlined in this thesis, several limitations require acknowledgment.

- (1) Resolution limitations: The  $\mu$ CT voxel size of 4.8  $\mu\text{m}$ , although sufficient for void characterization, is near the limit for resolving HGM wall thickness accurately ( $\sim 0.37 \mu\text{m}$ ). This limitation caused systematic underestimation of particle volume fractions and made quantification of microsphere wall integrity unreliable. Future research would be improved by higher resolution imaging or complementary characterization methods.
- (2) Dataset size: Annotate correlation analysis was conducted on five independent samples, which was sufficient for proof-of-concept, but limited statistical power and generalizability. A larger dataset that covers wider ranges of processing conditions and material compositions would increase the robustness of the discovered structure-property relationships.
- (3) Material system scope: The model was formulated and validated for HDPE-HGM syntactic foams. Although the approach is theoretically valid for other polymer-microsphere systems, validation for various matrix materials and types of fillers is required to ensure wider applicability.

### 6.4 Future Research Directions

The findings of this study identify several areas for future research and development. An immediate step that would be undertaken in this regard would be to increase the dataset significantly. With a greater and more diverse collection of samples varying in infill density, particle content, orientation, and void shape, there can be an improvement in the robustness and extensibility of the correlation analysis. A larger dataset would also reduce the statistical noise present in small-sample treatments and allow for more intense observation of minor trends, which would not be observable using limited data.

Going beyond the current structure-property correlations, future research can utilize machine learning models to predict the most critical mechanical properties, such as Ultimate Tensile Strength (UTS) and Young's Modulus (E), from microstructural properties. Regression-based approaches, such as multiple linear regression, ridge regression, lasso regression, and decision trees, can be used to identify feature importance and enhance prediction accuracy. Hybrid approaches that combine data-driven approaches with physics-based models can be developed to embed domain knowledge within the learning process. This would allow predictions to remain connected to the underlying material response mechanism while

leveraging the pattern recognition capability of machine learning. High resolution scans can be used, which although are longer to complete, increase the accuracy and prevent underestimation of voids and particles.

Moreover, this approach can be extended to other expansive shell methods, particularly expansive shells, which are polymeric, provided the contrast is sufficient post scanning these samples on a  $\mu$ CT setup. Lower contrast in  $\mu$ CT scanned images adds an additional layer of challenge, due to difficulty in identifying different phases as their grayscale intensities are similar, this is usually in the case of using same material as matrix and reinforcement, like SiC-SiC composites, or if the atomic number of the matrix and the reinforcement are similar, examples include titanium-titanium carbide composites and aluminum-aluminum oxide ( $\text{Al}_2\text{O}_3$ ) composites, both combining elements with closely matching atomic numbers. Before using this method of segmentation, increasing contrast by staining samples with solutions containing heavy elements (e.g., iodine, tungsten, molybdenum) that have a higher density and a greater X-ray absorption rate. Otherwise, contrast can be increased by dedicated contrast sliders found in photo editing software like Adobe Lightroom, online editors such as Canva and Adobe Express or using OpenCV library in Python, which has several built-in functions to increase contrast.

Thirdly, proposed framework in this study produces mesh files and morphological descriptors as outputs. These mesh files can be directly used as inputs for finite element analysis (FEA), enabling the simulation of material behaviour under different loading conditions. By linking the segmentation-derived meshes with FEA software, future research can connect microstructural data with macro-scale mechanical responses, creating a comprehensive digital process for material characterization and performance forecasting. This approach could significantly advance digital twin technology and *in silico* material design strategies.

## 6.5 Final Observations

This research outlines a comprehensive methodology for characterizing the morphology and spatial distribution of voids in HDPE-HGM syntactic foams, with the further aim of deriving preliminary structure-property correlations through statistical analysis. The integrated method, which combines  $\mu$ CT imaging, automated segmentation, quantitative morphological analysis, and correlation analysis within a tailored graphical user interface, enables the simultaneous analysis of void morphology, spatial distribution, and orientation features. This platform bridges microstructural imaging and mechanical performance prediction, providing a toolbox that can be easily integrated into industrial quality control and material design optimization.

Application of the methodology demonstrated correlations between microstructural features and mechanical properties. Elastic modulus (E) was strongly negatively correlated with void volume fraction, whereas ultimate tensile strength (UTS) was weakly correlated with total void content. The void shape had

a strong effect, circular voids were positively correlated with both UTS and E, while elongated voids were negatively correlated. An unexpected inverse correlation between the mean nearest neighbour distance and mechanical performance may imply that increased void spacing is an indicator of the presence of significant defects or defect clusters that disrupt load transfer. These results affirm that successful syntactic foam design needs to consider not only the void volume fraction but also the entire morphological terrain, including shape, size distribution, spatial distribution, and orientation. High infill densities yielded many small, well-oriented, and homogeneously distributed circular voids ideal for mechanical performance. In contrast, intermediate densities gave fewer, larger, and more irregular voids characteristic of less favourable processing conditions. Orientation alignment was high in all samples, particularly in fully dense samples, indicating processing-induced anisotropy with potential implications for directional performance. The creation of a user-friendly, inclusive GUI enables the use of sophisticated characterization methodologies for researchers with limited programming experience, incorporating all steps in the workflow from image segmentation through to quantitative analysis. Through the provision of real-time feedback, visualization of voids and particles in both static and dynamic formats, and exportation of mesh files in formats appropriate for finite element analysis and digital twin development, this software bridges the gap between microstructural analysis and mechanical property assessment and predictive modelling directly.

The integration of unsupervised machine learning, morphological analysis, and structure–property correlation demonstrated herein is a move away from manual characterization techniques. In addition to enhancing efficiency, this methodology reveals subtle microstructural features and correlations. As additive manufacturing transitions to fully digital, design-to-production processes, automated, data-driven characterization systems like this will be critical to ensuring the consistency, reliability, and performance optimization of syntactic foams and other complex composite materials.